

Validity of Inference Methods for Causal Trees: The Honest Role of Imbalanced, Small Leaves

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Contents

1	Introduction	1
2	Literature Review	2
3	Theoretical Overview	4
3.1	Estimator of treatment effects	4
3.2	Tree framework	4
3.3	Criterion function for adaptive estimation	5
3.4	Criterion function for honest estimation	6
3.5	Reasoning behind imbalanced splits	8
3.6	Asymptotic confidence intervals	9
4	Main Aspects of Cattaneo et al. [2025]	10
4.1	Assumptions on the data generating process	10
4.2	Uniform convergence	11
4.3	Probability of extreme splits	12
5	Tree Calibration	14
5.1	Bias-Variance Tradeoff	14
5.2	α - regularity	14
5.3	Tree depth vs. minimum leaf size	14
5.4	Minimum number of treated and control units per leaf	14
5.5	Choice of tree depth	15
5.6	Choice of package	15
6	Simulations	16
6.1	Simulation Design	16
6.2	Results	17
6.2.1	Adaptive estimation	17
6.2.2	Honest estimation	19
7	Conclusion	20
A	Coverage Ratios	24
A.1	Entire support	24
A.1.1	Adaptive estimation	24
A.1.2	Honest estimation with NaN coverages as NaN	27
A.1.3	Honest estimation with NaN coverages as 0	30
A.2	Extreme absolute deviations	33
A.2.1	Adaptive estimation	33
A.2.2	Honest estimation with NaN coverages as NaN	36

A.2.3	Honest estimation with NaN coverages as 0	39
B	Weighted Coverage Ratios	42
C	Confidence Interval Analysis	44
C.1	Entire support	44
C.1.1	Adaptive estimation	44
C.1.2	Honest estimation	47
C.2	Extreme absolute deviations	50
C.2.1	Adaptive estimation	50
C.2.2	Honest estimation	51
D	Relative Frequency Distribution of Leaf Means	52
D.1	Adaptive Estimation	52
D.2	Honest estimation with NaN coverages included	55
E	Relative Frequency Distribution of the Number of Samples per Leaf Mean	58
E.1	Adaptive estimation	58
E.2	Honest estimation with NaN coverages included	61
F	Mean Leaf Sizes	64
G	Share of Leaves with Coverage for Honest Estimation	68

1 Introduction

When it comes to detecting subgroups that differ with respect to their treatment effects, researchers need to be careful when doing this manually. To avoid spurious manual splits, algorithms like tree-based methods for causal inference that discover treatment effect heterogeneity are a popular alternative. Causal trees have first been introduced by Athey and Imbens [2016] and are based on Classification and Regression Trees (CART) algorithms which are used to detect heterogeneity in outcomes.

The resulting causal tree contains several leaves and within each leaf the units share the same treatment effect which is estimated via an appropriate estimator. An important question for researchers is whether the confidence intervals that are applied on the estimator are asymptotically valid. A necessary condition for this is that the size of the leaves grows over the entire support with the total sample size.

A recent paper from Cattaneo et al. [2025] claims the opposite. Centrally, the authors state that uniform convergence of causal tree estimators is arbitrarily slow. This means that at least for some regions of the covariate space, the treatment effect estimator does not converge to its true value as the sample size goes to infinity. The reason for this is seen in imbalanced splits which are close to the boundary of the support and lead to small, imbalanced leaves. This fundamental flaw has been described before (Breiman et al. [1984]) and is also referred to as end-cut preference (Ishwaran [2015]).

Moreover, causal trees just like decision trees are greedy since they do not look ahead when performing splits (Hastie et al. [2009, p. 292]). This exacerbates the effect of an early imbalanced split further, since an already small node may get even smaller at later splits. For the resulting small child nodes, the effective sample size as the authors claim may not even increase with the total sample size and a normal distributional approximation not hold. Consequently, standard confidence intervals i.e. inference would not be valid.

When introducing causal trees, Athey and Imbens [2016] crucially introduced the concept of honesty. By honesty the authors understand that not the same part of the data is used for construction of the tree as for the estimation of treatment effects. This means that the resulting leaves of the tree can be treated as exogenously given. Honest estimation of causal trees is supposed to preserve the validity of confidence intervals. However, as Cattaneo et al. [2025] claim, convergence rates for honest estimators only improve negligibly such that the tendency of the estimator for imbalanced splits continues to prevent asymptotically valid confidence intervals.

Aligned with the paper, we will not restrict the tree growing process more than necessary. This serves the purpose to not prevent the tree from splitting imbalancedly, but also to avoid misspecification bias. We will only restrict the tree growing process such that we promote well-defined standard errors. To that end, as a major contribution, we adjust the package (causalml) which we use to implement the causal tree. We add parameters to control the minimum number of treated and control units per leaf.

The data generating process is as well as non-restrictive as possible. We impose neither a

dependence of the features on the outcome nor on the treatment status or treatment effect. This will allow us to trace back any peculiarities in the splitting process to flaws in the algorithm. We eliminate the possibility for extensive splitting at the boundary due to high heterogeneity or boundary bias.

The main contribution of this paper is to examine the properties of confidence intervals of causal tree estimators in the light of the theoretical results of Cattaneo et al. [2025] over increasing sample sizes. We take a look at $n = 1000$, $n = 5000$ and $n = 25000$. This will be done via a simulation study. We compare the performance of adaptive (non-honest) and honest estimators. Centrally, we show that the coverage for honest estimators improves in sample size while it decays for adaptive estimators. The reason for the improvement lies in a lower mean absolute deviation of the treatment effect from its estimate and a high enough standard error.

For a weighted average coverage which weighs by the number of samples in the leaves we see that coverages are improved. For honest estimation and the two larger sample sizes and moderate depths, the metric is essentially equal to the nominal value.

Therefore, we can conclude that the consequences of the theoretical claims in the paper are mild for two reasons. First, the leaves which do not have asymptotically valid confidence intervals are by nature small and hence constitute only a small partition of the support. Second, the authors ignore the much higher baseline accuracy of honest estimators which is why the slow convergence matters less.

Furthermore, we will shed light on the behavior of the splitting process and illustrate that the tree splits more frequently close to the boundary which leads to smaller leaf sizes. Related to that, we will highlight that the reason for imbalanced splits lies in the fitting of noise which occurs specifically at the boundary. Furthermore, we will examine a major problem of adaptive and a major problem of honest estimation. Adaptive estimation suffers from massively inflated standard errors close the boundary which increases coverage. Honest estimation suffers from leaves with less than two treated or control units for which a valid coverage cannot be calculated.

The replication code for this project is publicly available on GitHub at https://github.com/holyflow/causal_tree_master_thesis.

2 Literature Review

Decision trees to predict outcomes have a long history and have first been introduced by Morgan and Sonquist [1963] and Breiman et al. [1984]. By splitting the support they create a discrete outcome space. They are known to possess an end-cut preference which is the property of favoring splits near the edge for noisy variables (Breiman et al. [1984], Ishwaran [2015], Zhang and Luo [2025]). This issue has been examined extensively in the literature. Building on top of the literature regarding decision trees, Cattaneo et al. [2025] examined the issue of end-cut preference and its effects on the convergence of the treatment effect estimator theoretically.

Importantly, the concept of honesty is not central to decision trees. Since the true outcome is known, a prediction from a decision tree can easily be evaluated via a validation set. Hence, it is

not necessary to estimate Y in another part of the data than the one which is used for construction of the tree. For causal inference the ground truth (the true treatment effect) is not known. This makes validation impossible and introduces the need to estimate the treatment effect on separate data to avoid overfitting. Therefore, the existing literature regarding end-cut preference has not dealt with the effect of honest estimation. Cattaneo et al. [2025] were the first to claim that honest estimation cannot significantly alleviate the problem induced by excessive splitting at the boundaries.

There are also claims in the literature that sample splitting following from honesty may reduce the model’s ability to detect effect variation. The justification is that, when treatment effects differ significantly across individuals and the data is rich enough to detect those differences, honest estimation can lead to less accurate models. Using the entire dataset and hence more information for splitting would make it easier to detect the heterogeneity (Hou and Fernández-Loría [2025]).

Athey and Imbens [2016] have translated decision trees to the causal inference framework. In research causal trees are usually not applied by its own but as a component of a causal forest. Causal forests have been introduced by Wager and Athey [2018] and Athey et al. [2019]. The advantage of Causal Forests is that a single tree often suffers from high variance and sharp decision boundaries. Because of that, it is often better to generate many different trees and average the predictions from many single optimized trees. This smooths the treatment effect estimates across the observations (Wager and Athey [2018]). The estimates of Causal Forests based on Wager and Athey [2018] are asymptotically normal and allow the calculation of asymptotically valid confidence intervals.

Causal forests are a part of the literature in Machine Learning for finding heterogeneous treatment effects in the data (Smith [2022]). They are one of the most popular methods in the field of Causal Machine Learning methods and are growing in use (Rehill [2025]).

Usually, the restrictions imposed on the tree growing process are quite strict. A popular restriction is called α -regularity which means that each split leaves at least a fraction of α of the data on both sides of the split (Wager and Athey [2018]). In our case of a constant treatment effect this assumption is not harmful. However, the treatment effect may be highly heterogeneous in which case misspecification bias could occur under α -regularity. Therefore, we will not impose such an assumption following the approach in Cattaneo et al. [2025].

We will evaluate the performance of causal trees via applying a simulation study. A simulation study to compare the coverage of different methods was for example done in Slade and Naylor [2020]. They have the key strength of providing the ability to understand the behavior of statistical methods because some “truth” (usually some parameter/s of interest) is known from the process of generating the data (Morris et al. [2019]). Knowing the ground truth is especially useful in the context of causal inference since treatment effects are not known when a method is applied on empirical data. That is why a simulation study enables to compare the performance of the method.

3 Theoretical Overview

3.1 Estimator of treatment effects

Potential outcomes are defined according to the causal model of Rubin [1974] for a binary treatment $W_i \in \{0, 1\}$.

$$Y_i^{\text{obs}} = Y_i(W_i) = \begin{cases} Y_i(0) & \text{if } W_i = 0, \\ Y_i(1) & \text{if } W_i = 1. \end{cases} \quad (1)$$

The Conditional Average Treatment Effect (CATE) is defined as the difference of the outcome of treated individuals minus the outcome of untreated individuals conditional on the covariates $\mathbf{X}_i$.

$$\tau(\mathbf{x}) = \mathbb{E}[Y_i(1) - Y_i(0) \mid \mathbf{X}_i = \mathbf{x}]. \quad (2)$$

Based on the above definition of CATE we can define our estimator as a difference-in-means (DIM) estimator according to Cattaneo et al. [2025]. For this purpose we use the sample analog of the above definition.

Suppose $\mathcal{T}$ is the tree used and $\mathcal{D}_\tau = \{(Y_i, W_i, \mathbf{X}_i^\top) : i = 1, 2, \dots, n_\tau\}$ with $n_\tau \leq n$, is the dataset used. Let $\mathbf{t}$ be the unique terminal node in $\mathcal{T}$ containing $\mathbf{x} \in \mathcal{X}$. We want to think of the leaves as small enough that the $(Y_i, \mathbf{X}_i)$ pairs have come from a randomized experiment. Then, the treatment effect within the leaves can be estimated as a difference in means (Wager and Athey [2018]):

The Difference-in-Means (DIM) estimator is

$$\hat{\tau}_{\text{DIM}}(\mathbf{x}; \mathcal{T}, \mathcal{D}_\tau) = \frac{1}{n_1(\mathbf{t})} \sum_{i: \mathbf{X}_i \in \mathbf{t}} W_i Y_i - \frac{1}{n_0(\mathbf{t})} \sum_{i: \mathbf{X}_i \in \mathbf{t}} (1 - W_i) Y_i, \quad (3)$$

where $n_w(\mathbf{t}) = \sum_{i=1}^{n_\tau} \mathbb{1}(\mathbf{X}_i \in \mathbf{t}, W_i = w)$, for $w = 0, 1$, are the “local” sample sizes. We set $\hat{\tau}_{\text{DIM}}(\mathbf{x}; \mathcal{T}, \mathcal{D}_\tau) = 0$ whenever $n_0(\mathbf{t}) = 0$ or $n_1(\mathbf{t}) = 0$.

Since we abstract from selection bias, an Inverse Probability Weighting (IPW) estimator is not relevant.

3.2 Tree framework

At a broader level a tree is being constructed as follows. The original node of the tree consists of the entire sample and is split into two child nodes which contain a subset of the data. By recursive binary splitting this procedure is continued until a certain stopping criterion is reached. This is either a certain tree depth or a minimum leaf size. The sample has then been split into a multitude of subsets i.e. final nodes to which we also refer as leaves. To separate the data into two child nodes at each parent node of the tree a criterion function is applied (Hastie et al.

[2009, p. 309 ff.]

All observations within a leaf are assigned a treatment effect that corresponds to the Average Treatment Effect (ATE) within that leaf. Conditioning on the covariates within that leaf we can refer to this ATE as the Conditional Average Treatment Effect (CATE).

3.3 Criterion function for adaptive estimation

At each node of the tree the algorithm tries to maximize an objective function to which we also refer as a criterion function. The goal is to discover as much treatment heterogeneity as possible.

The main part of Athey and Imbens [2016] deals with deriving criteria based on the Mean Squared Error (MSE). We will first focus on the adaptive case i.e. the case where the same sample is used for creating the tree and estimating treatment effects. As usual evaluation of the MSE is done based on test set. The MSE for a given treatment effect $\hat{\tau}$ within a leaf looks as follows.

$$\begin{aligned} \text{MSE}_{\tau}(\mathcal{S}^{\text{te}}, \mathcal{S}^{\text{tr}}, \Pi) &\equiv \frac{1}{\#(\mathcal{S}^{\text{te}})} \sum_{i \in \mathcal{S}^{\text{te}}} \left\{ (\tau_i - \hat{\tau}(\mathbf{X}_i; \mathcal{S}^{\text{tr}}, \Pi))^2 \right\} \\ &= \frac{1}{\#(\mathcal{S}^{\text{te}})} \sum_{i \in \mathcal{S}^{\text{te}}} \left(\tau_i^2 - 2\tau_i \hat{\tau}(\mathbf{X}_i; \mathcal{S}^{\text{tr}}, \Pi) + \hat{\tau}^2(\mathbf{X}_i; \mathcal{S}^{\text{tr}}, \Pi) \right) \end{aligned} \quad (4)$$

Problematically, this criterion is infeasible because we do not observe the counterfactual outcome for individuals and for that reason we also do not observe the individual treatment effects τ_i . This is also referred to as the fundamental problem of causal inference (Holland [1986]). Therefore, some adjustments of the above criterion are necessary.

When formulating their MSE criterion the authors adjust the Expected Mean Squared Error (EMSE) by $-E[\tau_i^2]$ which does not depend on $\mathbf{X}_i$ and hence does not affect how the splits are made. The authors also divide by $N^{\text{tr}} = \#(\mathcal{S}^{\text{tr}})$. Once the number of elements in the training and test split is set dividing by N^{tr} or N^{te} does not change which split Π minimizes the error.

However, we still need an estimator for τ_i in $-2\tau_i \hat{\tau}(\mathbf{X}_i; \mathcal{S}^{\text{tr}}, \Pi)$. For that we can use $\hat{\tau}_i$ as an unbiased estimator such that $\mathbb{E}_{\mathcal{S}^{\text{te}}} [\tau_i \mid i \in \mathcal{S}^{\text{te}} : i \in \ell(\mathbf{x}, \Pi)] = \mathbb{E}_{\mathcal{S}^{\text{te}}} [\hat{\tau}(\mathbf{x}; \mathcal{S}^{\text{te}}, \Pi)]$. By applying the sample analogs we receive the following adjusted MSE:

$$\begin{aligned} \widehat{\text{MSE}}_{\tau}(\mathcal{S}^{\text{te}}, \mathcal{S}^{\text{tr}}, \Pi) \\ = -\frac{2}{N^{\text{tr}}} \sum_{i \in \mathcal{S}^{\text{te}}} \hat{\tau}(\mathbf{X}_i; \mathcal{S}^{\text{te}}, \Pi) \cdot \hat{\tau}(\mathbf{X}_i; \mathcal{S}^{\text{tr}}, \Pi) + \frac{1}{N^{\text{tr}}} \sum_{i \in \mathcal{S}^{\text{te}}} \hat{\tau}^2(\mathbf{X}_i; \mathcal{S}^{\text{tr}}, \Pi) \end{aligned}$$

Instead of evaluating the MSE using a test set we can evaluate the error in-sample which leads to $-\widehat{\text{MSE}}_{\tau}(\mathcal{S}^{\text{tr}}, \mathcal{S}^{\text{tr}}, \Pi)$ as the criterion.

$$-\widehat{\text{MSE}}_{\mathbf{X}_i, \tau}(\mathcal{S}^{\text{tr}}, \mathcal{S}^{\text{tr}}, \Pi) = \frac{1}{N^{\text{tr}}} \sum_{i \in \mathcal{S}^{\text{tr}}} \hat{\tau}^2(\mathbf{X}_i; \mathcal{S}^{\text{tr}}, \Pi) \quad (5)$$

From this criterion function we can directly derive what Cattaneo et al. [2025] call variance maximization. When $-\widehat{\text{MSE}}_\tau(\mathcal{S}^{\text{tr}}, \mathcal{S}^{\text{tr}}, \Pi)$ is maximized and tries to find the best split we essentially maximize the variance of the estimator between the child nodes. This is done by having as different as possible treatment effects in the nodes, since we sum up squared treatment effects. The algorithm maximizes the criterion in each node and not the whole tree at once, it is greedy. Because of that, when to decide where to make a split we want to get the sum of the criterion over both child nodes as large as possible. Said differently, we want to explain as much heterogeneity in the treatment effects as possible by splitting one group with one treatment effect into two different groups with two treatment effects. The two child nodes L, R have a summed up error (Sum of Squared Treatment effects) of

$$\text{SST}_{\text{after}} = \sum_{i \in t_L} \hat{\tau}_L^2 + \sum_{i \in t_R} \hat{\tau}_R^2 = n(t_L) \hat{\tau}_L^2 + n(t_R) \hat{\tau}_R^2$$

However, where and if to make the split should be evaluated with respect to the variance/heterogeneity in τ_i that is already explained in the parent node. We can express the error in the parent node in terms of the two treatment effects in the child node.

$$\text{SST}_{\text{before}} = \sum_{i \in t} \hat{\tau}_t^2 = n(t) \hat{\tau}_t^2 = n(t) \left(\frac{n(t_L) \hat{\tau}_L + n(t_R) \hat{\tau}_R}{n(t)} \right)^2$$

Subtracting $\text{SST}_{\text{before}}$ from $\text{SST}_{\text{after}}$ yields

$$\begin{aligned} \text{Gain} &= [n(t_L) \hat{\tau}_L^2 + n(t_R) \hat{\tau}_R^2] - n(t) \hat{\tau}_t^2 \\ &= n(t_L) \hat{\tau}_L^2 + n(t_R) \hat{\tau}_R^2 - n(t) \left(\frac{n(t_L) \hat{\tau}_L + n(t_R) \hat{\tau}_R}{n(t)} \right)^2 \\ &= n(t_L) \hat{\tau}_L^2 + n(t_R) \hat{\tau}_R^2 - \frac{(n(t_L) \hat{\tau}_L + n(t_R) \hat{\tau}_R)^2}{n(t)} \\ &= \frac{n(t_L) n(t_R)}{n(t)} (\hat{\tau}_L - \hat{\tau}_R)^2 \end{aligned} \tag{6}$$

The last step is achieved through algebraic simplification by expanding the square and creating a common denominator. This yields that the criterion which is supposed to be maximized can be written as a weighted difference of the squared treatment effects.

3.4 Criterion function for honest estimation

In the paper Athey and Imbens [2016] the authors centrally apply a procedure which they call honesty. It means to perform the splitting of features on one part of the sample and to estimate treatment effects on the other part of the sample. The downside of this is reduced efficiency since the tree is built on a lower sample size. The advantage is that estimation of treatment effects is now exogenous. This implies that the algorithm does not risk to fit the tree based on spurious extreme values (noise). Hence, the authors argue that the approach eliminates bias.

The splitting criterion is modified and considers that the estimates are now unbiased. Moreover, the authors derive a penalty term that is supposed to prevent high variances in leaf estimates that stem from small leaf sizes from only using part of the sample for estimation.

To derive the modified criterion, we first rewrite the original MSE criterion and take the expectation to receive the Expected Mean Squared Error (EMSE).

$$\begin{aligned}
-\text{EMSE}_\tau(\Pi) &= -\mathbb{E}_{(\tau_i, \mathbf{X}_i), \mathcal{S}^{\text{est}}} \left[(\tau_i - \tau(\mathbf{X}_i; \Pi))^2 - \tau_i^2 \right] \\
&\quad - \mathbb{E}_{\mathbf{X}_i, \mathcal{S}^{\text{est}}} \left[(\hat{\tau}(\mathbf{X}_i; \mathcal{S}^{\text{est}}, \Pi) - \tau(\mathbf{X}_i; \Pi))^2 \right] \\
&= \mathbb{E}_{\mathbf{X}_i} \left[\tau^2(\mathbf{X}_i; \Pi) \right] - \mathbb{E}_{\mathcal{S}^{\text{est}}, \mathbf{X}_i} \left[\mathbb{V} \left(\hat{\tau}^2(\mathbf{X}_i; \mathcal{S}^{\text{est}}, \Pi) \right) \right],
\end{aligned} \tag{7}$$

As an estimator for the within-leaf variance $\mathbb{V}(\hat{\tau}^2(\mathbf{X}_i; \mathcal{S}^{\text{est}}, \Pi))$ from the estimation sample, we can estimate its equivalent on the training sample. This ensures that we do not violate the honesty assumption by using data from the estimation sample for the construction of the tree. However, since we want to minimize the variance of the final leaves which will be a function of the estimation sample, we still scale with the number of samples in the leaf of the estimation sample.

$$\hat{\mathbb{V}}(\hat{\tau}(\mathbf{x}; \mathcal{S}^{\text{est}}, \Pi)) \equiv \frac{S_{\text{tr}}^2(\ell(\mathbf{x}; \Pi))}{N^{\text{est}}(\ell(\mathbf{x}; \Pi))}$$

To compute the variance over all leaves, we have to weight with the proportion each leaf has compared to the total sample size of the estimation sample which is $\frac{N^{\text{est}}(l)}{N^{\text{est}}}$.

$$\begin{aligned}
\hat{\mathbb{E}} \left[\hat{\mathbb{V}}(\hat{\tau}(\mathbf{x}; \mathcal{S}^{\text{est}}, \Pi)) \right] &= \sum_{l \in \Pi} \left(\frac{N^{\text{est}}(l)}{N^{\text{est}}} \right) \cdot \left(\frac{S_{\text{tr}}^2(l)}{N^{\text{est}}(l)} \right) \\
&= \sum_{l \in \Pi} \frac{S_{\text{tr}}^2(l)}{N^{\text{est}}} \\
&= \frac{1}{N^{\text{est}}} \sum_{l \in \Pi} S_{\text{tr}}^2(l)
\end{aligned} \tag{8}$$

Furthermore, we need an estimator for $\mathbb{E}_{\mathbf{X}_i} [\tau^2(\mathbf{X}_i; \Pi)]$. From the definition of the variance we know that $\text{Var}(\hat{\tau}) = \mathbb{E}[\hat{\tau}^2] - (\mathbb{E}[\hat{\tau}])^2$ and since $\hat{\tau}$ is an unbiased estimator of τ^2 we have $(\mathbb{E}[\hat{\tau}])^2 = \tau^2$. It follows

$$\tau^2 = \mathbb{E}[\hat{\tau}^2] - \text{Var}(\hat{\tau})$$

Because τ^2 is the true population mean and thus a fixed constant we have $\tau^2 = E[\tau^2]$. As an estimator for $\mathbb{E}[\hat{\tau}^2]$ we use the estimated squared treatment effect from the training sample and as an estimator of $\text{Var}(\hat{\tau})$ we use the estimated variance from the training sample. This

yields

$$\hat{\mathbb{E}} [\tau^2(\mathbf{x}; \Pi)] = \hat{\tau}^2(\mathbf{x}; \mathcal{S}^{\text{tr}}, \Pi) - \frac{S_{\mathcal{S}^{\text{tr}}}^2(\ell(\mathbf{x}; \Pi))}{N^{\text{tr}}(\ell(\mathbf{x}; \Pi))}. \quad (9)$$

Combining the estimators for $\hat{\mathbb{V}}(\hat{\tau}(\mathbf{x}; \mathcal{S}^{\text{est}}, \Pi))$ and $\mathbb{E}_{\mathbf{X}_i} [\tau^2(\mathbf{X}_i; \Pi)]$ yields as the modified criterion for honest splitting

$$\begin{aligned} -\widehat{\text{EMSE}}_{\tau}(\mathcal{S}^{\text{tr}}, N^{\text{est}}, \Pi) &\equiv \frac{1}{N^{\text{tr}}} \sum_{i \in \mathcal{S}^{\text{tr}}} \hat{\tau}^2(\mathbf{X}_i; \mathcal{S}^{\text{tr}}, \Pi) \\ &\quad - \left(\frac{1}{N^{\text{tr}}} + \frac{1}{N^{\text{est}}} \right) \cdot \sum_{\ell \in \Pi} S_{\mathcal{S}^{\text{tr}}}^2(\ell(\mathbf{x}; \Pi)). \end{aligned} \quad (10)$$

We see that differing to the criterion for adaptive estimation we have

$$\left(\frac{1}{N^{\text{tr}}} + \frac{1}{N^{\text{est}}} \right) \cdot \sum_{\ell \in \Pi} S_{\mathcal{S}^{\text{tr}}}^2(\ell(\mathbf{x}; \Pi))$$

as a penalty term to prevent overfitting. The term prevents the tree to grow too granularly since then the variance term would increase and the criterion not be maximized.

Remark. An essential part of Athey and Imbens [2016] is to apply cross-validation to prevent overfitting. Such cross-validation would lead to splitting D_{struct} further to estimate a pruning parameter. However, since we are concerned about imbalanced splits which are supposed to occur with or without cross-validation, we abstract from it following the approach in Cattaneo et al. [2025].

3.5 Reasoning behind imbalanced splits

Weighted variance splitting rules that are known from Classification and Regression Trees are part of the class of weighted splitting rules. They possess an end-cut preference property (Ishwaran [2015]). We have explicitly derived that the algorithm from Athey and Imbens [2016] evaluates a split based on a gain function that assesses how much more variance in the treatment effect can be explained by inducing a split (see equation (6)). It is also based on a variance based splitting rule. The problem regarding imbalanced splits can therefore be seen as equivalent to the issue that is prevalent in CART algorithms.

Given that, also the reason for imbalanced splits is equivalent for tree methods that are predictive models and for models for causal inference. For tree based predictors split points near the boundary isolate very few points. Hence, the sample mean of such a tiny child has a high variance, so even pure noise can create a large difference between child means and an inflated impurity reduction. ”Taking the maximum over many candidate cut-off points amplifies this selection bias—extremal thresholds are disproportionately likely to ‘win’ ” (Zhang and Luo [2025]). In Machine Learning we refer to this issue as overfitting.

Translated to the case of treatment effects, this means that splits close to the boundary can lead to leaves with a strongly deviating treatment effect from the one in the parent node. That is because such leaves isolate few points and are thus more likely to be distorted by noise. The algorithm is then more likely to split close to the boundary since it believes to have found treatment effect heterogeneity.

Centrally, this problem may be significantly less prevalent under honest estimation. Let us say that the algorithm decides to split close to the boundary based on the splitting subsample due to noisy variables in the resulting small leaves. Then, the variables from the estimation subsample do not have to be noisy as well. However, since the leaf size will remain smallish there is still a large potential that the treatment effect estimates within the leaf are distorted by noise.

3.6 Asymptotic confidence intervals

A desired property of the leaf size would be that it scales with the total sample size. By examining the coverage ratio this is essentially what we examine. The reason is that by the Law of Large Numbers the treatment effect estimate within each leaf for a total sample size large enough would be normally distributed. From this asymptotic distribution, we can derive an asymptotic confidence interval by applying the properties of the normal distribution. If the leaf size scales with the total sample size and hence the asymptotic normality holds, we should see coverage ratios close to the nominal value.

In line with Hansen [2022] let $\widehat{C}$ be the asymptotic confidence interval. Then,

$$\widehat{C} = [\hat{\tau} - cs(\hat{\tau}), \quad \hat{\tau} + cs(\hat{\tau})] \quad (11)$$

is an asymptotic confidence interval for the treatment effect estimate within a leaf. $s(\hat{\tau})$ is the standard error of the estimate. The parameter c is the quantile of the standard normal distribution which leads to $c = \Phi^{-1}\left(1 - \frac{0.05}{2}\right) = 1.96$ because we choose a standard 95% confidence interval. The asymptotic confidence interval is accordingly

$$\widehat{C} = [\hat{\tau} - 1.96s(\hat{\tau}), \quad \hat{\tau} + 1.96s(\hat{\tau})] \quad (12)$$

and it holds

$$P(\theta \in C) \rightarrow 0.95 \quad (13)$$

4 Main Aspects of Cattaneo et al. [2025]

4.1 Assumptions on the data generating process

The authors formulate several conditions on the data generating process under which their results are supposed to hold.

Assumption 1 (Data Generating Process). $\mathcal{D} = \{(Y_i, W_i, \mathbf{X}_i^\top) : 1 \leq i \leq n\}$ is a random sample, where $Y_i = W_i Y_i(1) + (1 - W_i) Y_i(0)$, $\mathbf{X}_i = (X_{i,1}, \dots, X_{i,p})^\top$, and the following conditions hold for all $w = 0, 1$ and $i = 1, 2, \dots, n$.

- (i) $(Y_i(0), Y_i(1), \mathbf{X}_i) \perp W_i$, and $\xi = \mathbb{P}(W_i = 1) \in (0, 1)$.
- (ii) $Y_i(w) = \mu_w(\mathbf{X}_i) + \varepsilon_i(w)$, with $\mathbb{E}[\varepsilon_i(w)|\mathbf{X}_i] = 0$ and $\mathbf{X}_i \perp \varepsilon_i(w)$. (14)
- (iii) $\mu_w(\mathbf{x}) = c_w$ for all $\mathbf{x} \in \mathcal{X}$, where c_w is some constant and $\mathcal{X}$ is the support of $\mathbf{X}_i$.
- (iv) $X_{i,1}, \dots, X_{i,p}$ are independent and continuously distributed.
- (v) There exists $\alpha > 0$ such that $\mathbb{E}[\exp(\lambda \varepsilon_i(w))] < \infty$ for all $|\lambda| < 1/\alpha$ and $\mathbb{E}[\varepsilon_i^2(w)] > 0$.

Assumption (i) expresses the independence of treatment with covariates and potential outcomes and thus assumes a randomized experiment (Cattaneo et al. [2025]). Besides that, we enforce an overlap assumption i.e. a positive probability for every individual of getting or not getting the treatment. In (ii) the authors assume a standard regression model with additive, exogenous and independent errors.

Assumption (iii) assumes that the outcome y_i does not depend on $\mathbf{X}_i$, neither directly through the features nor through a treatment effect that depends on $\mathbf{X}_i$. The latter implies a constant treatment effect. This assumption is certainly the most crucial one. While it may not hold in reality, it allows us to keep the model simple since the splits partition only the noise of the data. The simplicity is characterized by the fact that the splits should in expectation occur at the same frequency over the whole support of the node if the splitting algorithm was flawless. The residuals are seen as exogenous and homoskedastic and $\mathbf{X}_i$ does not induce any heterogeneity in the treatment effects.

Moreover, such a constant treatment effect model is as the authors claim a canonical member of any interesting class of data generating processes, because the constant function belongs to all classical smoothness function classes, as well as to the set of functions with bounded total variation. From this it follows that the model under Assumption (i) allows us to define lower bounds on the worst-case probability of a more than marginal deviation of the estimate of $\tau(\mathbf{x})$ from its actual value (Cattaneo et al. [2025]). Let $\mathbb{P}_1$ be the probability measure under Assumption 1 and $\mathbb{P}$ be a probability measure in any data generating class $\mathcal{P}$ that includes $\mathbb{P}_1$. Then, we have the following inequality.

$$\sup_{\mathbb{P} \in \mathcal{P}} \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{X}} |\hat{\tau}(\mathbf{x}) - \tau(\mathbf{x})| > \epsilon \right) \geq \mathbb{P}_1 \left(\sup_{\mathbf{x} \in \mathcal{X}} |\hat{\tau}(\mathbf{x}) - \tau(\mathbf{x})| > \epsilon \right) \quad (15)$$

The constant treatment effect assumption also removes issues related to standard (boundary or other smoothing) bias which would be introduced for a treatment effect that depends on the covariates (Cattaneo et al. [2025]). The non-constant treatment effect would be approximated by piece-wise constant functions (differences-in-means) which introduces bias.

Assumption (iv) postulates that the features $\mathbf{x}$ are independent and continuously distributed. Importantly, trees are invariant with respect to monotone transformations (Wager and Walther [2015], Su [2025], Cattaneo et al. [2025]). Every random variable can be monotonically transformed by its own cumulative distribution function (cdf) and by the universality of the uniform every cdf follows a uniform. This implies that we can just assume that the covariates are uniformly distributed. Assumption (v) further states that potential outcome errors are sub-exponential, or equivalently, they satisfy a Bernstein moment condition (Cattaneo et al. [2025]).

4.2 Uniform convergence

The authors define as their central contribution a theorem on the convergence properties of causal tree estimators. While parametric estimators converge with a rate of $\sqrt{n}$, non-parametric estimators like in the given case typically converge at a much slower rate (Hansen [2009]).

The central theorem of the authors regarding uniform convergence looks as follows:

Theorem 1 (Uniform Accuracy). *Suppose Assumption 1 holds, and the underlying causal tree has at least one split (i.e., at least two terminal nodes). Then, for $l \in \{\text{DIM}, \text{IPW}, \text{SSE}\}$ and all $b \in (0, 1)$,*

$$\liminf_{n \rightarrow \infty} \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{X}} |\hat{\tau}_l^{\text{NSS}}(\mathbf{x}) - \tau(\mathbf{x})| \geq C_1 n^{-b/2} \sqrt{\log \log n} \right) \geq b/e,$$

where the positive constant C_1 only depends on the distribution of $(\varepsilon_i(0), \varepsilon_i(1), d_i)$, and

$$\liminf_{n \rightarrow \infty} \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{X}} |\hat{\tau}_l^{\text{HON}}(\mathbf{x}) - \tau(\mathbf{x})| \geq C_2 n^{-b/2} \right) \geq C_3 b,$$

where the positive constants C_2 and C_3 only depend on the distribution of $(\varepsilon_i(0), \varepsilon_i(1), d_i)$, and the sample splitting scheme via $\liminf_{n \rightarrow \infty} \frac{n_T}{n_\tau}$ and $\limsup_{n \rightarrow \infty} \frac{n_T}{n_\tau}$. The precise definitions of the constants are given in the supplemental appendix of Cattaneo et al. [2025].

The first equation defines lower bounds on the uniform convergence for adaptive estimators and the second equation defines uniform convergence for honest estimators. Since C_1 , C_2 and C_3 are positive constants and hence do not depend on n , they do not affect the rate of convergence of the estimators. For the adaptive case we have a rate of convergence which is not faster than $n^{-b/2} \sqrt{\log \log n}$ and for the honest case a rate of convergence not faster than $n^{-b/2}$. Since $b \in (0, 1)$ it holds that $n^{-b/2} > \sqrt{n}$ and therefore the convergence rate is slower than in a parametric case.

4.3 Probability of extreme splits

The central reason for the slow uniform convergence is shown by Cattaneo et al. [2025] in an equation that characterizes the probability of extreme splits. This equation also explains the meaning of n^{-b} in Theorem 1. The authors provide their argument based on a decision tree of depth 1 (a so called decision stump). They show as follows that with non-vanishing probability the first split will concentrate near a boundary of the parent node.

Let $\hat{t}$ be the index at which the split is performed (CART split index), $\hat{j}$ be the split variable at the root node and $l \in \{\text{DIM}\}$. We notice that the first split coincides for no-sample-splitting and honest constructions. For each $a, b \in (0, 1)$ with $a < b$ and $j \in \{1, 2, \dots, p\}$ we have

$$\begin{aligned} \liminf_{n \rightarrow \infty} \mathbb{P} \left(n^a \leq \hat{t} \leq n^b, \hat{j} = j \right) \\ = \liminf_{n \rightarrow \infty} \mathbb{P} \left(n - n^b \leq \hat{t} \leq n - n^a, \hat{j} = j \right) \geq \frac{b - a}{2pe} \end{aligned} \quad (16)$$

Since we want to illustrate the probability of small, imbalanced child nodes, we can let $a \rightarrow 0$. As an imbalanced split we can classify splits for which the splits are below the 10th percentile or above the 90th percentile. For that reason b has to be chosen such that

$$\frac{n^b}{n} \leq 0.1 \quad (\text{condition for an imbalanced split}) \quad (17)$$

In that case, for b that satisfies the condition

$$\mathbb{P} \left(n^a \leq \hat{t} \leq n^b, \hat{j} = j \right) \geq \frac{b}{2pe}$$

is the probability of an imbalanced split on the left side of covariate j . Similarly,

$$\mathbb{P} \left(n - n^b \leq \hat{t} \leq n - n^a, \hat{j} = j \right) \geq \frac{b}{2pe}$$

is the probability of an imbalanced split on the right side of covariate j . A split can at the same time occur to the left or to the right of the data and thus the two probabilities are disjoint. Adding up the two probabilities results in the following lower bound probability.

$$\liminf_{n \rightarrow \infty} \mathbb{P} (\text{"imbalanced split"}) = \mathbb{P} \left(\hat{t} \leq n^b \vee n - n^b \leq \hat{t} \right) \geq \frac{b}{pe} \quad (18)$$

As an illustration we can choose $n = 100.000$, $b = 0.6$ such that $\frac{100.000^{0.6}}{100.000} = 0.1$.

This results to

$$\inf \mathbb{P} \left(\hat{t} \leq 100.000^{0.6} = 1000 \vee 100.000 - 100.000^{0.6} = 99000 \leq \hat{t} \right) \gtrsim \frac{0.2208}{p}$$

The result implies that for the case of a single covariate, the probability of an imbalanced

split is 0.2208. If we would consider a model where the treatment effect is constant, no meaningful splitting could occur. An optimal splitting behavior would then be splitting with equal probability along the entire covariate space. Such a splitting behavior would result in a probability of 0.02 for the imbalanced split from above and would thus be very different to the splitting behavior that is underlying our model.

Furthermore, we see that the probability of small child nodes decays inversely linear to p . Hence, while for one covariate the probability is large it decreases rapidly as the number of covariates increases. This is the case because at each node a split occurs only along values of a single covariate (Wager and Walther [2015]). Consequently, the probability that a specific variable is chosen at an arbitrary split is $\frac{1}{p}$.

The authors also show that the issue of an imbalanced split is inherited from decisions stumps to trees of arbitrary depth (Cattaneo et al. [2025, p. 33 app.]). Due to recursive splitting the K -th parent node satisfies for the optimal split index $\hat{i}$ and $a \rightarrow 0$ that

$$\liminf_{n \rightarrow \infty} \mathbb{P}(\hat{i} \leq n^b) = \liminf_{n \rightarrow \infty} \mathbb{P}(n - n^b \leq \hat{i}) \geq \frac{b}{2pe} \quad (19)$$

Accordingly, the K -th parent node satisfies

$$P(\text{"imbalanced split"}) \geq \frac{b}{pe}$$

We can derive the limit probability that at least one imbalanced split occurs in a tree of depth k . The probability of at least one extreme split can be rewritten as follows

$$\begin{aligned} \liminf_{n \rightarrow \infty} \mathbb{P}(\text{"at least 1 imbalanced split"}) &= 1 - \liminf_{n \rightarrow \infty} \mathbb{P}(\text{"no imbalanced split"}) \\ &= 1 - \left(1 - \liminf_{n \rightarrow \infty} \mathbb{P}(\text{"imbalanced split"})\right)^k \end{aligned}$$

Since

$$\liminf_{n \rightarrow \infty} \mathbb{P}(\text{"imbalanced split"}) \geq \frac{b}{pe} \Leftrightarrow 1 - \liminf_{n \rightarrow \infty} \mathbb{P}(\text{"imbalanced split"}) \leq 1 - \frac{b}{pe}$$

it holds

$$\liminf_{n \rightarrow \infty} \mathbb{P}(\text{"at least 1 imbalanced split"}) \geq 1 - \left(1 - \frac{b}{pe}\right)^k \quad (20)$$

This result is extensive. It means that for every additional tree depth the probability of having at least one imbalanced split increases.

5 Tree Calibration

5.1 Bias-Variance Tradeoff

When it comes to determining the parameters of the model the key metric is the bias-variance tradeoff which is determined by the magnitude of the parameters i.e. the flexibility of the model (James et al. [2023, p. 32 ff.]). A more flexible model typically has a higher variance while a less flexible model has a higher bias. In our case a more flexible model is a deeper tree which has smaller leaves i.e. a higher variance of the CATE estimates but a smaller bias.

5.2 α - regularity

A popular assumption is to force child nodes to contain at least a share α of data points contained in the parent node, so called α - regularity (Wager and Walther [2015], Wager and Athey [2018]). However, such a condition makes splits along the edges of noise variables less desirable (Wager and Walther [2015]). Since we want to examine imbalanced splits and not rule them out per sé we will not impose α - regularity.

5.3 Tree depth vs. minimum leaf size

The two key parameters that define tree depth and need to be adjusted in the simulation part are tree depth and minimum sample size per leaf. There is clearly a relationship between these parameters. A deeper tree tends to have smaller leaf sizes and smaller minimum leaf sizes permit deeper trees.

Since we do not want to limit the scope for imbalanced splits we prefer not to enforce a minimum leaf size. We are interested in detecting as many extremely small leaves as possible and do not want to stop as soon as a minimum leaf size in one leaf appears. The according minimum leaf size would be 1.

However, we want to be able to calculate the standard error of the treatment effect estimate within each leaf to calculate valid confidence intervals. For that we need at least two treated and two control observations within each leaf. This implies that `min_samples_leaf` needs to be at least equal to 4.

5.4 Minimum number of treated and control units per leaf

The required `min_samples_leaf` parameter can be set in `causalml`. However, the package `causalml` does not contain parameters to set the minimum number of treated and control units per leaf. Because of that, an adjustment of the package became necessary. To achieve that the tree does not split in a way that results in less than two treated or control units I have adjusted the `criterion.py` function. To be able to determine the minimum number of treated and the minimum number of control units I have added the parameters `min_treated_leaf` and `min_control_leaf` per leaf. Splits are evaluated based on the impurity improvement they generate. For splits below

the number specified in these parameters, I have set the impurity improvement for splits that do not fulfill the parameters to minus infinity.

5.5 Choice of tree depth

There are several factors that determine tree depth. In the context of Machine Learning an excessively high hyperparameter like tree depth is referred to as overfitting which needs to be prevented. For that reason, too complex models should be prevented ((James et al. [2023, p. 22])). The tree size governs the model's complexity and the optimal tree size should be adaptively chosen from the data (Hastie et al. [2009]). To determine the optimal tree size, usually, a grown tree is pruned by applying a cost-complexity criterion via cross-validation.

Cattaneo et al. [2025] rely in their prove on a constant treatment effect because as they say such a model "is a canonical member of any interesting class of data generating processes". Therefore, in our simulation we will also consider such a model. In a real world setting where we would like to get accurate estimates of the then constant treatment effect, we would decide to renounce on any split and just compute treatment effects based on the whole dataset. In our case, however, where we want to determine in what way small leaves occur such an approach would be counterproductive. If there is no problem with imbalanced splits and slow convergence, the causal tree would still detect the true treatment effect for a reasonable high sample size. As a result, treatment effect heterogeneity or the absence of it does not affect our parameter choice.

In an actual real world scenario, in any case a researcher will not know the data generating process. While she or he knows the number of observations and may have an idea of the number of relevant features, the amount of treatment effect heterogeneity may be unknown. This is a further reason to not restrict the tree growing process and to grow the trees to their full extent until they are fully grown.

A tree, as far as we are concerned, is fully grown when the coverage ratio does not change anymore substantially. To consider that higher sample sizes need deeper trees, we increase the tree depth by five over each increasing sample size. This procedure will allow us to observe how the coverage and the effect of imbalanced splits behaves over all relevant tree depths.

5.6 Choice of package

When implementing a Causal Tree in Python, there is essentially the choice between two packages, `econml` and `causalml`. The relevant module is on the one hand `econml.grf` with its `CausalForest` method and `n_estimators = 1`. On the other hand, it is the `CausalTreeRegressor` from `causalml.inference.tree`.

The essential difference is that `econml` assumes a local linear model in each leaf which is solved via a local moment equation (see `EconML Documentation`). This means that for every observation over all candidate leaves, there exists a vector $\beta(x)$ and a vector $\tau(x)$ that constitute the local linear model. Since applying an MSE criterion based on such a model for every candidate leaf is computationally expensive, the criterion is replaced via a gradient based

approximation.

In contrast, `causalml` assumes a constant model within each candidate leaf which is defined by a simple difference in the means of treatment and control group since it is based on the approach from (Athey and Imbens [2016]) as it is mentioned in the CausalML Documentation. Because this is computationally less expensive, there is no need for a gradient based approximation of the MSE based splitting criterion.

The approach pursued by Athey and Imbens [2016] and adopted by Cattaneo et al. [2025] assumes that the treatment effect within a leaf is simply characterized by a difference-in-means estimators. To not diverge from this approach, we will therefore use `causalml` in our simulations.

Furthermore, attempts to implement the required parameters have failed in `econml` but not in `causalml`. This was driven by the fact that `causalml` makes use of the number of treated and control units within a leaf immanently, to calculate the sample means of each group. In contrast, `econml` does not actively track these subgroup sizes since it optimizes via finding an optimal score over all leaves, based on matrix algebra and gradient approximations. This was a further reason to proceed with `causalml`.

6 Simulations

6.1 Simulation Design

The underlying data generating process is aligned with the assumptions on the Data Generating Process (DGP) from Cattaneo et al. [2025]. There is only one feature in the data generating process to keep the DGP and the analysis as simple as possible. With only one feature, we can be sure that at each split point the algorithm splits the support of the particular feature. The DGP can be characterized as follows:

$$\begin{aligned}
X_i &\sim \mathcal{U}(0, 100) \\
\epsilon_{0i} &\sim \mathcal{N}(0, 100) \\
\epsilon_{1i} &\sim \mathcal{N}(0, 100) \\
Y_{0i} &= c_0 + \epsilon_{0i} \\
Y_{1i} &= c_1 + \epsilon_{1i} \\
W_i &\sim \text{Bernoulli}(0.5) \\
Y_i &= W_i Y_{1i} + (1 - W_i) Y_{0i}
\end{aligned} \tag{21}$$

c_0 is set equal to 10 and c_1 is set equal to 30 which results in a constant treatment effect of 20.

We examine three different sample sizes starting from $n = 1000$ and then multiply by five twice which leads to sample sizes of $n = 5000$ and $n = 25000$. This ensures to capture the asymptotic behavior of the algorithm which is characterized by its speed of convergence. We multiply by five such that the average leaf size for a given depth grows by the same proportion. This makes metrics like the coverage more comparable between sample sizes. The conditions on the Causal Tree algorithm have been described in section 4 and are driven by imposing as

few restrictions as possible while still enabling the calculation of confidence intervals at least for adaptive estimation.

The number of Monte Carlo simulations is influenced by the fact that we are interested in the behavior of confidence intervals of leaves that are located in the tails of the distribution. This brings up two problems. First, if we want to examine the coverage of confidence intervals around the bounds of the distribution, we can naturally only come back to data on one site of the bound since for the other site of the bound the support is not defined. This reduces the number of available leaves. Second, we would like to examine the behavior of bins that are as close as reasonable to the boundary and thus as narrow as possible. Both facts imply that we need a high number of Monte Carlo simulations to dispose of enough data within the bins. Attempts with 10.000 Monte Carlo simulations have suffered from results in the tails that were driven by sampling variation. Hence, we move forward with 25.000 simulations.

6.2 Results

6.2.1 Adaptive estimation

The coverage ratios for adaptive estimation lie far below the nominal coverage of 0.95. The coverage does not improve for larger sample sizes, it even decreases further (see section A.1.1). For an example depth of 10 the average coverage is 0.613 for $n = 1000$, 0.582 for $n = 5000$ and 0.558 for $n = 25000$. The reason for this lies in the fact that as the sample size increases the number of noisy observations also increases. Therefore, it gets even easier for the algorithm to find a small set of observations that has a significant variation in the treatment effect just because of random noise. Such small leaves then have a treatment effect which deviates from the constant treatment effect and a coverage of 0.

In section B we do not compute a simple sample mean to average the coverages. Instead, we weight each coverage value by its sample size such that we reduce the effect of very small leaves which are more likely to not have a valid coverage value, but only contain the treatment effect estimate for a tiny partition of the support. The weighted average is significantly better than the simple average. It lies for moderate depths above 80 %. Nevertheless, it does not substantially improve for larger samples sizes either. This indicates that the problem for adaptive estimation does not only come from imbalanced splits which cause tiny leaves. For a sample size of $n = 25000$ such leaves would only receive very small weight. Instead, the problem is that treatment effect estimation and tree construction happen on the same part of the data. Because of that, the estimates can be distorted by noise on which the leaves were constructed.

For both coverage types we can take a further look why the metrics do not improve for larger sample sizes. It is important to recall that the coverage of a particular treatment effect is determined by two factors. First, the absolute distance of the treatment effect estimate from the actual treatment effect. Second, the width of the confidence interval which is determined for a nominal coverage of 95% by $1.96 \cdot$ standard error. Only if the latter number is higher than the absolute deviation of the estimate from the actual effect, we consider the treatment

effect to be covered. Section C shows that the standard error is massively inflated for $n = 1000$ and it decreases for higher sample sizes because the average leaf sizes increase. Although the mean absolute deviation of the treatment effect from its estimate also decreases, this leads to a worsening overall coverage since the confidence intervals are getting narrower. The effect of a decrease in the scaled standard error is stronger than the effect of decreasing mean absolute deviation.

The coverage improves for deeper trees. The reason for this can also be seen in section C as the scaled standard error increases on average much more than the mean absolute deviations over higher depths. This comes with smaller leaf sizes over increased depths as we can see in section F.

Section D.1.1 shows that the algorithm splits more often in regions close to the boundary of the support. As section E.1.1 demonstrates, this results already for low depths in an overrepresentation of small average leaf sizes. In Tables 5 to 7 of section F, we see that the average leaf size decreases as the leaf means get closer to the boundary for all depths. Section D.1.1 also shows the counter effect that for a depth of 1 and 2 there are more splits in the middle of the support. This leads to the effect that also high average leaf sizes (close to the total sample size) are overrepresented (section E.1.1). For these high average leaf sizes in the middle of the support, we see a coverage that is close to the nominal value as a Gaussian distributional assumption that constitutes our confidence interval is more likely to be valid.

After a peak in the middle of the support, the coverages decrease as the leaf means move away from the center. Interestingly, for depths larger than one, the small leaf sizes close to the boundary do not lead to coverage ratios that decrease monotonically. On the contrary, the coverage ratios increase as the leaf means approach the boundary (see section A.2.1).

As section C.2.1 shows, the scaled standard error and the mean absolute deviations increase close to the boundary. The scaled standard error increases on average more than the mean absolute deviations. For smaller depths both lines intersect since the scaled standard error is at first smaller than the mean absolute deviations. Both lines increase since for the smaller leaves closer to the boundary the treatment effect estimate tends to deviate more from the actual treatment effect and at the same time we observe a higher scaled standard error.

The stronger increase in the scaled standard error leads to an improving coverage as the coverages approach the boundary. Nevertheless, the increased coverage is not a desirable outcome as it does not stem from a more exact Gaussian distributional assumption on the treatment effect estimate since the average leaf sizes are small. Instead, the improved coverage stems from inflated confidence intervals which lead to an increased chance for a Type-II Error.

The increasing absolute distance of the treatment effect estimate also means that as the leaf means get closer to the boundary and smaller the leaves contain more noisy variables. This aligns with the explanations in section 2.5 that the reason for more frequent splits close to the boundary lies in (in absolute values) larger treatment effects in small noisy leaves (overfitting).

The results of non-increasing (and even decreasing) coverage as the total sample size increases align with the theoretical illustrations in Cattaneo et al. [2025]. Since the speed of

convergence is not fast enough, the mean absolute deviations do not diminish fast enough. At the same time, the standard error also decreases and the coverage decays for larger sample sizes.

While for moderate depths the coverage increases in sample size close to the center of the support it worsens close to the boundary. This comes from the imbalanced splitting behavior at the boundary. Therefore, Cattaneo et al. [2025] was right to state that a Gaussian distributional assumption does not hold in general as the effective sample size in boundary leaves does not even increase for the approximation to apply in many regions of $\mathcal{X}$.

6.2.2 Honest estimation

For honest estimation the coverages improve drastically (section A.1.2). For lower depths we see coverages that are closer to the nominal value, although for larger depths the coverage is still substantially different from the nominal value. Unlike under adaptive estimation, the coverages improve for larger sample sizes. As section C.1.2 illustrates, this comes from a scaled standard error that lies more and more above the mean absolute deviations for larger sample sizes. A gap between these metrics which gets larger unlike under adaptive estimation explains why the coverage improves. In general, the mean absolute deviations are much lower than for adaptive estimation.

Section B shows that the weighted average coverages lie much closer to the nominal value than the simple average. Especially for $n = 5000$ and $n = 25000$ for moderate depths we see weighted average coverages which are more or less equal to 95 %. This demonstrates that simple averages which are different from the nominal value are driven by the effect of tiny leaves.

The effect of deeper trees on coverage is opposite to adaptive estimation. The reason is that the mean absolute deviation increases more than the scaled standard error at least for part of the support.

Close to the boundary the coverages decrease. This has to be driven by an increase in the absolute deviation of the estimate from the treatment effect, compared to the scaled standard error and thus by noise in the leaves. Since in section C.1.2 the scaled standard error appears to lie consistently above the absolute distance, this was examined in more detail in section C.2.2. The scaled standard error starts to oscillate close to the boundary and indeed lies below the absolute deviation which explains the decrease in coverage.

Problematically, the calculation of coverages for honest estimation suffers from a fundamental problem. Since there are many leaf sizes close to the minimum of 4, the estimation data populates many leaves with less than 4 observations. This makes a calculation of a standard error and hence a confidence interval impossible. Section G shows the effect on the percentage of leaves that contain a coverage value. The percentage increases mildly with the number of observations but even with $n = 25000$ for a depth of 10 only around 72 % of leaves are assigned a coverage value. This highlights again a consequence of imbalanced splits which also concerns the existence of a coverage value of a particular leaf.

The problem of a lack of a valid coverage value does not only lie in the absence of the possibility of inference. Since leaves that do not contain a coverage value are small leaves

which are likely to not attain coverage, we systematically overestimate the coverage for honest estimation by just discarding these leaves. Sections A.1.3 and A.2.3 show the coverages if we decide to set the coverages of leaves with too few observations equal to 0 instead. As a consequence, we underestimate the coverage such that we know that the ground truth must lie somewhere in between these two coverage values.

To get a glimpse on the severity of the problem, we can compare the coverage for $n = 25000$ and a depth of 10 for both cases. For the case of coverages with NA's the coverage is 0.9 and for the case of coverages with 0's the coverage is 0.65 which amounts to a difference of 25 %. For high depths (10 for $n = 1000$, 15 for $n = 5000$ and 20 for $n = 25000$) coverage for adaptive estimation even surpasses coverage for honest estimation with enforced 0's.

Section C.2 demonstrates that the absolute deviation of the treatment effect estimate for honest estimation lies significantly below the one for adaptive estimation. For a depth of 10 the metric has an average around 60 for honest estimation and around 100 for adaptive estimation. This means that we see significant improvement in the accuracy of the estimator for honest estimation. Since the scaled standard error is approximately the same for both estimation types, this results in a better coverage for honest estimation at least for moderate depths. Cattaneo et al. [2025] claims to have proved that causal trees fail to benefit from honesty. Here, we see that honesty significantly improves the coverage characteristics.

According to Cattaneo et al. [2025], honesty delivers at best "logarithmic improvements in sample size" compared to adaptive estimation. The authors claim that the convergence speed of the treatment effect estimator only improves at a logarithmic rate for honest estimation. For us this is relevant as it affects the mean absolute deviations of the treatment effect from its estimate and therefore the confidence intervals. For an example depth of 3 and adaptive estimation we see an improvement from $n = 1000$ to $n = 5000$ of around 5.2 and from $n = 5000$ to $n = 25000$ of 4.8. For honest estimation we see an improvement of around 6 from $n = 1000$ to $n = 5000$ and of 4.1 from $n = 5000$ to $n = 25000$. Indeed, we see a similar improvement in the mean absolute deviations for both estimation types.

However, the authors ignore the different baseline of the mean absolute deviations. The metric starts for a depth of 3 at around 87.3 for adaptive and 41.5 for honest estimation. The consequence is that coverage improves for honest estimation and does not decay since the scaled standard error remains far enough above the mean absolute deviations.

7 Conclusion

The honest truth of Cattaneo et al. [2025] was supposed to lie in only negligible improvements in the estimation accuracy in sample size for honest estimation. This is true if we only look at the rate of improvement across different sample sizes. However, if we were truly honest, we would also have to consider another point: the baseline of the estimation accuracy. If we want to reach a certain accuracy, at least as important as the speed is where we started.

Our analysis has demonstrated major improvements of honest estimation. The coverage

improves and does not worsen due to a significantly lower mean absolute deviation than under adaptive estimation. However, major problems close to the boundary prevail or even new problems are created. First of all coverage decays. Moreover, many leaves get so tiny that we cannot infer a valid coverage since a standard error cannot be calculated. And although we see improvements compared to adaptive estimation, the coverage remains off from the nominal coverage. Only if we calculate the coverage by not considering leaves without a valid coverage the coverage for $n = 25000$ does not fall below 90% for moderate depths. This value, however, is overestimated as we do not consider tiny leaves which do not receive a valid coverage.

Tiny leaves, however, only constitute a small partition of the support which is why we have also considered a weighted coverage metric for which coverages are significantly improved. Here, we see the strength of honest estimation which provides at least for moderate depths weighted coverages that are very close to the nominal value.

Although the data generating process was very simple, the implications of our results for real-world settings are broad. Extensive splitting at the boundary occurs due to fitting of noise and is not only limited to the data generating process. This also applies for the observed inflated standard errors due to small leaf sizes and the consequences on coverage. A higher number of covariates will however decrease the problem as the algorithm splits more than one feature which may prevent repetitive imbalanced splits on one feature.

If honesty leads to better results than adaptive estimation, may depend on the real-world setting. There may such high treatment effect variation that using the whole sample for construction of the tree may be beneficial. Nevertheless, if there is reason to assume that the variation is rather low, honesty will most likely lead to better results as in our case.

For honest estimation the coverage decreases as the trees get deeper. Hence, the usual approach to penalize unnecessary deep trees via cross-validation is desirable. What needs to be considered is that due to the simplified data generating process all possible tree depths were already overfitted. This is why coverage has monotonically decreased over increasing tree depths. In real-world settings, the tree depth for which coverage is optimized will be larger than 1. However, for such a case imbalanced splits will still occur increasingly for deeper trees such that this will lower coverage and thus the optimal tree depth.

The problems are diminished by applying an ensemble method like a Causal Forest to estimate the treatment effects. Each tree of a forest is built on a subsample of the data. Together with honest estimation, this allows to receive the treatment effect estimate as a (weighted) average over the trees. Naturally, this reduces the effect of noisy leaves in single trees and leaves close to the boundary exhibit more stable coverages.

Nevertheless, there may be cases where a Causal Tree is preferable. A Causal Tree splits the support discretely depending on treatment heterogeneity. This leads to better interpretability and can enable the setting of policy rules. A solution for keeping the characteristics of a Causal Tree, while still profiting from the distributional advantages of Causal Forests, could be the method introduced from Rehill [2024]: a Distilled Causal Tree.

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A Coverage Ratios

A.1 Entire support

A.1.1 Adaptive estimation

Coverage for $n = 1000$ and adaptive estimation by leaf mean (bin width = 0.5)

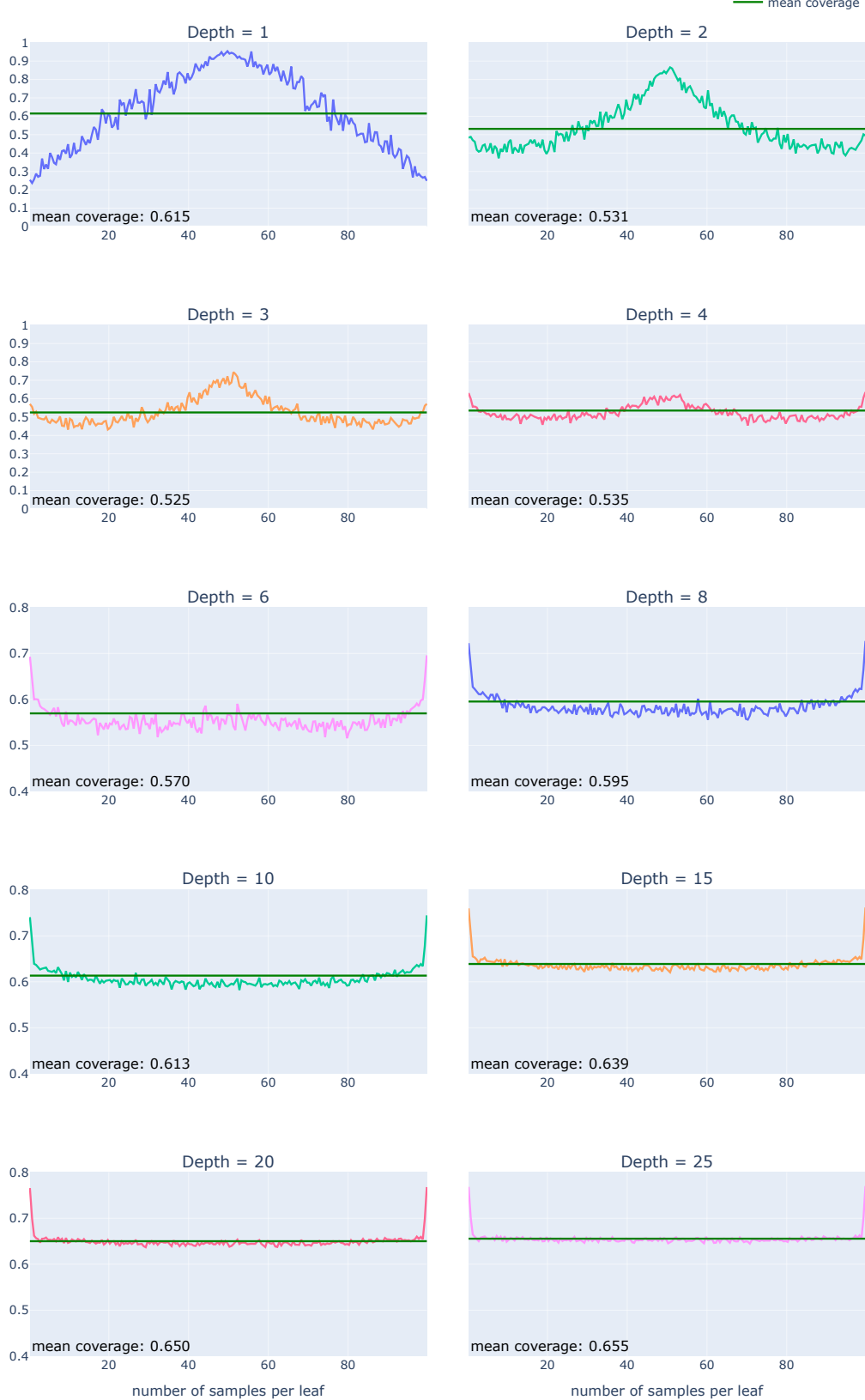


Figure 1: Coverage ratios for the whole support of the covariate and adaptive estimation (n=1000).

Coverage for $n = 5000$ and adaptive estimation by leaf mean (bin width = 0.5)

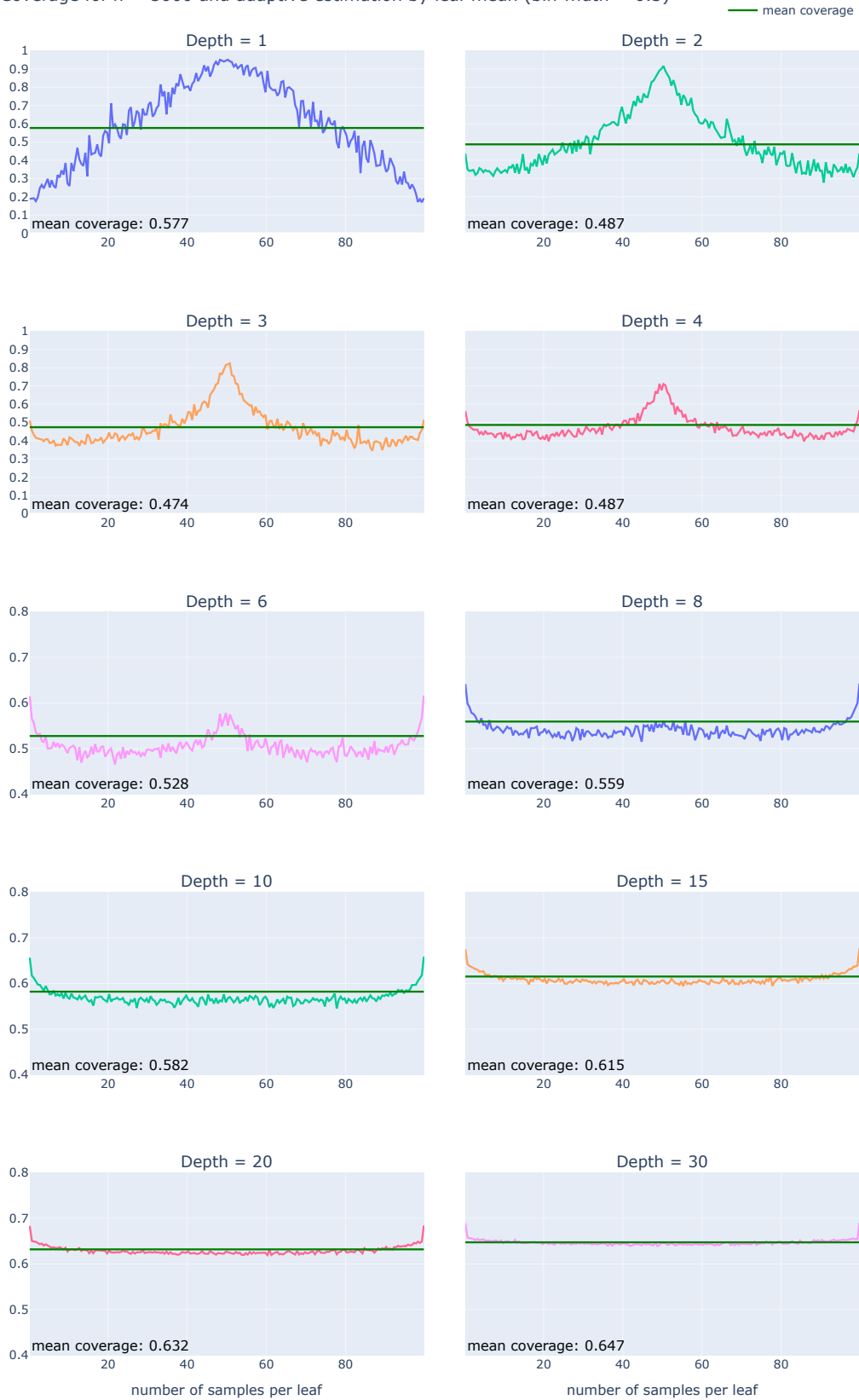


Figure 2: Coverage ratios for the whole support of the covariate and adaptive estimation (n=5000).

Coverage for $n = 25000$ and adaptive estimation by leaf mean (bin width = 0.5)

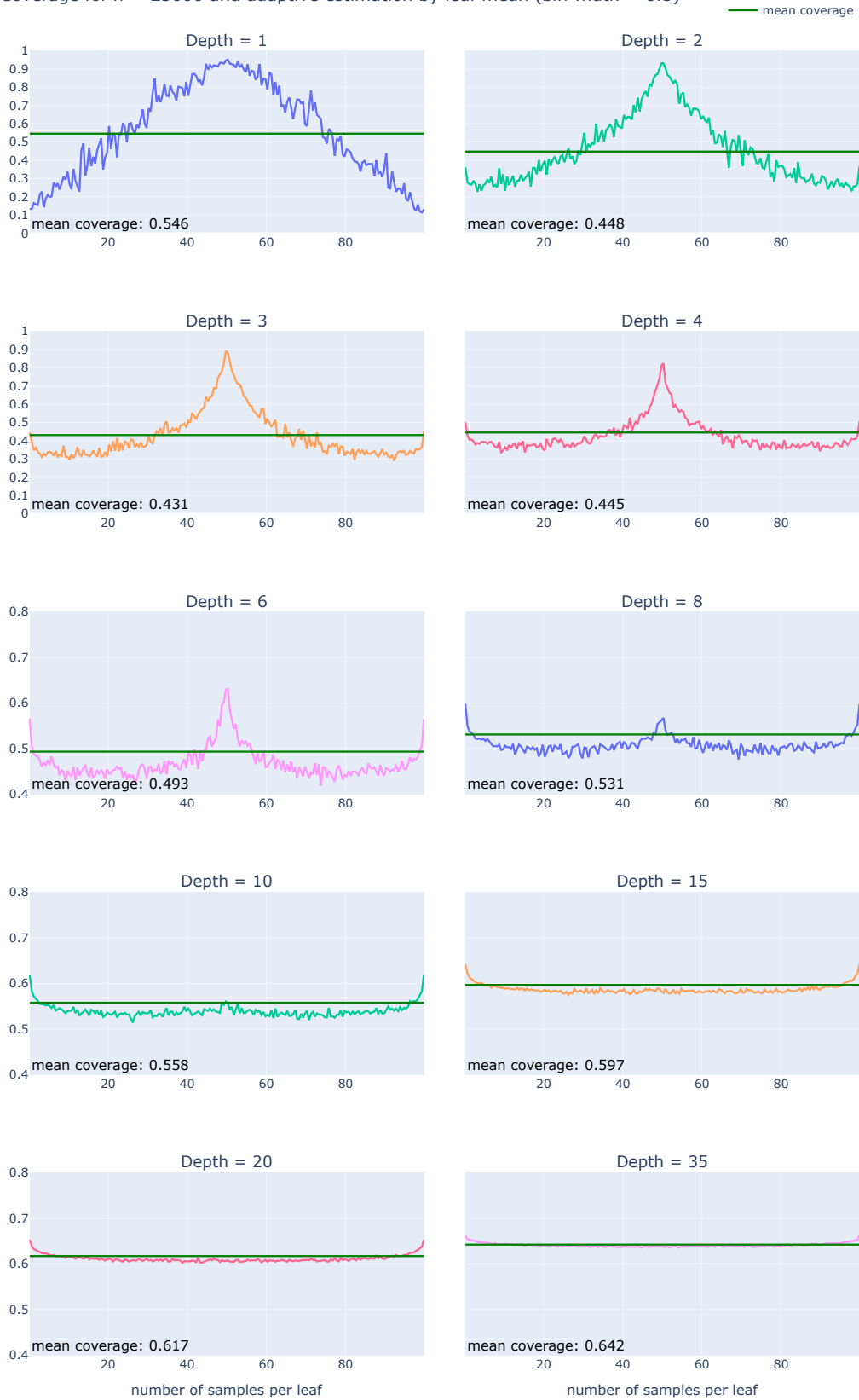


Figure 3: Coverage ratios for the whole support of the covariate and adaptive estimation ($n=25000$).

A.1.2 Honest estimation with NaN coverages as NaN

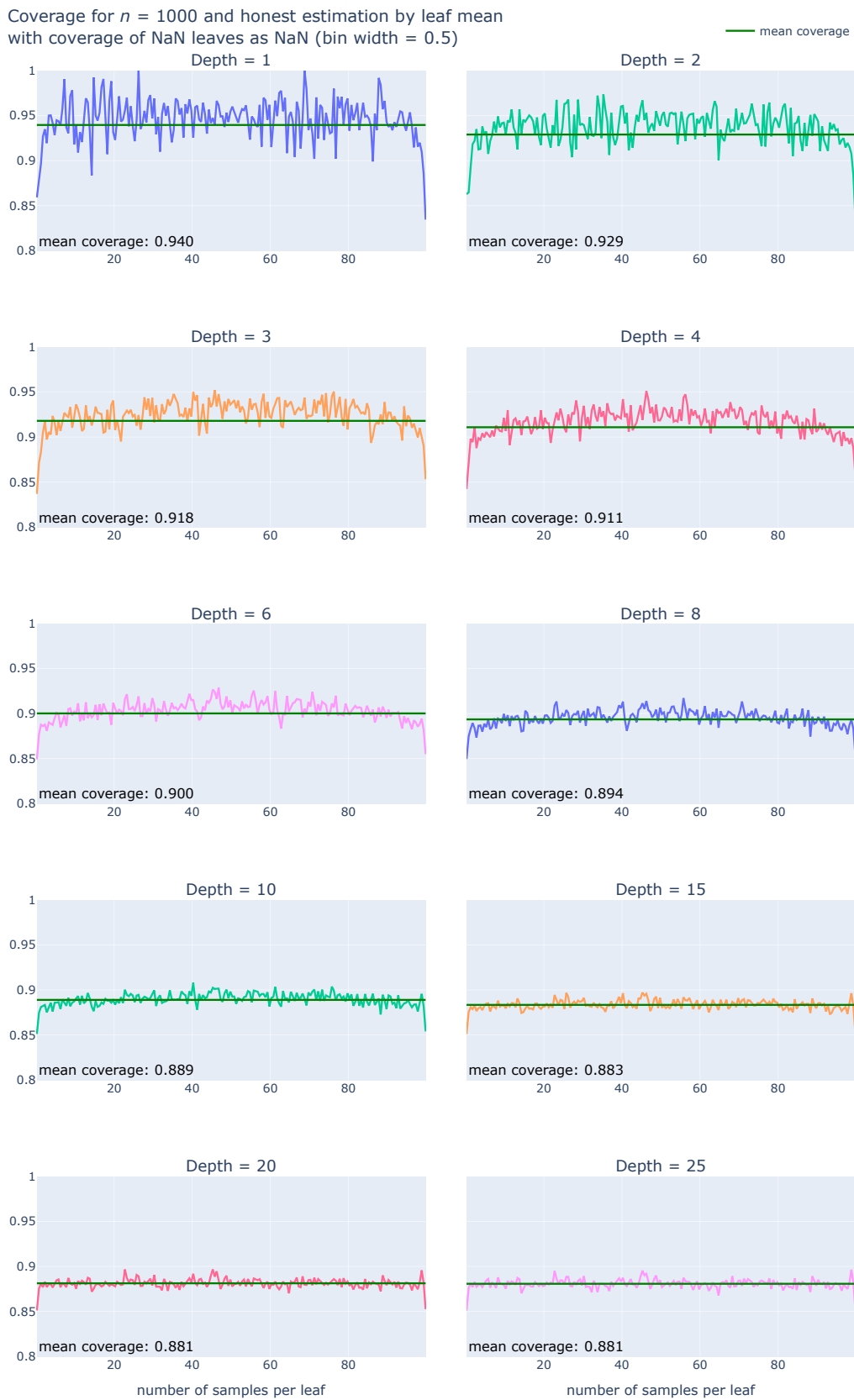


Figure 4: Coverage ratios for the whole support of the covariate and honest estimation with coverages that cannot be estimated as NaN ($n=1000$).

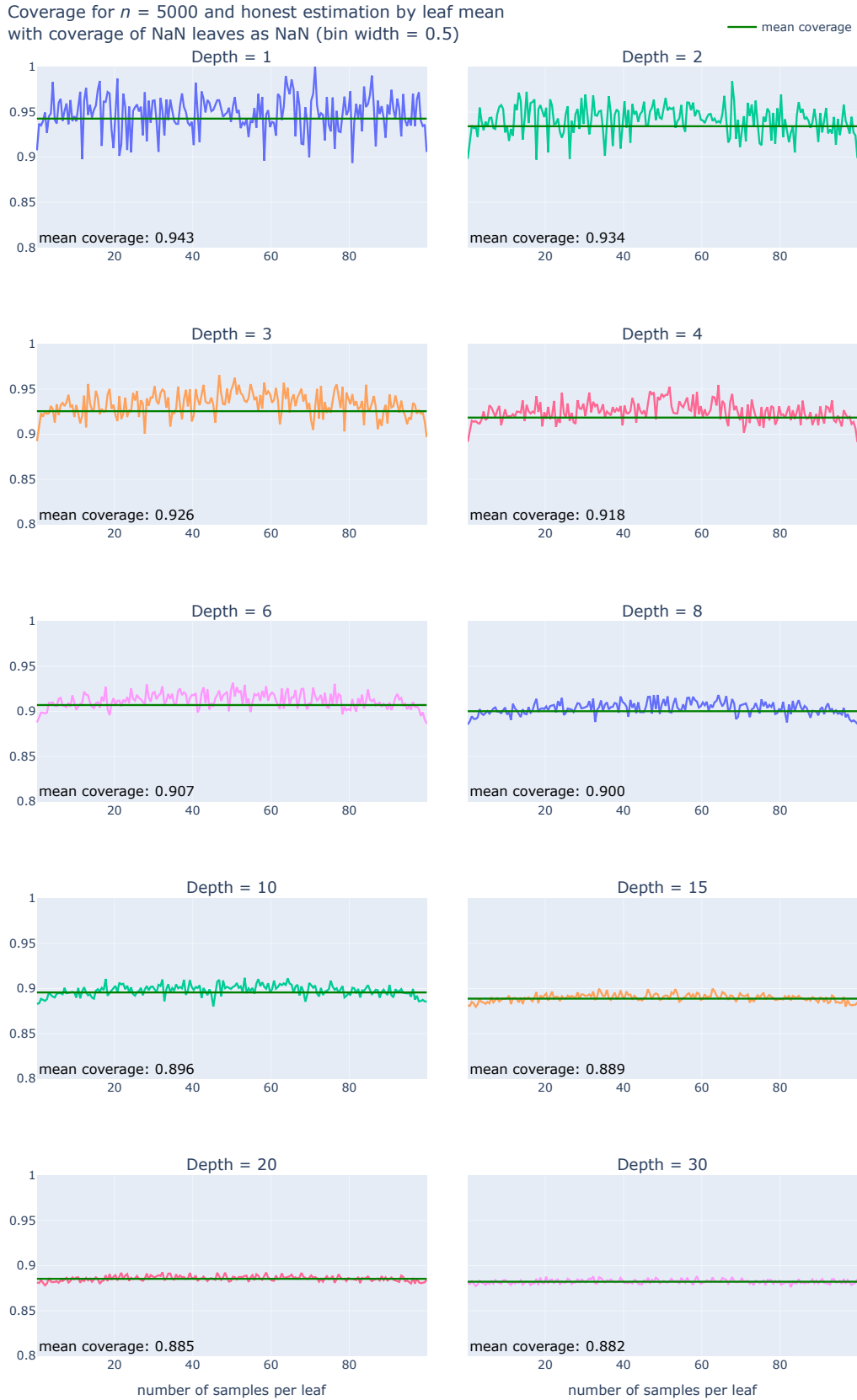


Figure 5: Coverage ratios for the whole support of the covariate and honest estimation with coverages that cannot be estimated as NaN ($n=5000$).

Coverage for $n = 25000$ and honest estimation by leaf mean
with coverage of NaN leaves as NaN (bin width = 0.5)

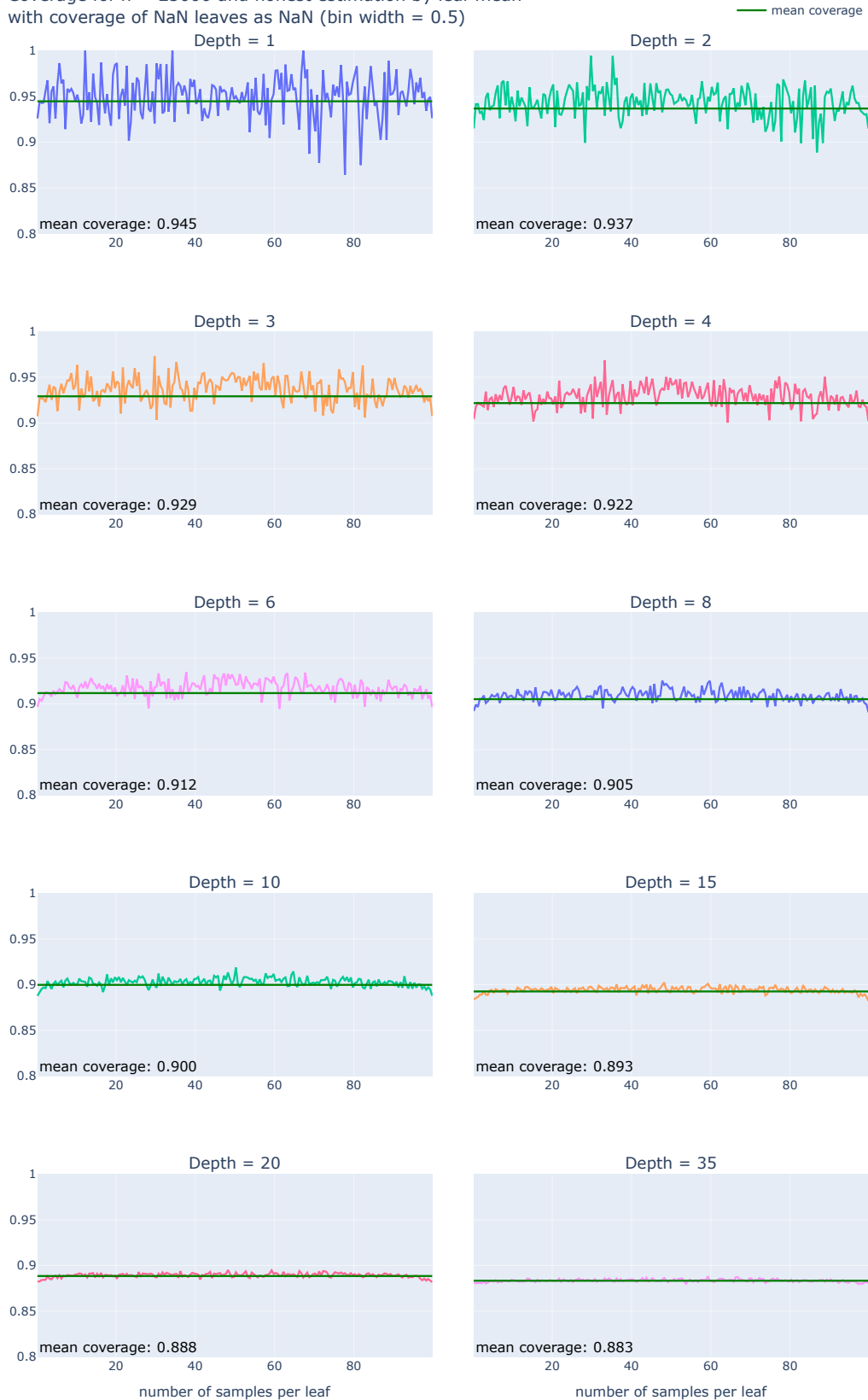


Figure 6: Coverage ratios for the whole support of the covariate and honest estimation with coverages that cannot be estimated as NaN ($n=25000$).

A.1.3 Honest estimation with NaN coverages as 0

Coverage for $n = 1000$ and honest estimation by leaf mean
with coverage of NaN leaves as 0 (bin width = 0.5)

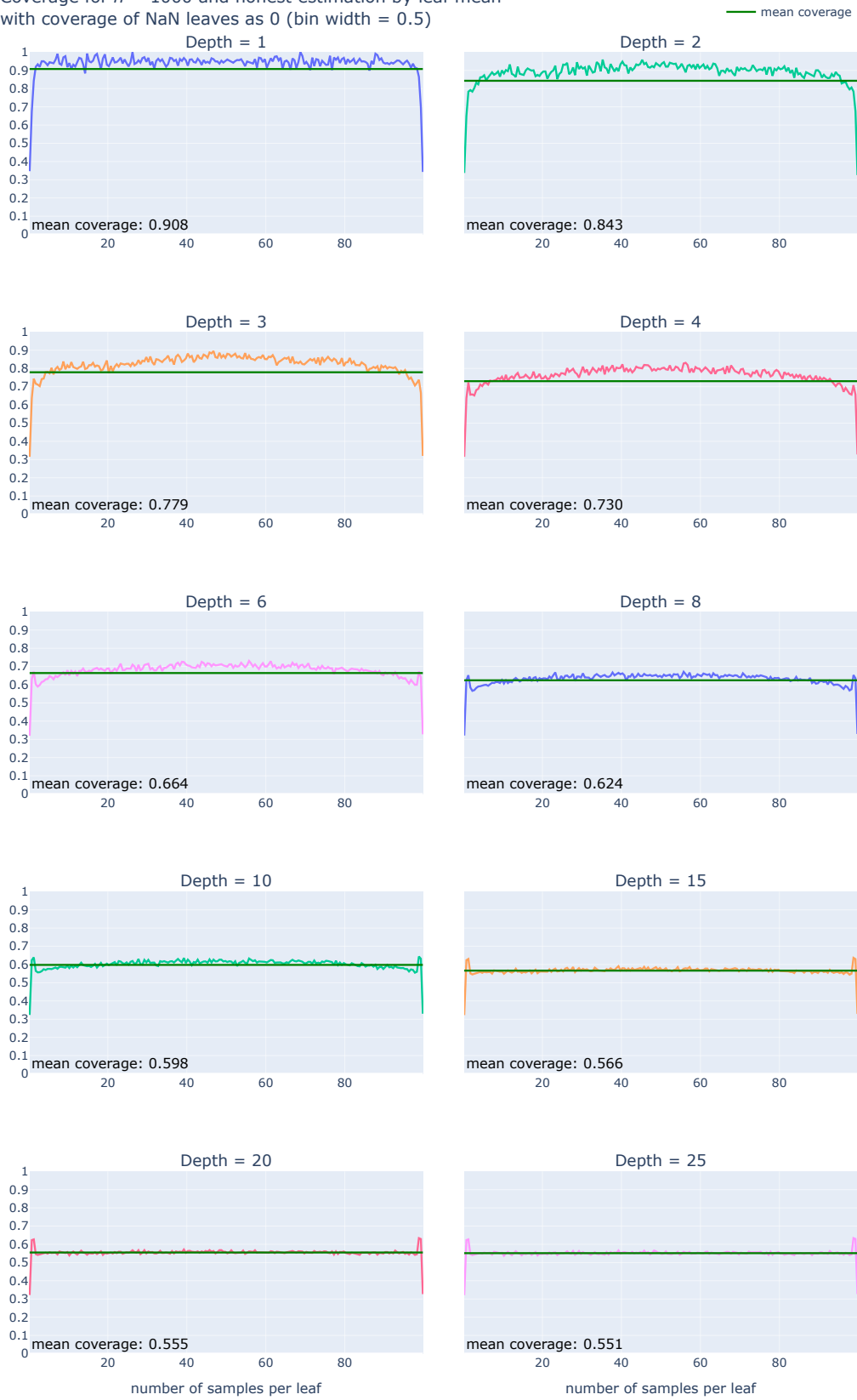


Figure 7: Coverage ratios for the whole support of the covariate and honest estimation with coverages that cannot be estimated as 0 ($n=1000$).

Coverage for $n = 5000$ and honest estimation by leaf mean
with coverage of NaN leaves as 0 (bin width = 0.5)

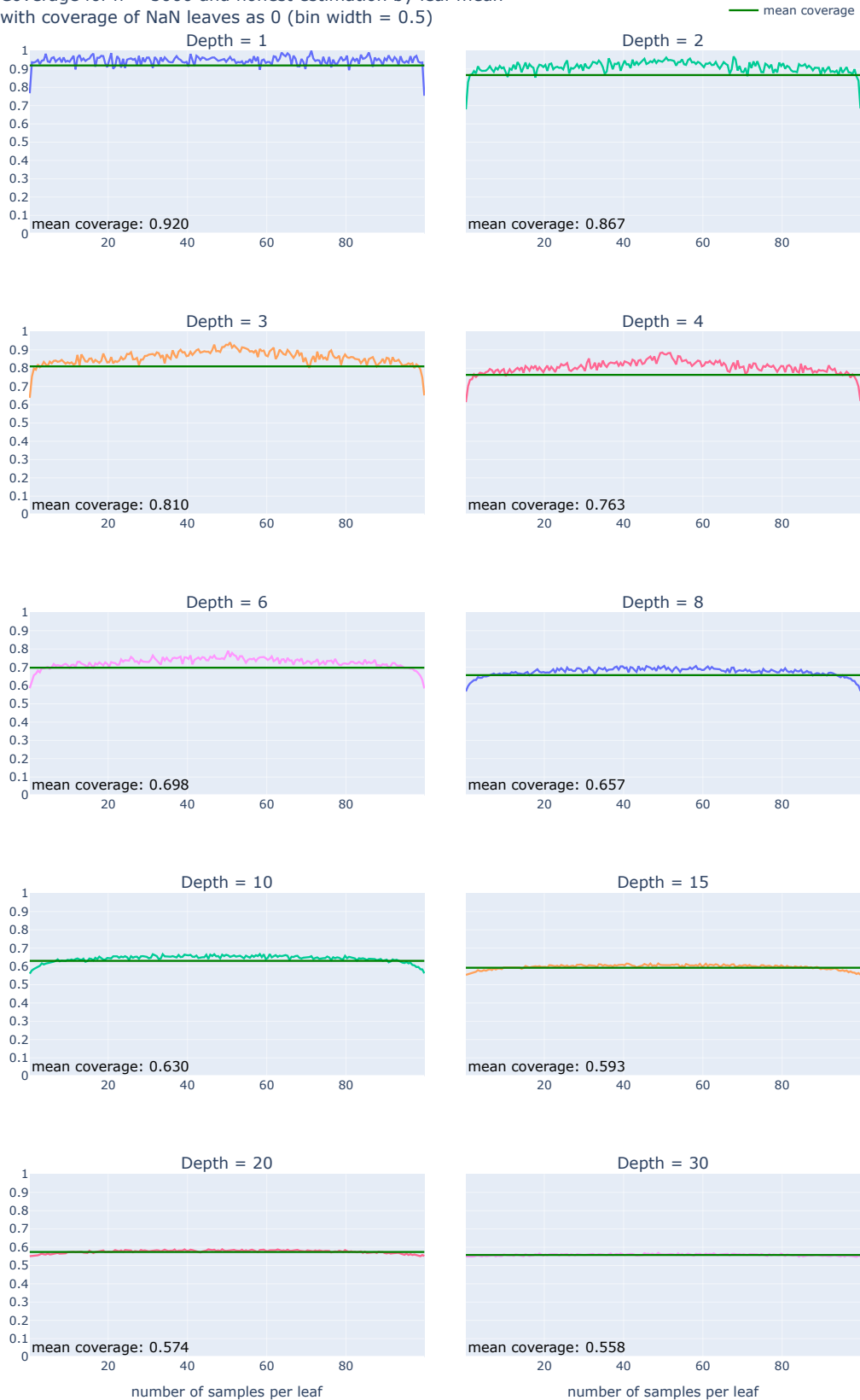


Figure 8: Coverage ratios for the whole support of the covariate and honest estimation with coverages that cannot be estimated as 0 ($n=5000$).

Coverage for $n = 25000$ and honest estimation by leaf mean
with coverage of NaN leaves as 0 (bin width = 0.5)

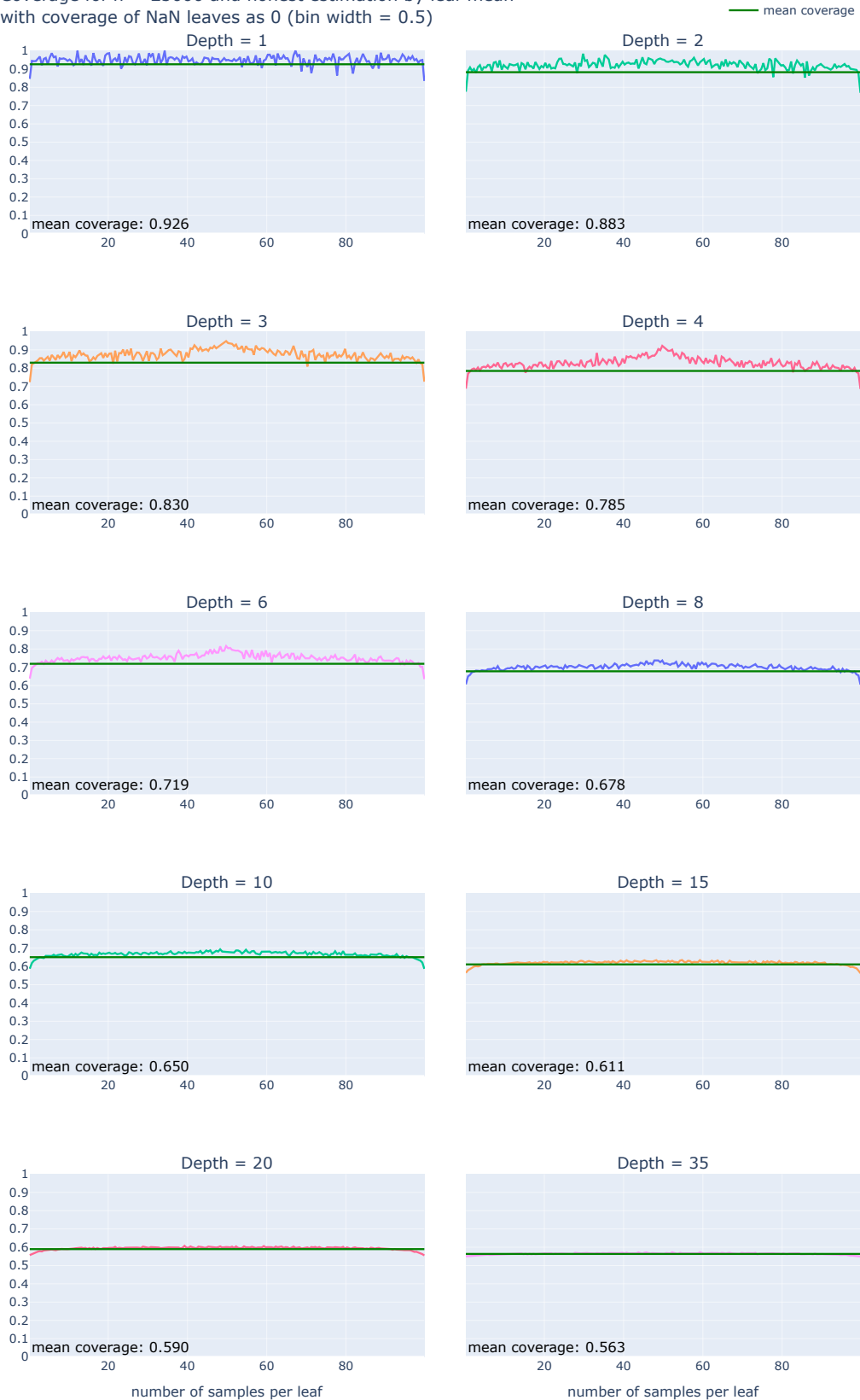


Figure 9: Coverage ratios for the whole support of the covariate and honest estimation with coverages that cannot be estimated as 0 ($n=25000$).

A.2 Extreme absolute deviations

A.2.1 Adaptive estimation

Coverage for $n = 1000$ and adaptive estimation by extreme absolute deviations (bin width = 0.2)

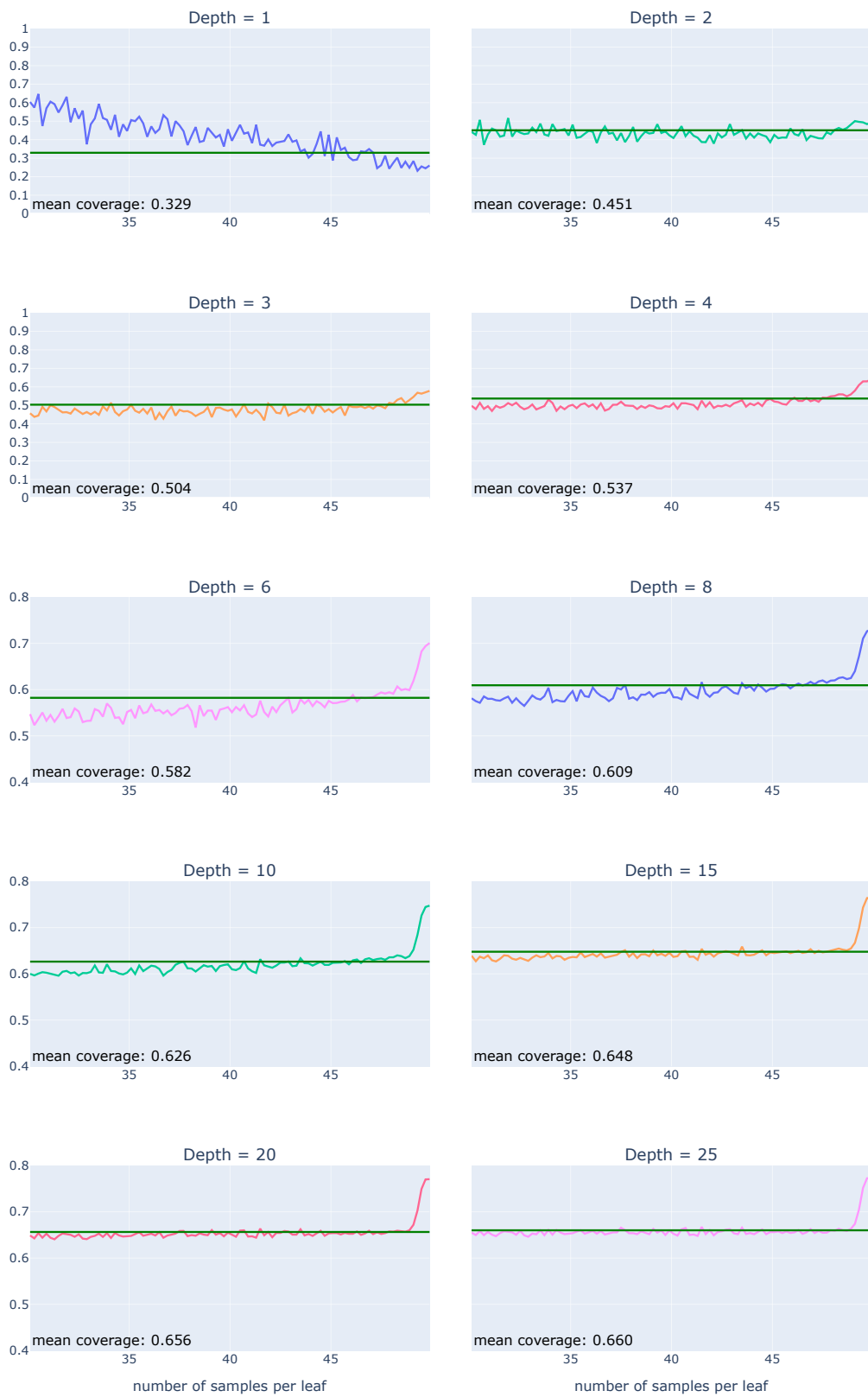


Figure 10: Coverage ratios for absolute deviations from the population mean of the covariate and adaptive estimation ($n=1000$).

Coverage for $n = 5000$ and adaptive estimation by extreme absolute deviations (bin width = 0.2)

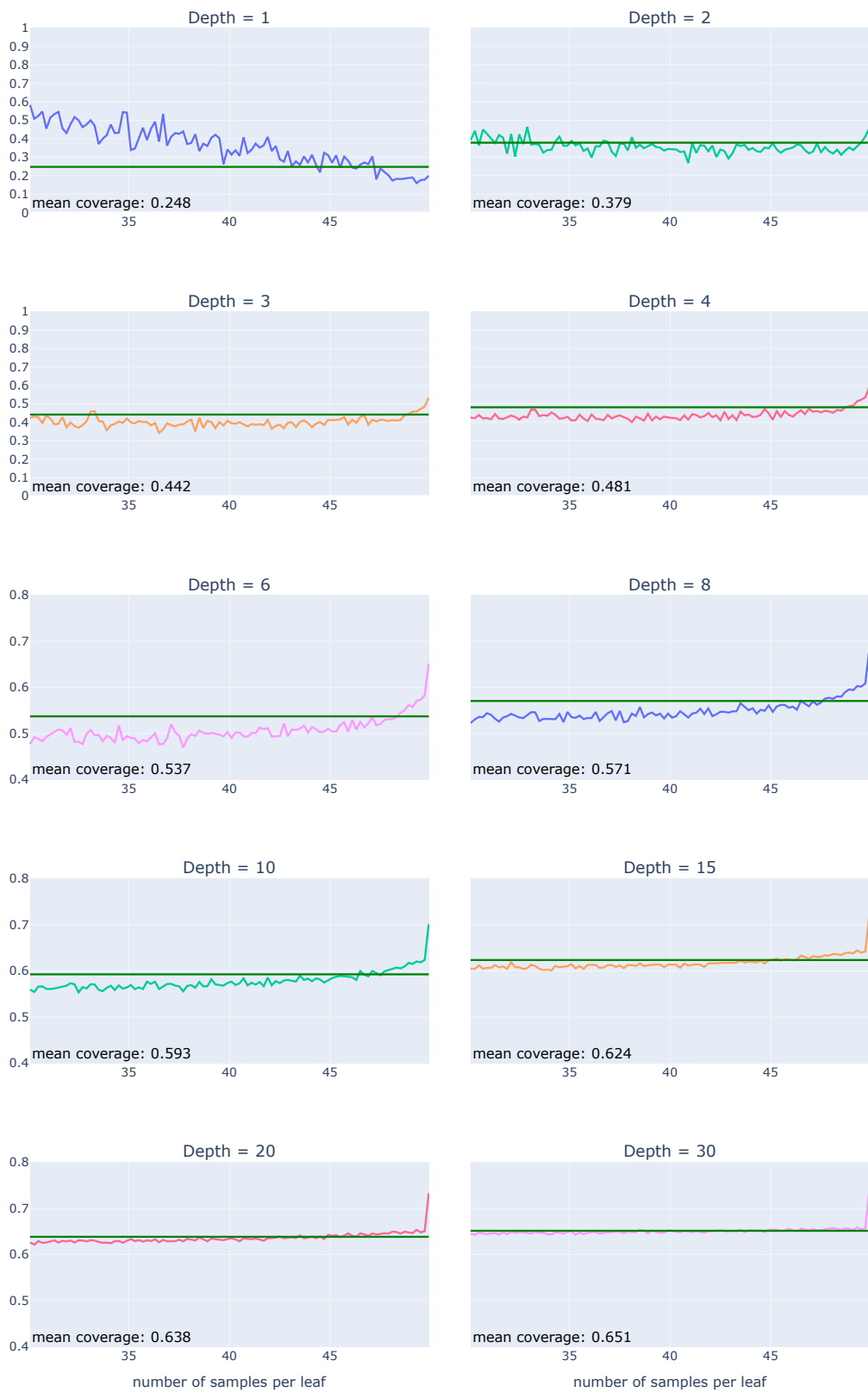


Figure 11: Coverage ratios for absolute deviations from the population mean of the covariate and adaptive estimation ($n=5000$).

Coverage for $n = 25000$ and adaptive estimation by extreme absolute deviations (bin width = 0.2)

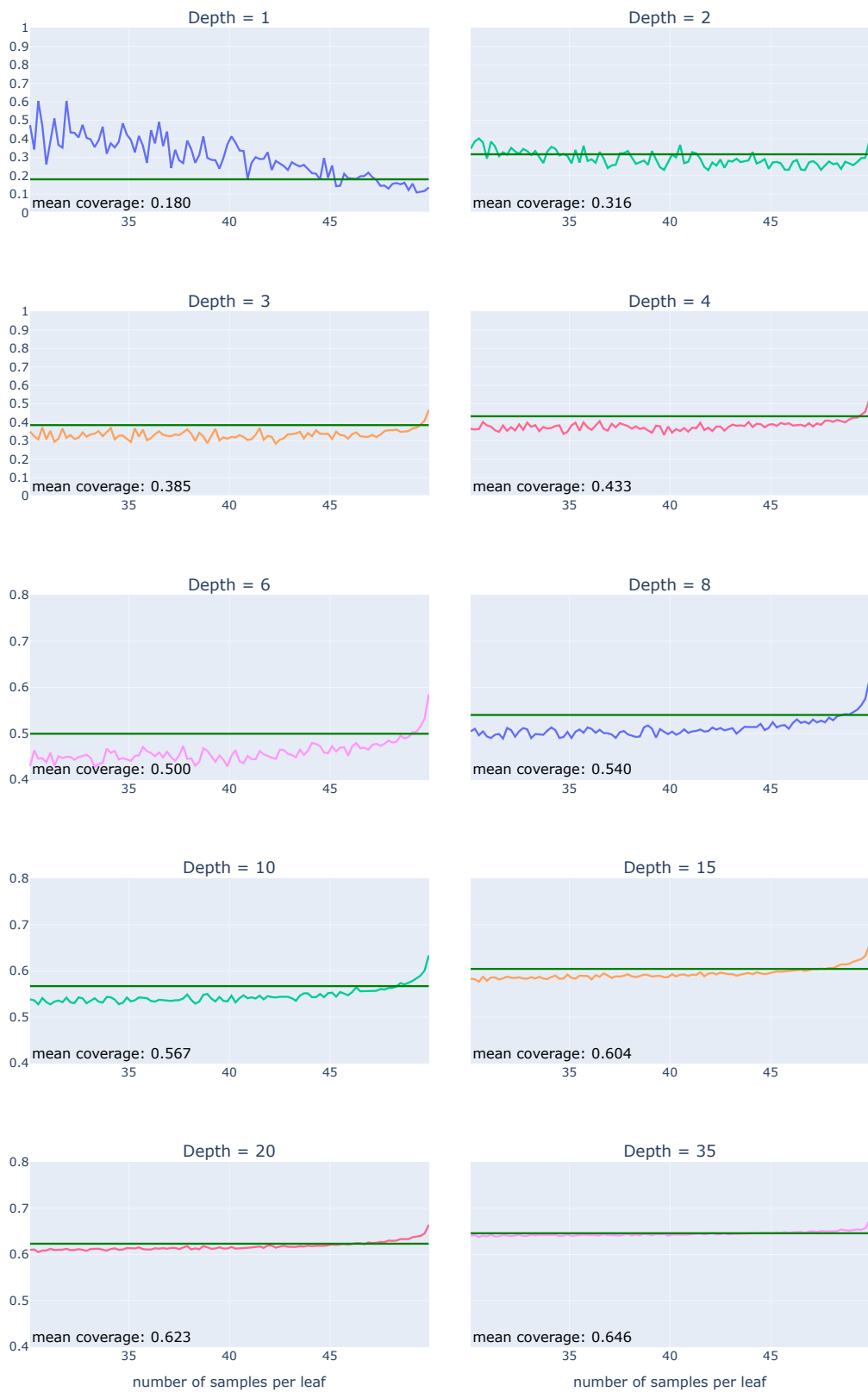


Figure 12: Coverage ratios for absolute deviations from the population mean of the covariate and adaptive estimation ($n=25000$).

A.2.2 Honest estimation with NaN coverages as NaN

Coverage for $n = 1000$ and honest estimation by extreme absolute deviations with coverage of NaN leaves as NaN (bin width = 0.02)

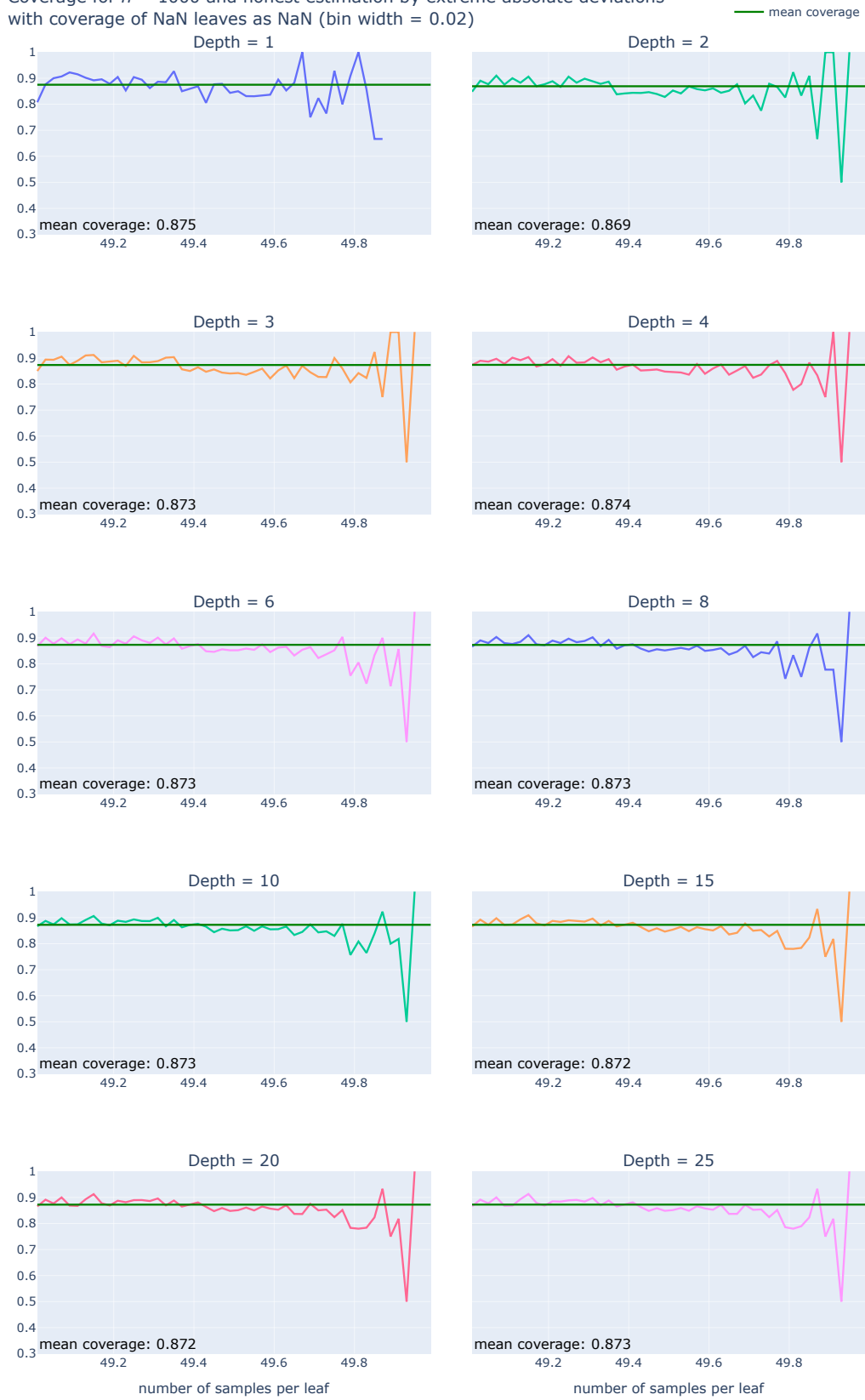


Figure 13: Coverage ratios for absolute deviations from the population mean of the covariate and honest estimation with coverages that cannot be estimated as nan ($n=1000$).

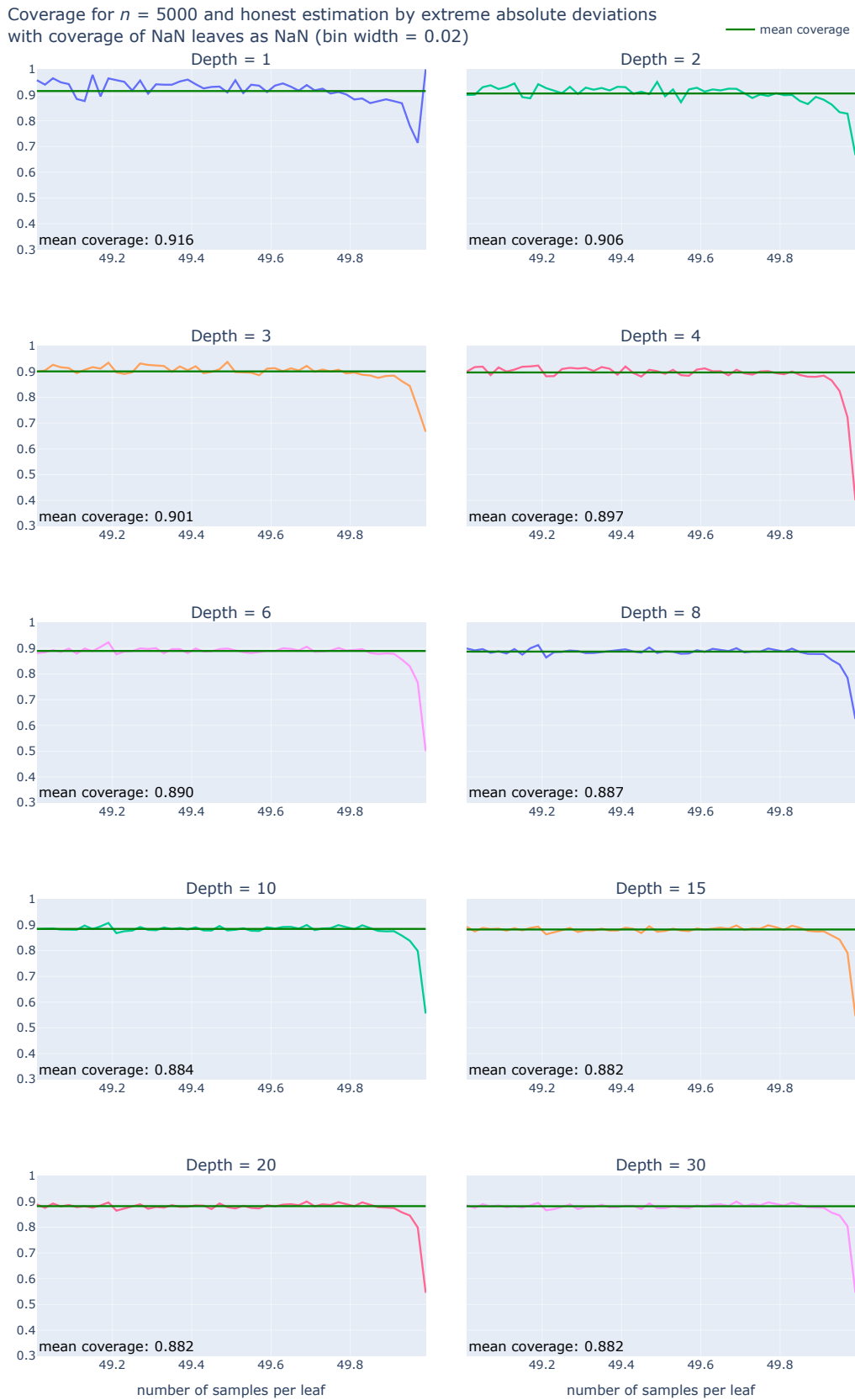


Figure 14: Coverage ratios for absolute deviations from the population mean of the covariate and honest estimation with coverages that cannot be estimated as nan ($n=5000$).

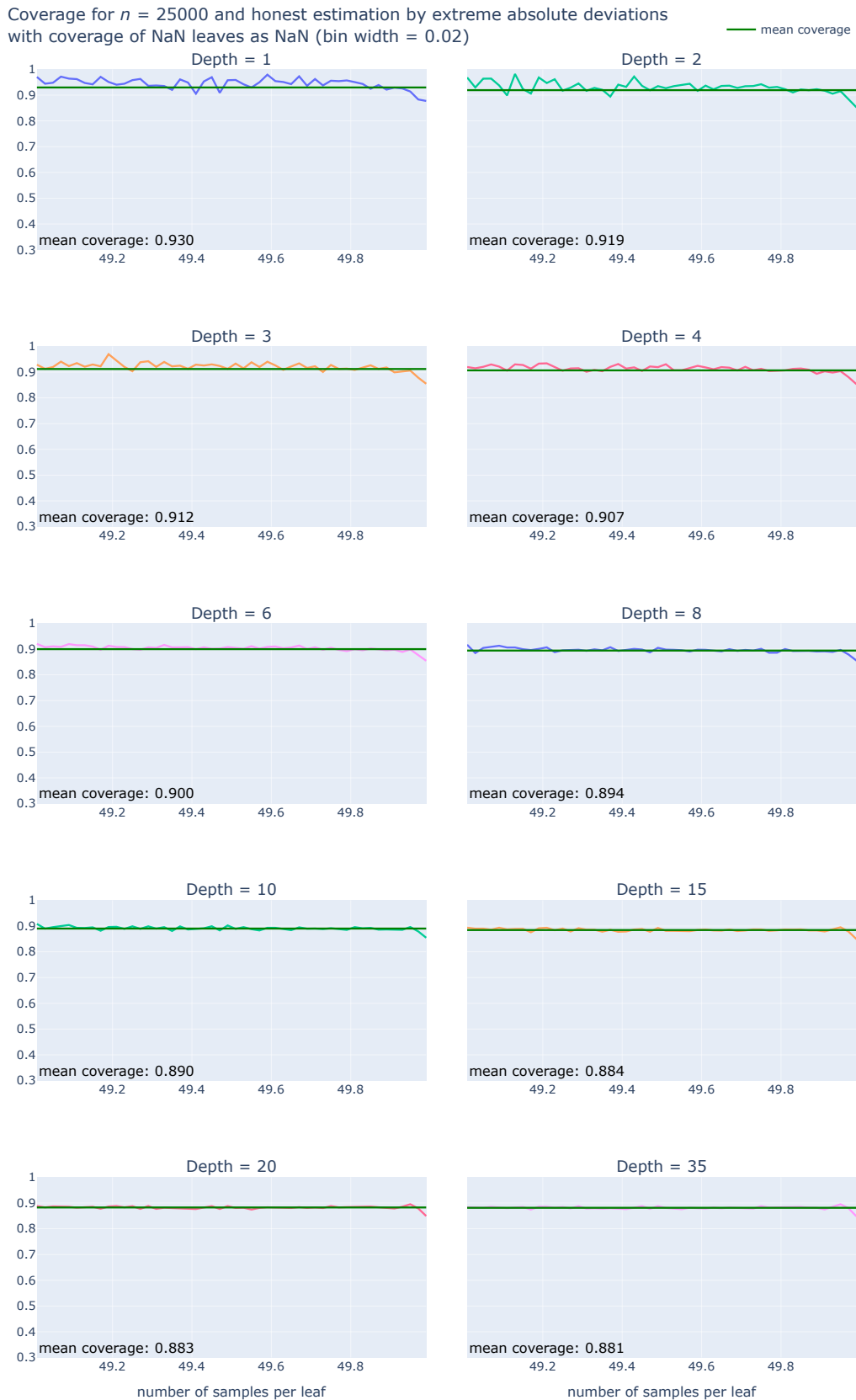


Figure 15: Coverage ratios for absolute deviations from the population mean of the covariate and honest estimation with coverages that cannot be estimated as nan ($n=25000$).

A.2.3 Honest estimation with NaN coverages as 0

Coverage for $n = 1000$ and honest estimation by extreme absolute deviations with coverage of NaN leaves as 0 (bin width = 0.1)

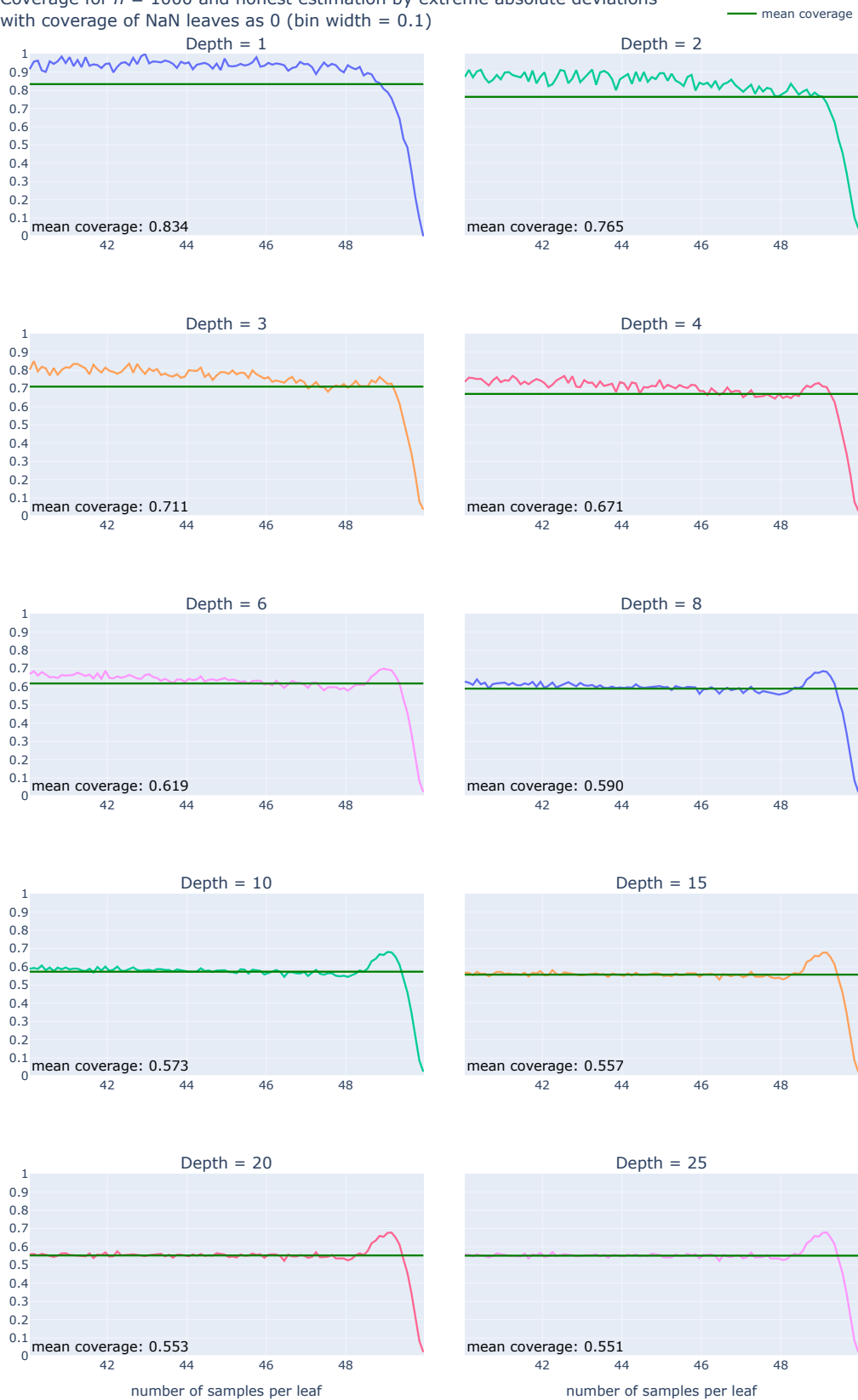


Figure 16: Coverage ratios for absolute deviations from the population mean of the covariate and honest estimation with coverages that cannot be estimated as 0 ($n=1000$).

Coverage for $n = 5000$ and honest estimation by extreme absolute deviations
with coverage of NaN leaves as 0 (bin width = 0.1)

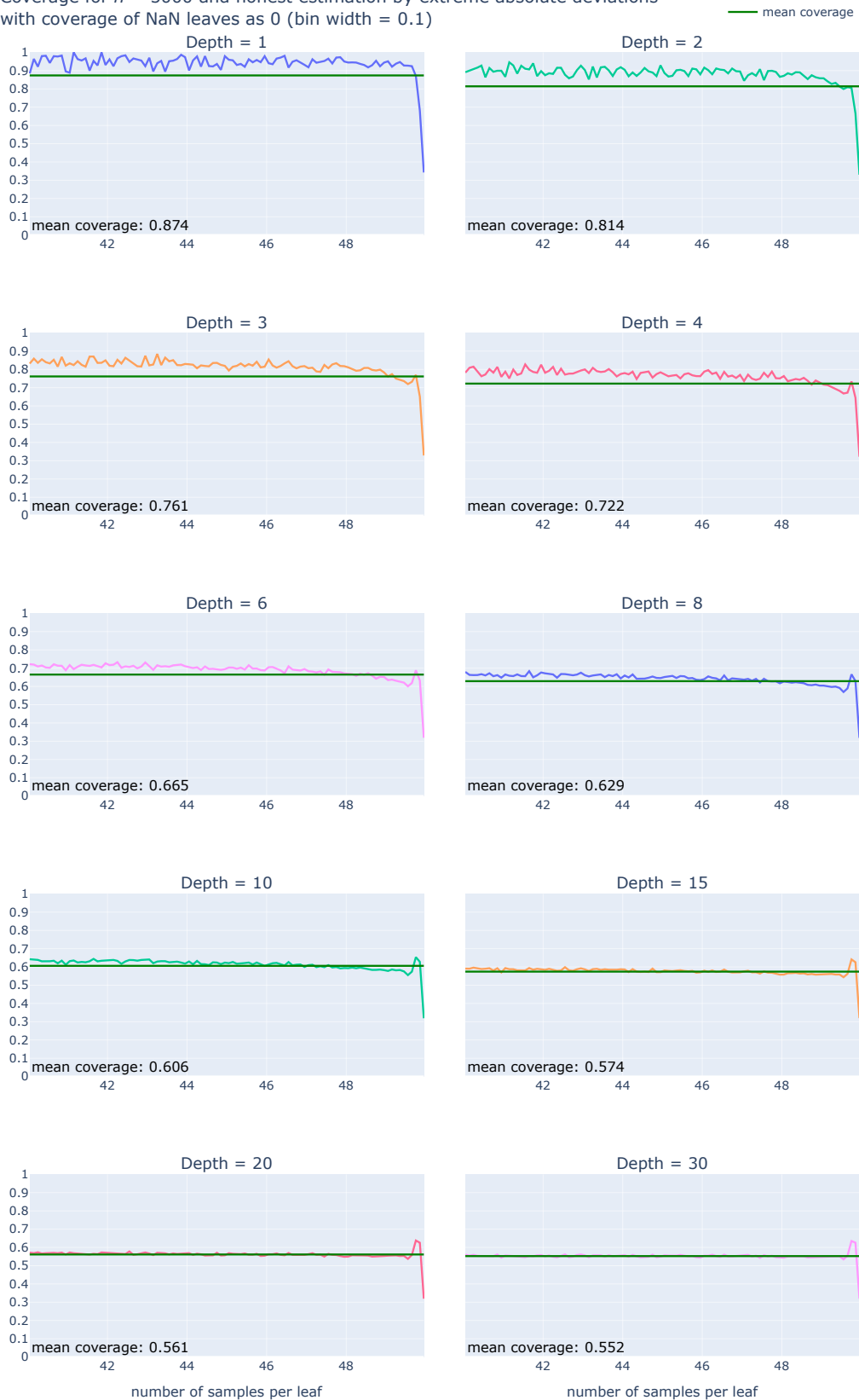


Figure 17: Coverage ratios for absolute deviations from the population mean of the covariate and honest estimation with coverages that cannot be estimated as 0 ($n=5000$).

Coverage for $n = 25000$ and honest estimation by extreme absolute deviations with coverage of NaN leaves as 0 (bin width = 0.1)

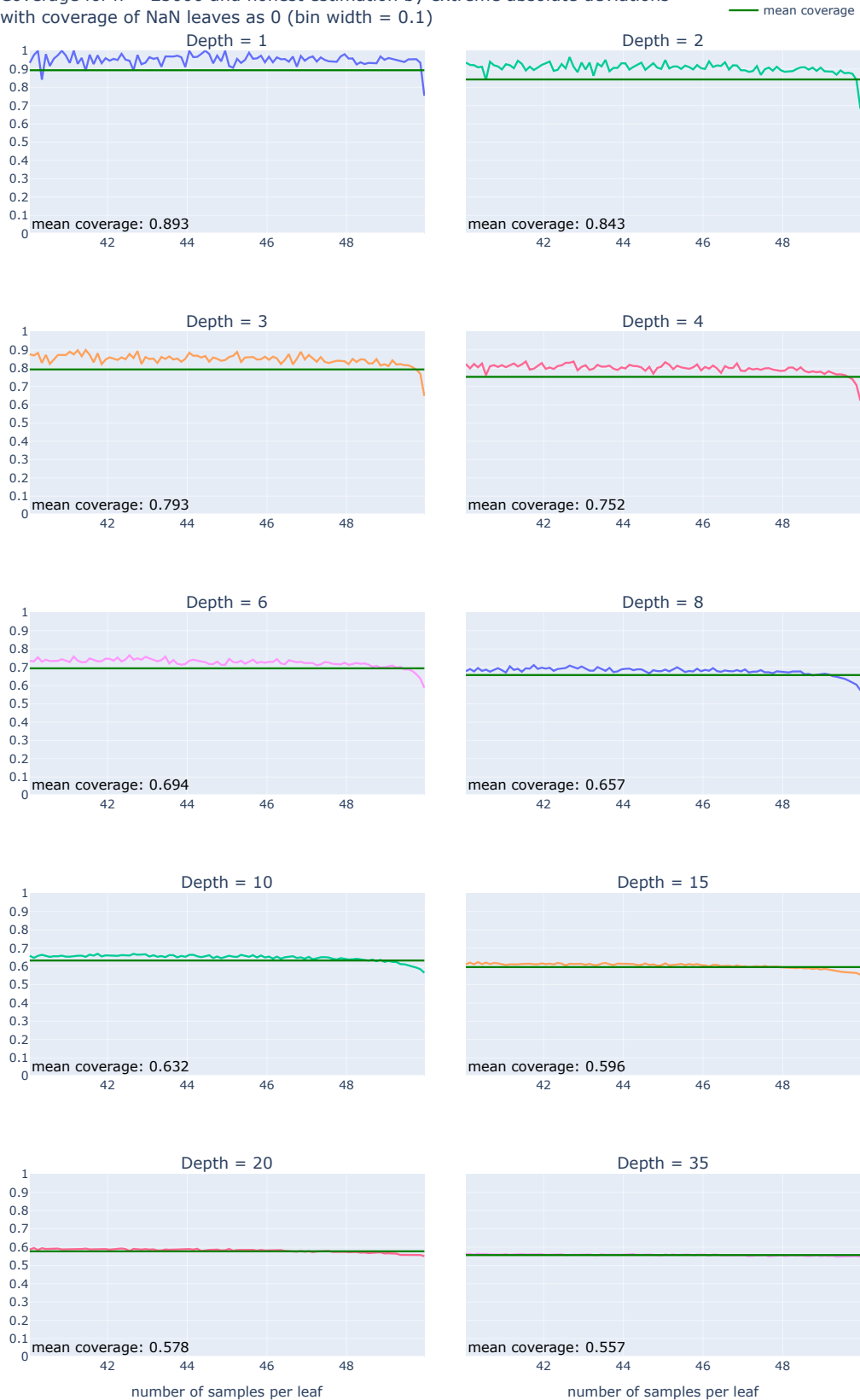


Figure 18: Coverage ratios for absolute deviations from the population mean of the covariate and honest estimation with coverages that cannot be estimated as 0 ($n=25000$).

B Weighted Coverage Ratios

Table 1: Weighted average coverage by depth and splitting type (n=1000)

	Adaptive	Honest (NaN as 0)
Depth		
1	0.844	0.949
2	0.825	0.944
3	0.819	0.938
4	0.812	0.929
6	0.810	0.902
8	0.801	0.869
10	0.788	0.834
15	0.742	0.770
20	0.702	0.741
25	0.679	0.732

Table 2: Weighted average coverage by depth and splitting type (n=5000)

	Adaptive	Honest (NaN as 0)
Depth		
1	0.854	0.950
2	0.843	0.950
3	0.827	0.949
4	0.825	0.947
6	0.823	0.940
8	0.826	0.928
10	0.828	0.915
15	0.822	0.872
20	0.799	0.830
25	0.770	0.794
30	0.742	0.769

Table 3: Weighted average coverage by depth and splitting type (n=25000)

	Adaptive	Honest (NaN as 0)
Depth		
1	0.864	0.949
2	0.852	0.949
3	0.838	0.950
4	0.831	0.950
6	0.827	0.948
8	0.827	0.946
10	0.830	0.942
15	0.838	0.926
20	0.841	0.903
35	0.805	0.829

C Confidence Interval Analysis

C.1 Entire support

C.1.1 Adaptive estimation

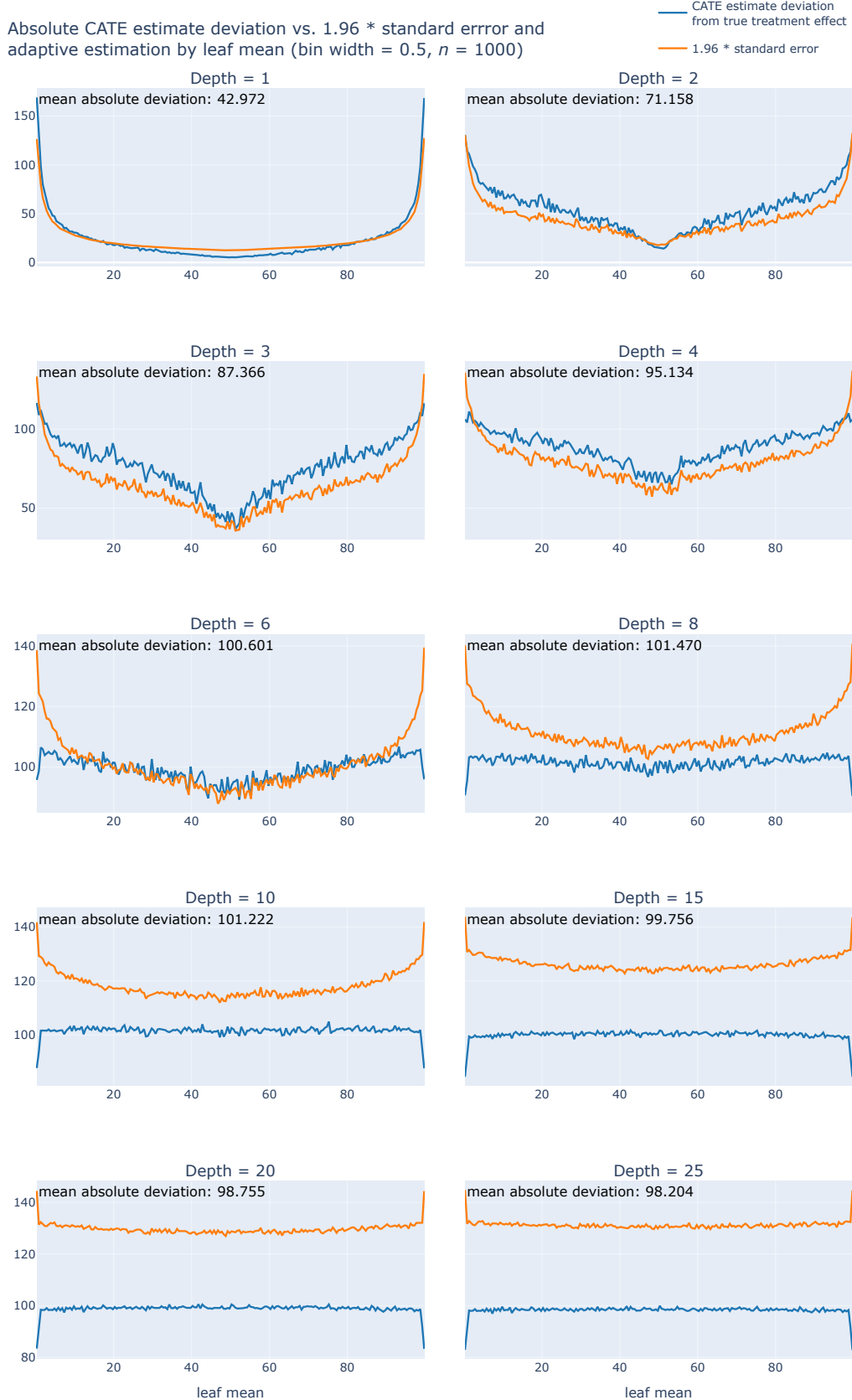


Figure 19: Deviation of the treatment effect estimate from the true treatment effect compared to $1.96 \times$ standard error for adaptive estimation ($n=1000$).

Absolute CATE estimate deviation vs. $1.96 * \text{standard error}$ and adaptive estimation by leaf mean (bin width = 0.5, $n = 5000$)

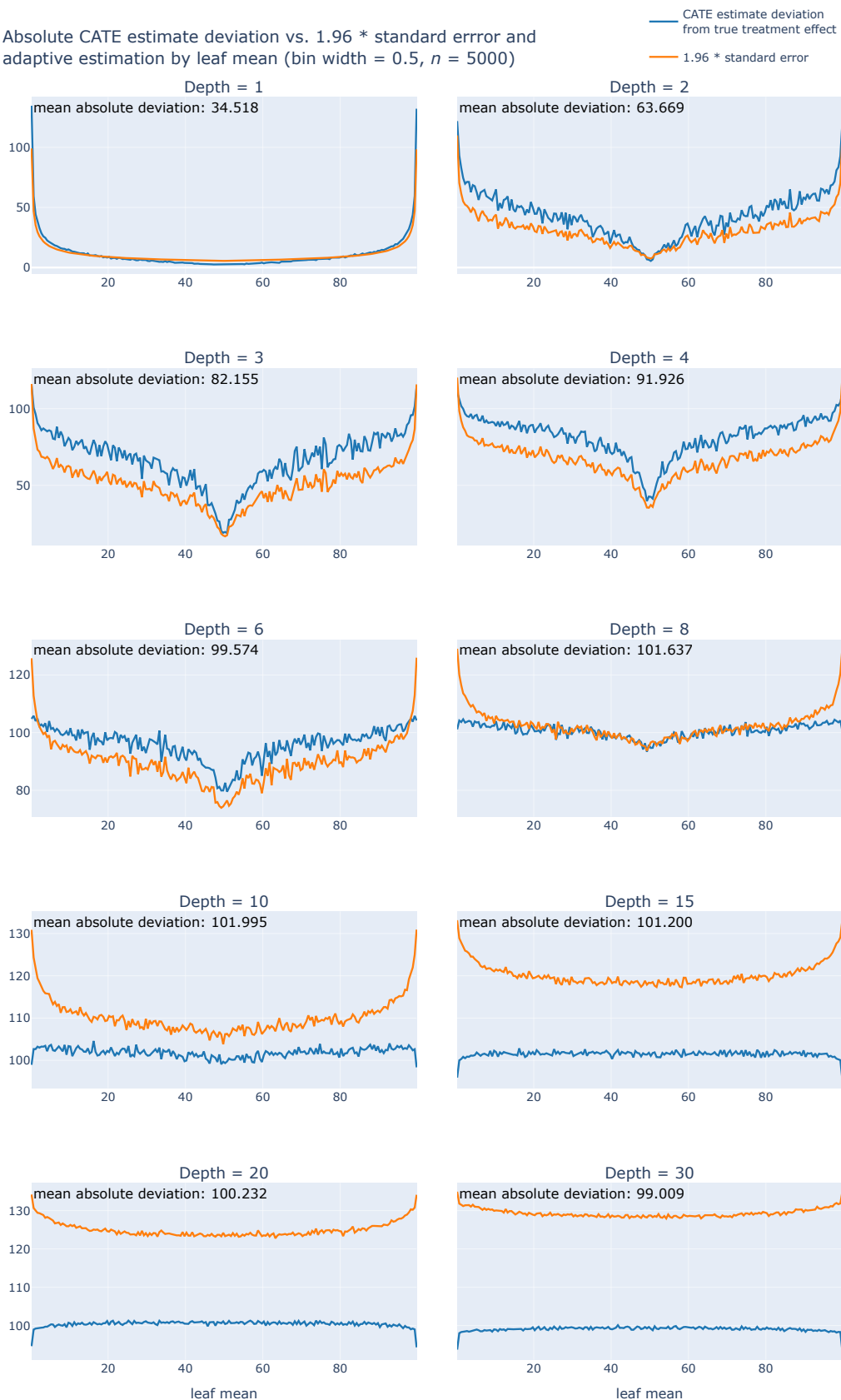


Figure 20: Deviation of the treatment effect estimate from the true treatment effect compared to $1.96 * \text{standard error}$ for adaptive estimation ($n=5000$).

Absolute CATE estimate deviation vs. $1.96 \times$ standard error and adaptive estimation by leaf mean (bin width = 0.5, $n = 25000$)

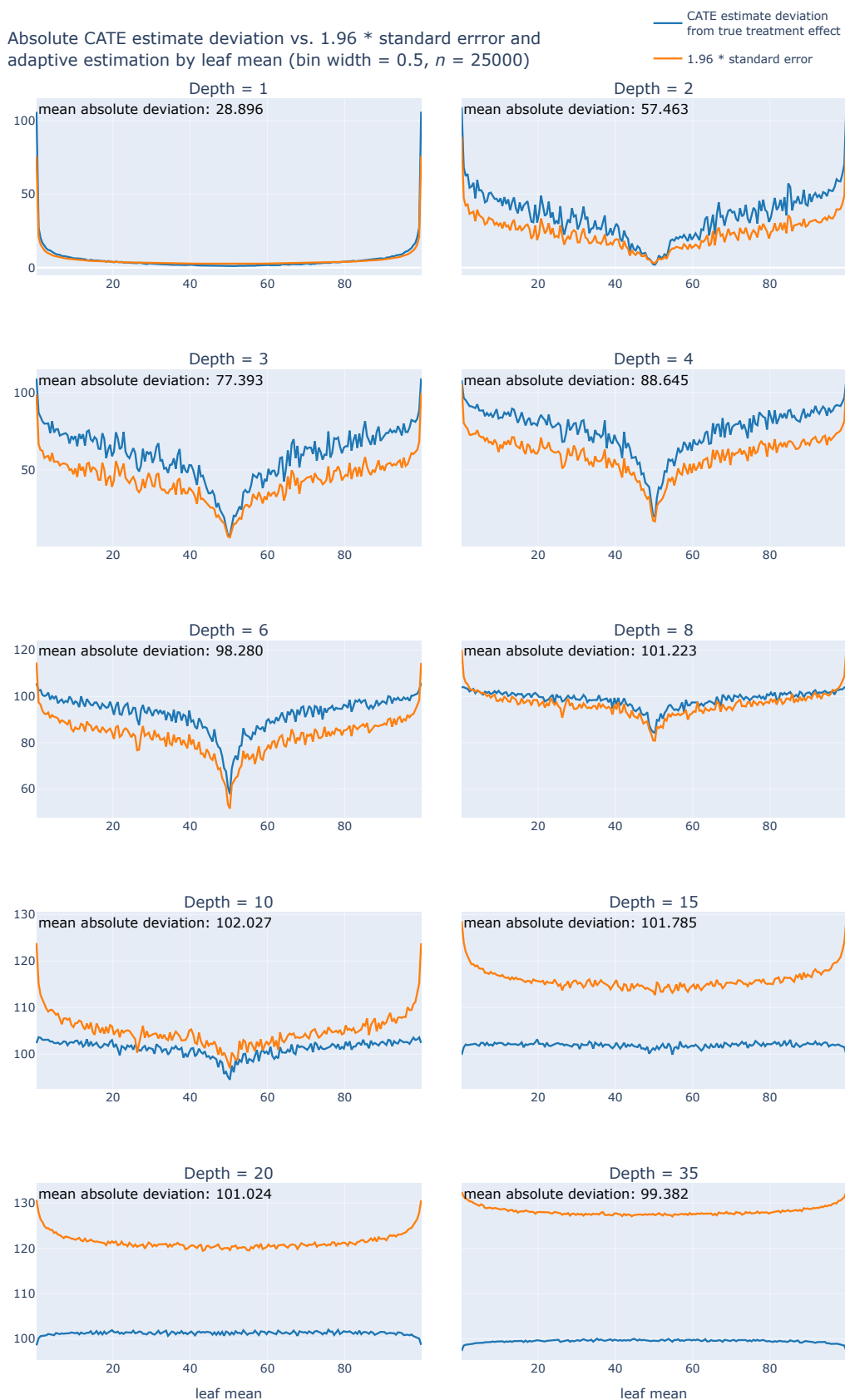


Figure 21: Deviation of the treatment effect estimate from the true treatment effect compared to $1.96 \times$ standard error for adaptive estimation ($n=25000$).

C.1.2 Honest estimation

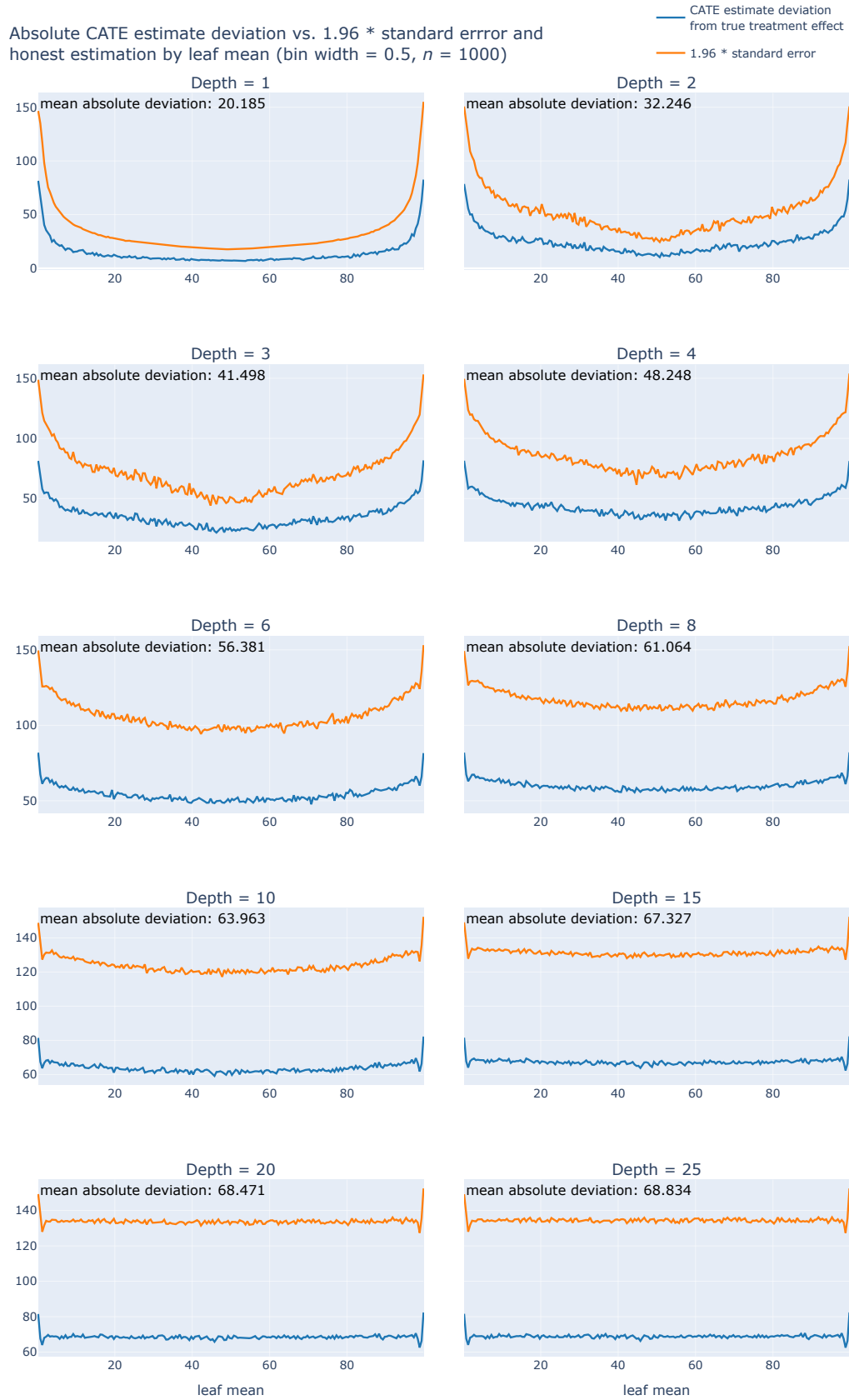


Figure 22: Deviation of the treatment effect estimate from the true treatment effect compared to $1.96 \times$ standard error for honest estimation ($n=1000$).

Absolute CATE estimate deviation vs. $1.96 \times$ standard error and honest estimation by leaf mean (bin width = 0.5, $n = 5000$)

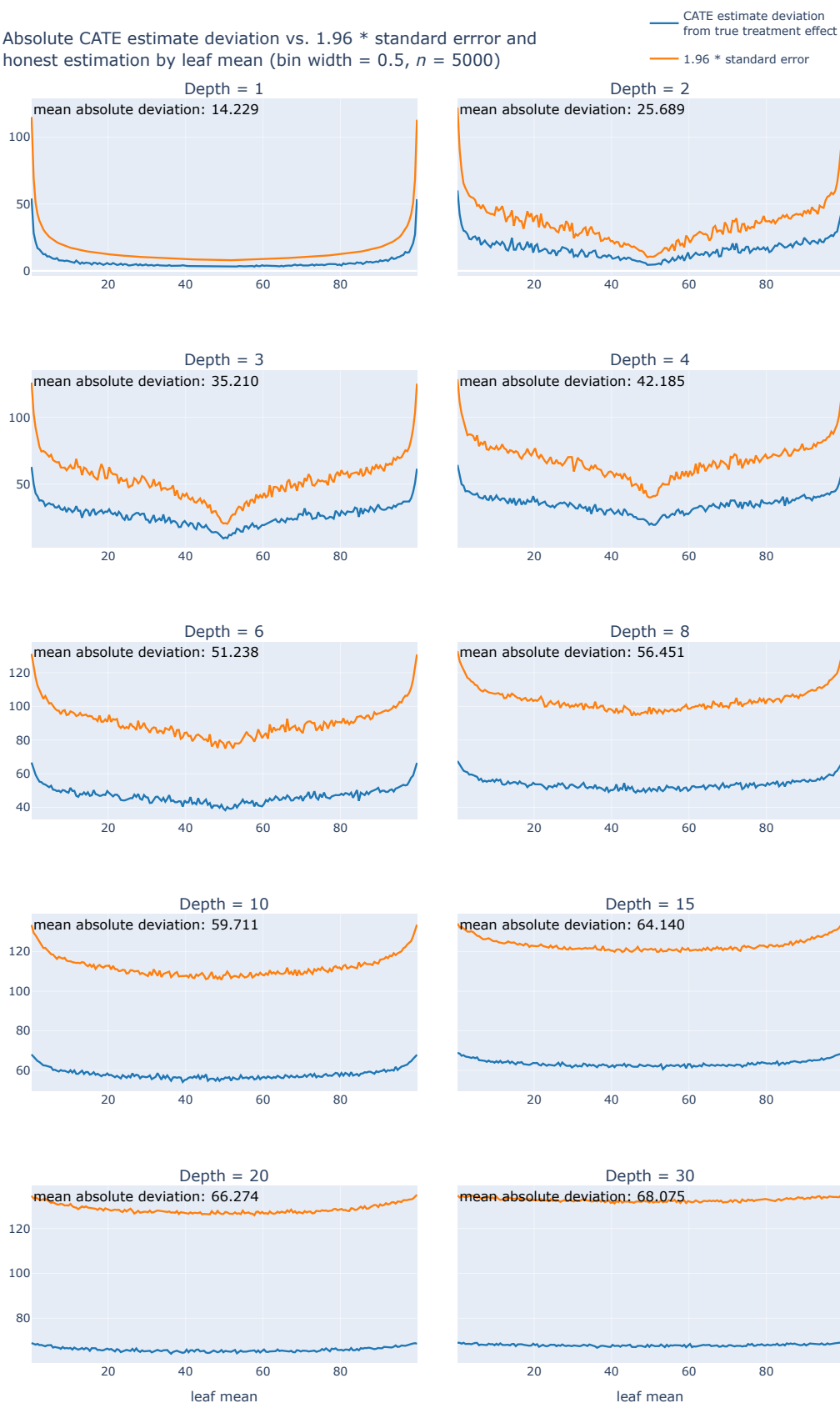


Figure 23: Deviation of the treatment effect estimate from the true treatment effect compared to $1.96 \times$ standard error for honest estimation ($n=5000$).

Absolute CATE estimate deviation vs. $1.96 \times$ standard error and honest estimation by leaf mean (bin width = 0.5, $n = 25000$)

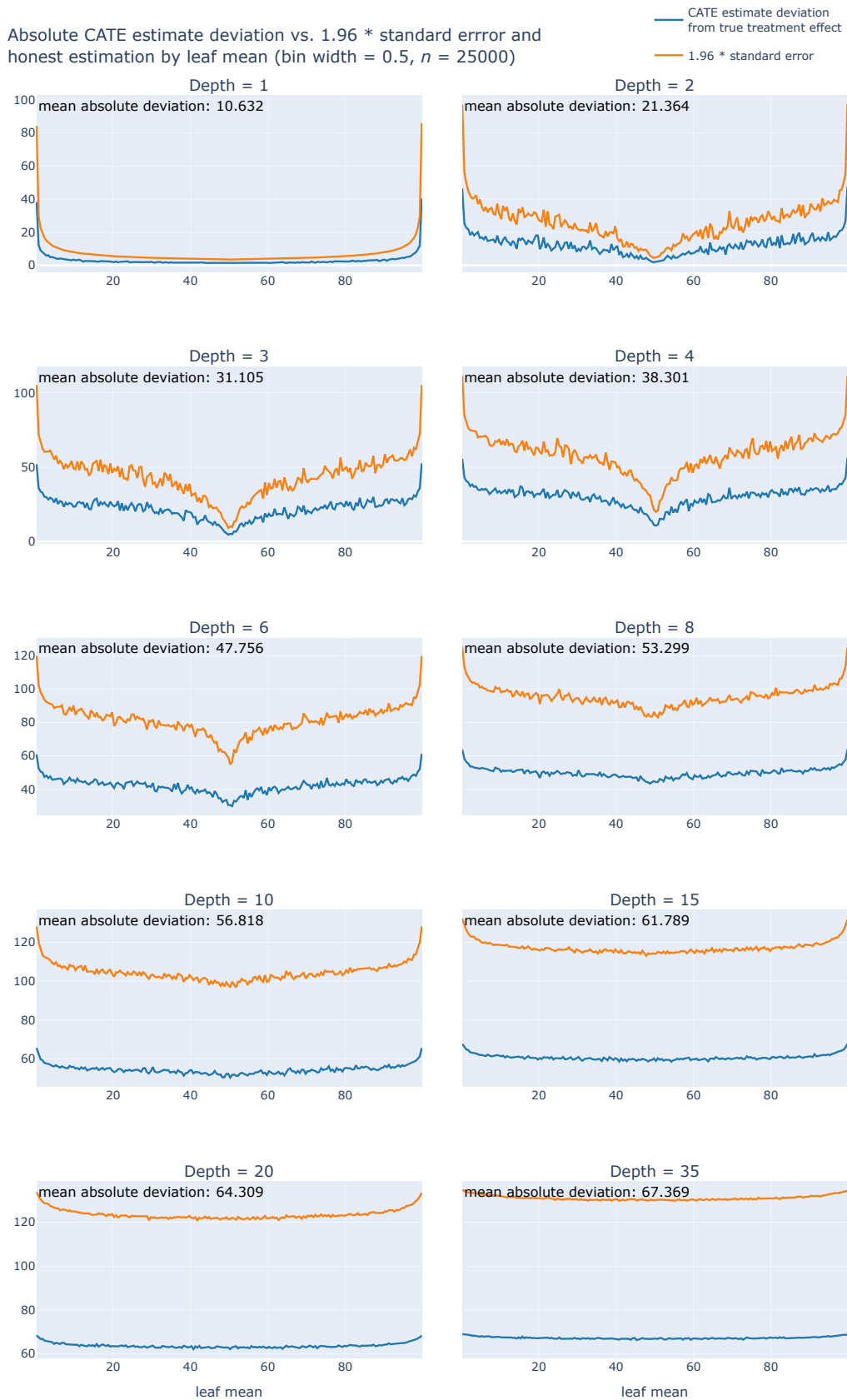


Figure 24: Deviation of the treatment effect estimate from the true treatment effect compared to $1.96 \times$ standard error for honest estimation ($n=25000$).

C.2 Extreme absolute deviations

C.2.1 Adaptive estimation

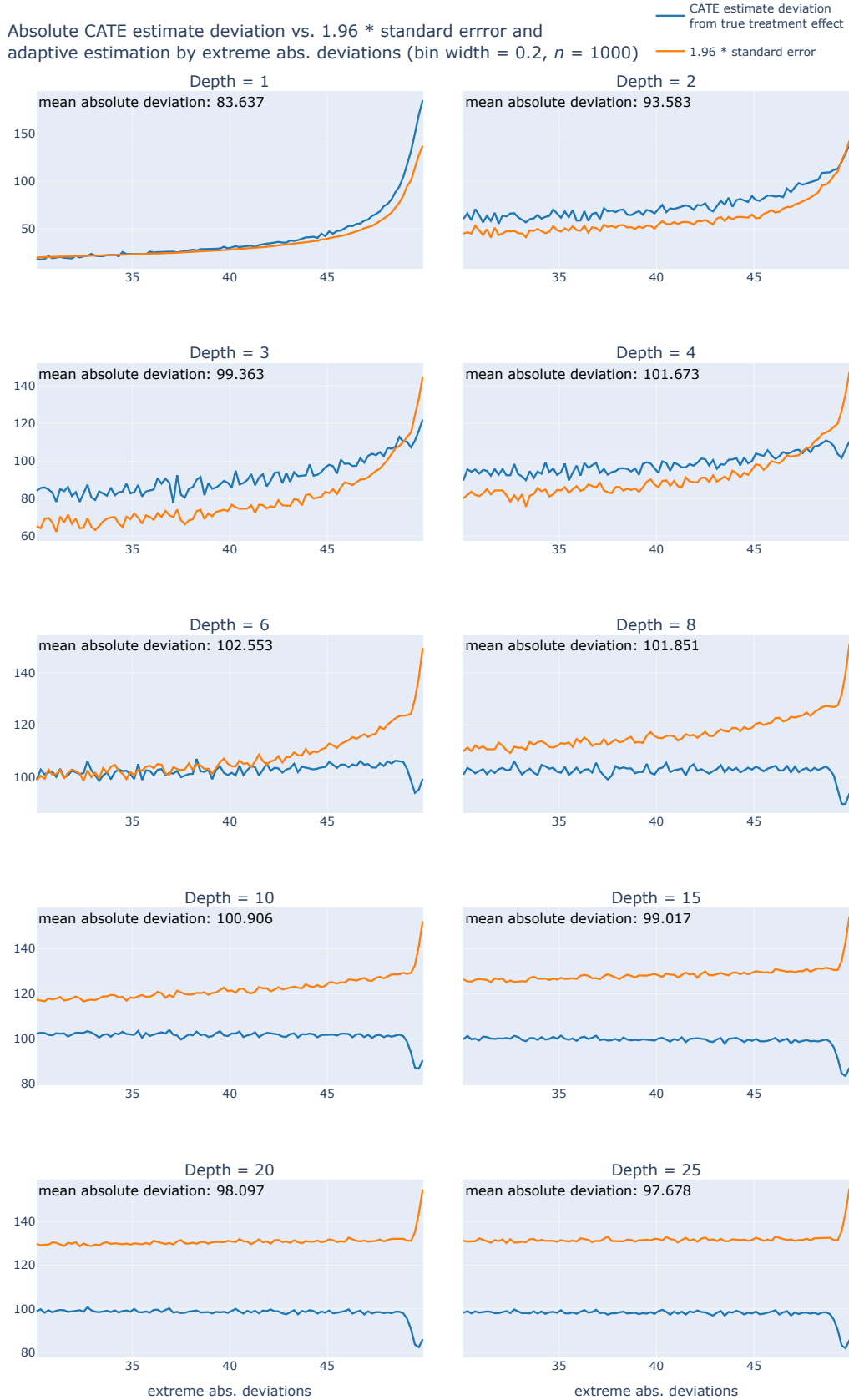


Figure 25: Deviation of the treatment effect estimate from the true treatment effect compared to $1.96 \times$ standard error for absolute deviations from the population mean of the covariate and adaptive estimation ($n=1000$).

C.2.2 Honest estimation

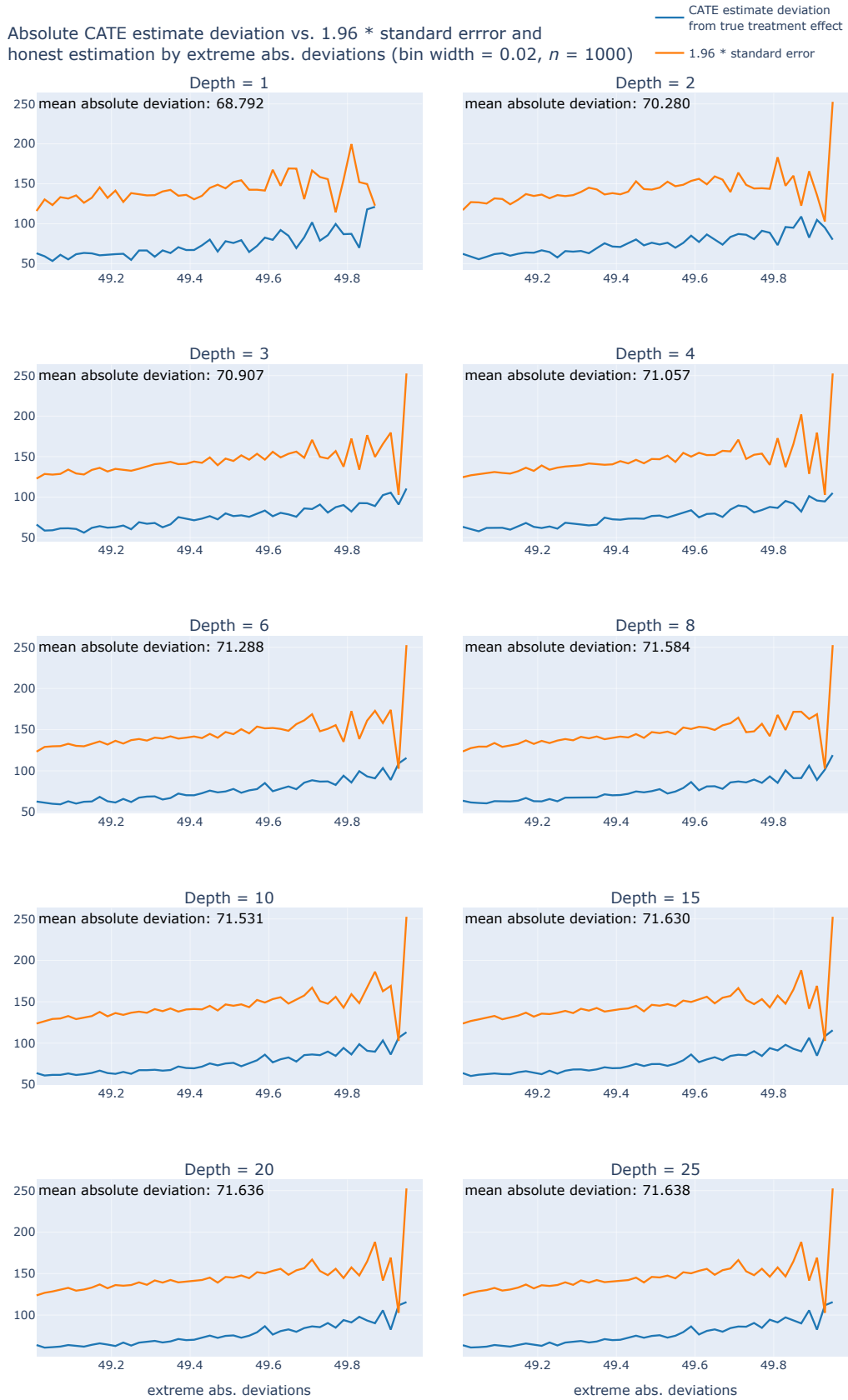


Figure 26: Deviation of the treatment effect estimate from the true treatment effect compared to $1.96 \times$ standard error for absolute deviations from the population mean of the covariate and honest estimation ($n=1000$).

D Relative Frequency Distribution of Leaf Means

D.1 Adaptive Estimation

Relative frequency distribution of leaf means for $n = 1000$ and adaptive estimation (bin width=0.5)

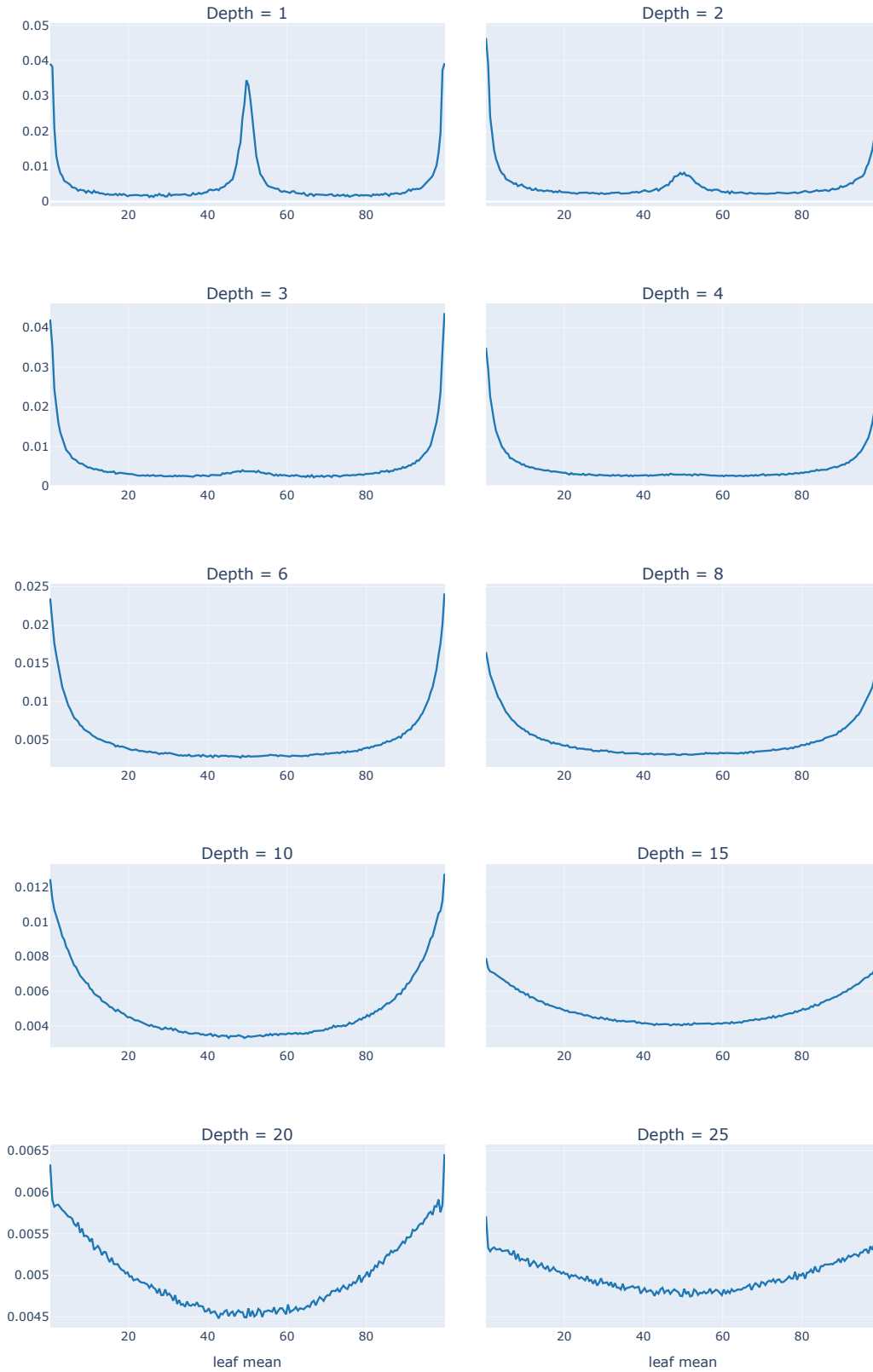


Figure 27: Relative frequency distribution of leaf means for adaptive estimation ($n=1000$).

Relative frequency distribution of leaf means for $n = 5000$ and adaptive estimation (bin width=0.5)

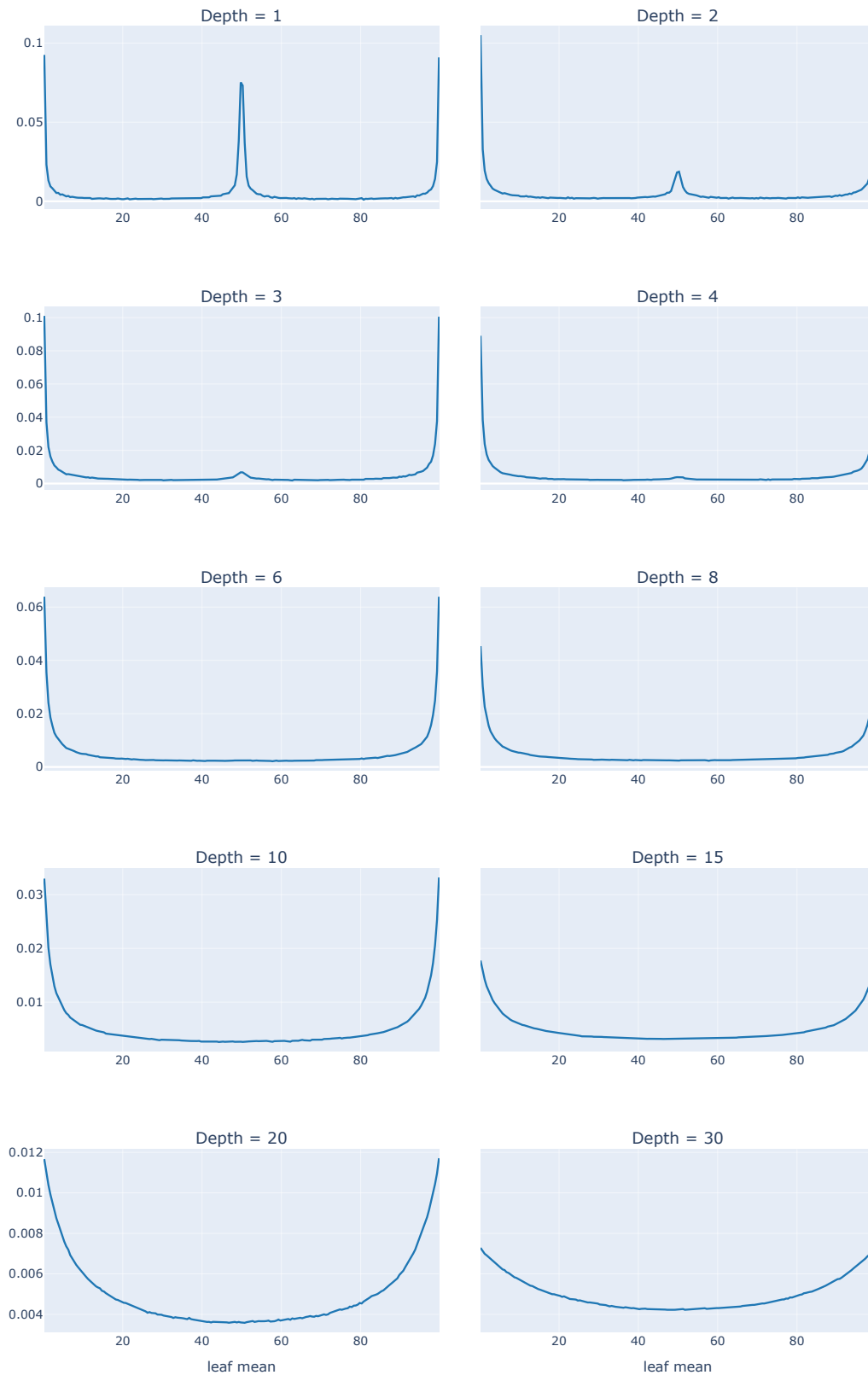


Figure 28: Relative frequency distribution of leaf means for adaptive estimation ($n=5000$).

Relative frequency distribution of leaf means for $n = 25000$ and adaptive estimation (bin width=0.5)

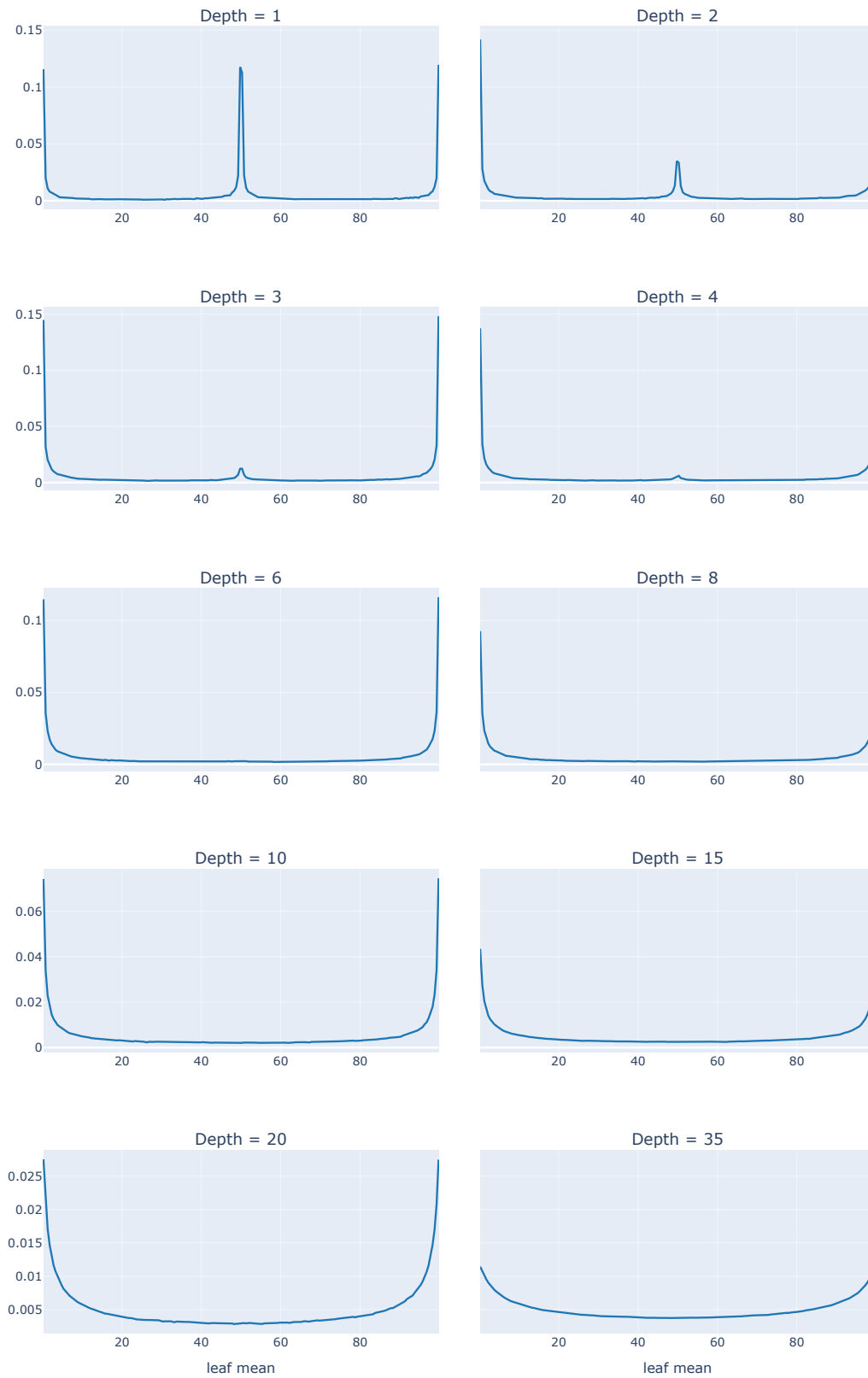


Figure 29: Relative frequency distribution of leaf means for adaptive estimation ($n=25000$).

D.2 Honest estimation with NaN coverages included

Relative frequency distribution of leaf means for $n = 1000$
and honest estimation with nan coverages included (bin width=0.5)

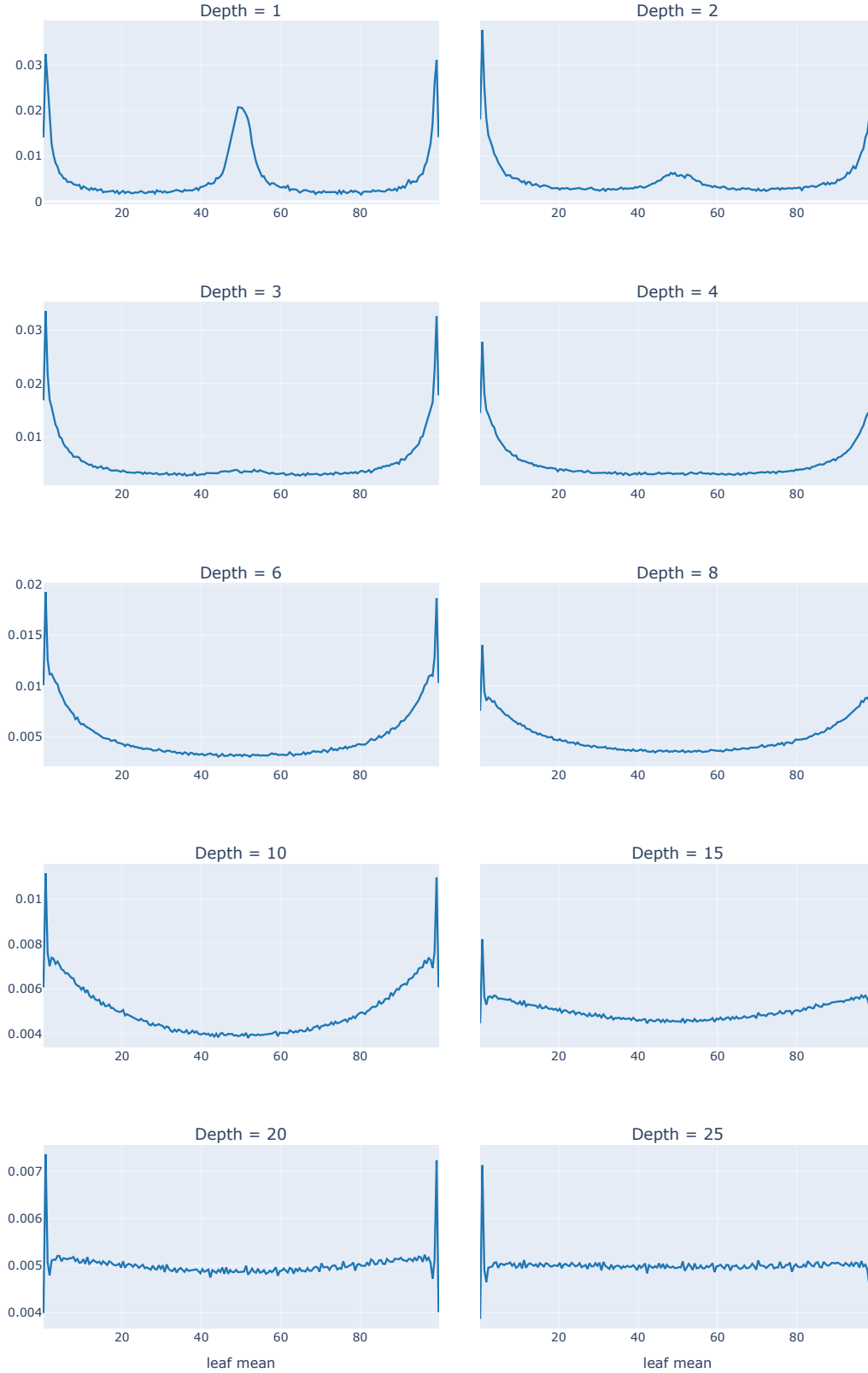


Figure 30: Relative frequency distribution of leaf means for honest estimation with leaf means that have coverage as nan ($n=1000$).

Relative frequency distribution of leaf means for $n = 5000$
and honest estimation with nan coverages included (bin width=0.5)

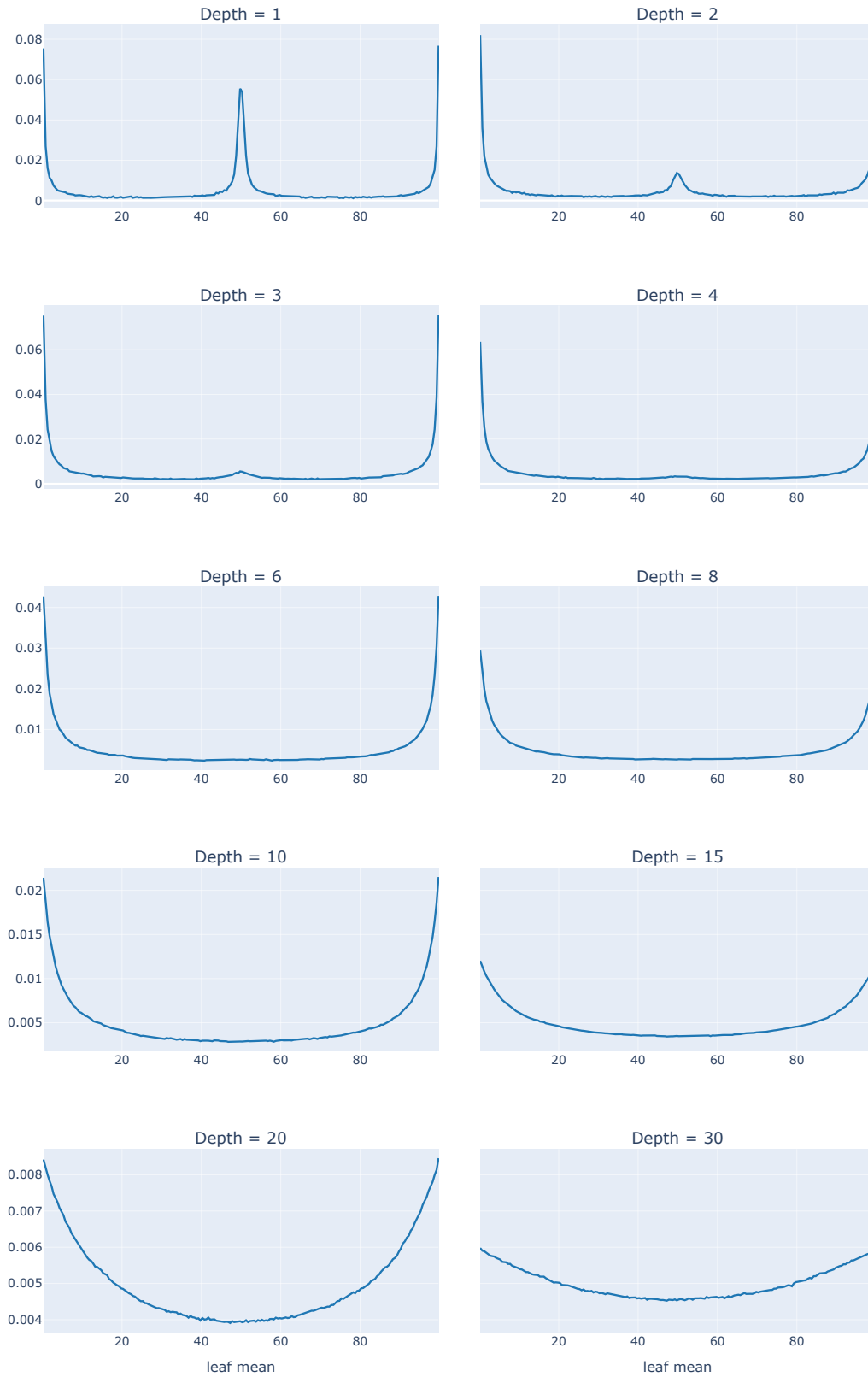


Figure 31: Relative frequency distribution of leaf means for honest estimation with leaf means that have coverage as nan ($n=5000$).

Relative frequency distribution of leaf means for $n = 25000$
and honest estimation with nan coverages included (bin width=0.5)

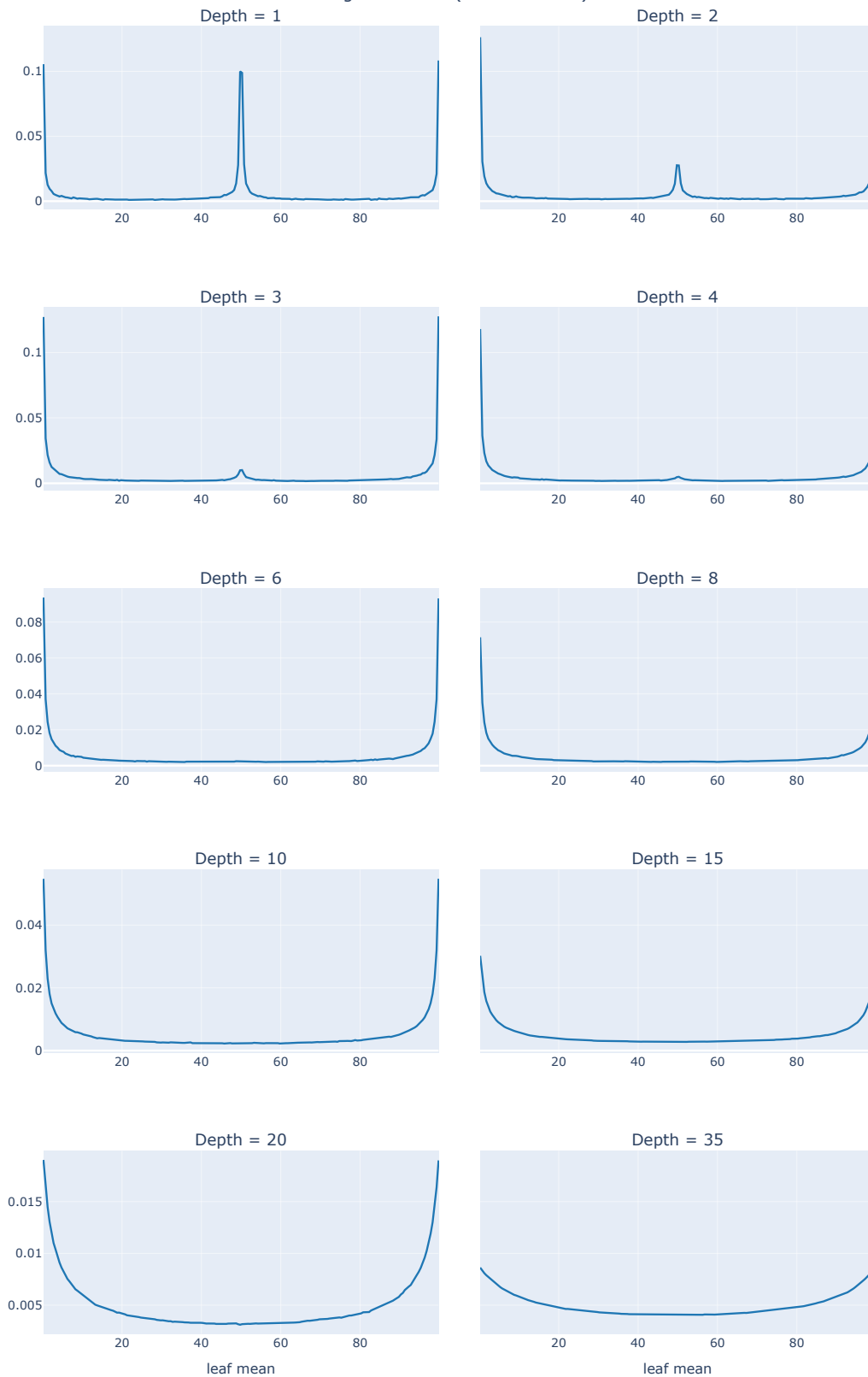


Figure 32: Relative frequency distribution of leaf means for honest estimation with leaf means that have coverage as nan ($n=25000$).

E Relative Frequency Distribution of the Number of Samples per Leaf Mean

E.1 Adaptive estimation

Relative frequency distribution of the number of samples per leaf for $n = 1000$ and adaptive estimation (bin width=0.5)

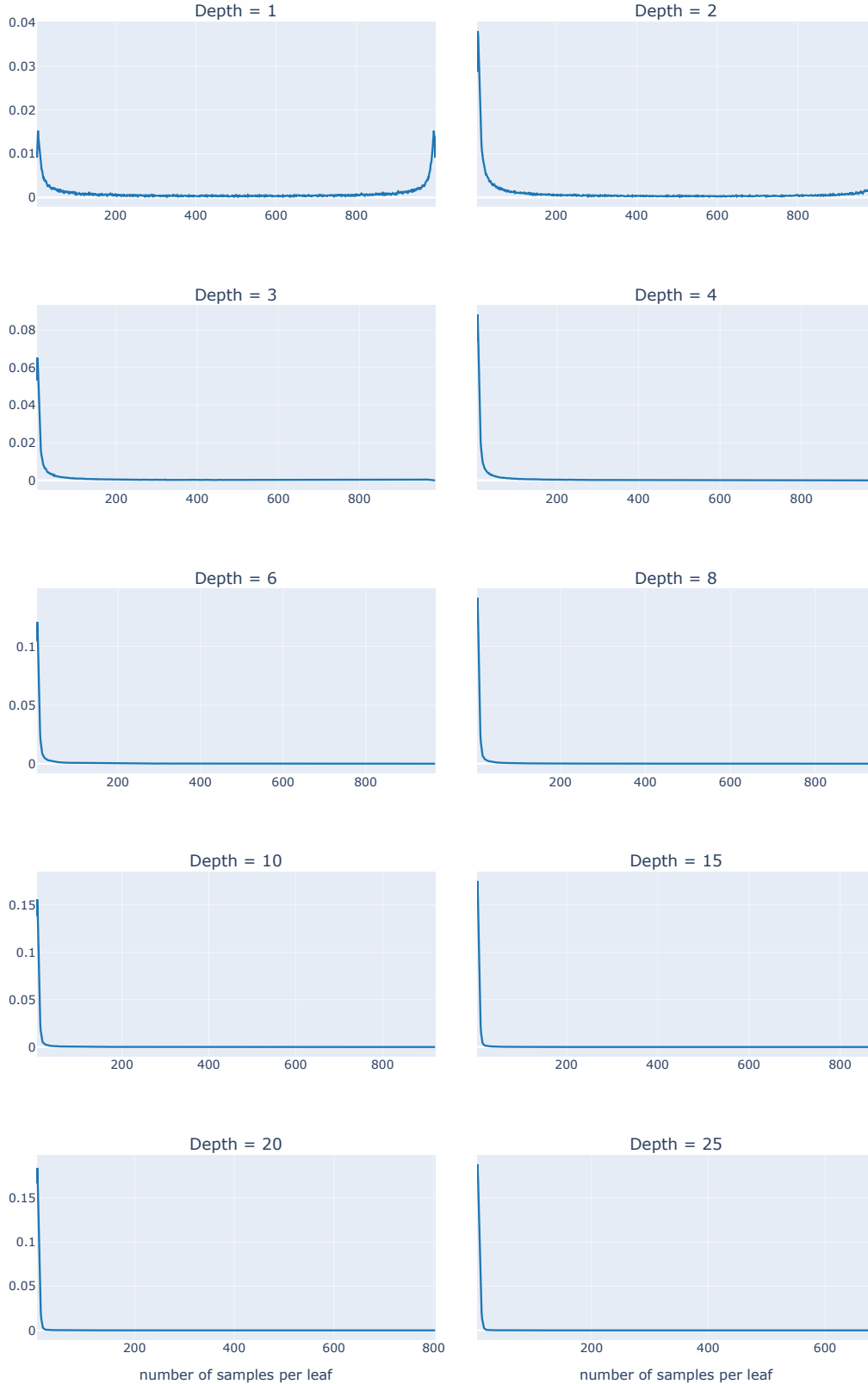


Figure 33: Relative frequency distribution of leaf means for adaptive estimation ($n=1000$).

Relative frequency distribution of the number of samples per leaf for $n = 5000$ and adaptive estimation (bin width=0.5)

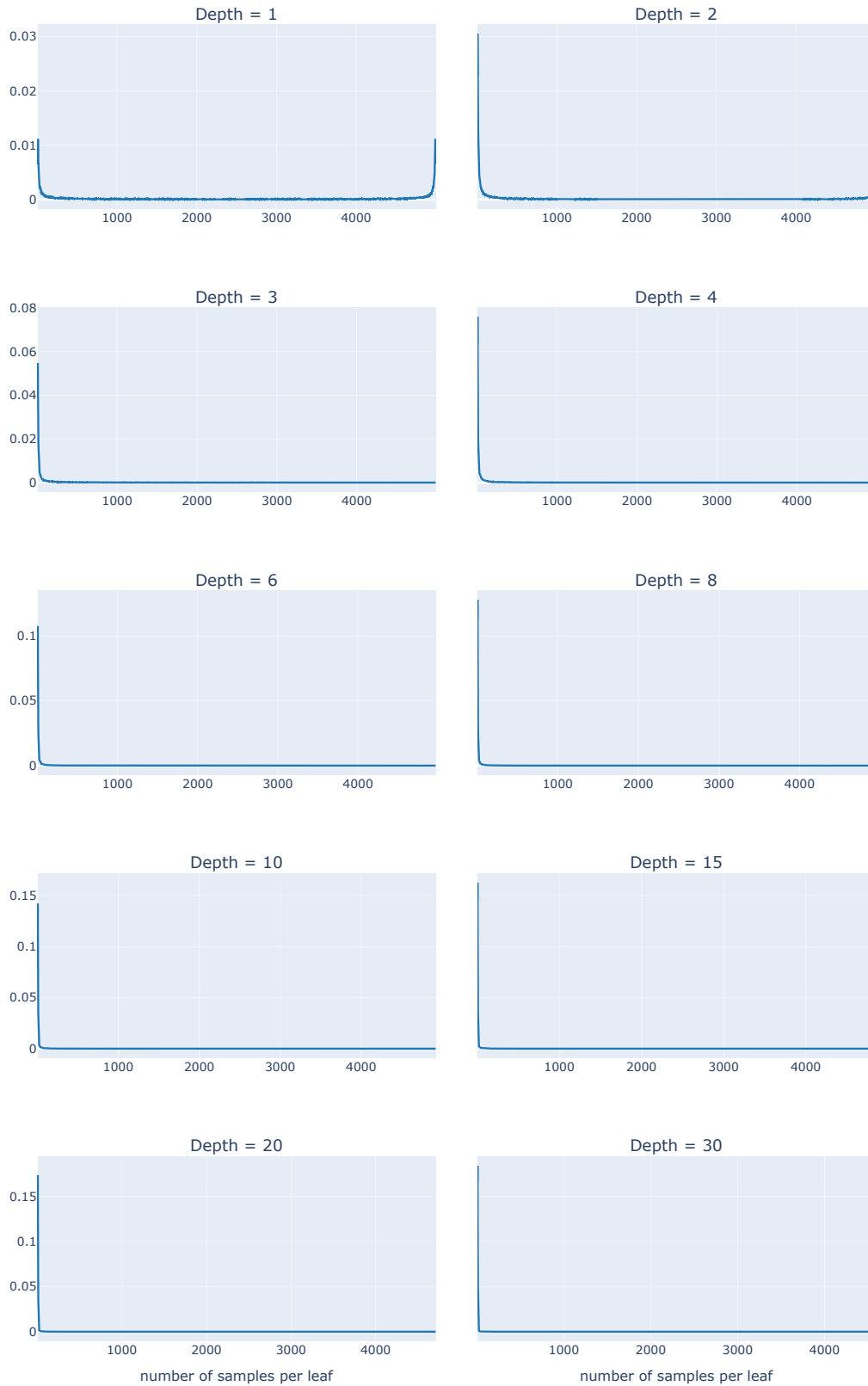


Figure 34: Relative frequency distribution of leaf means for adaptive estimation ($n=5000$).

Relative frequency distribution of the number of samples per leaf for $n = 25000$ and adaptive estimation (bin width=0.5)

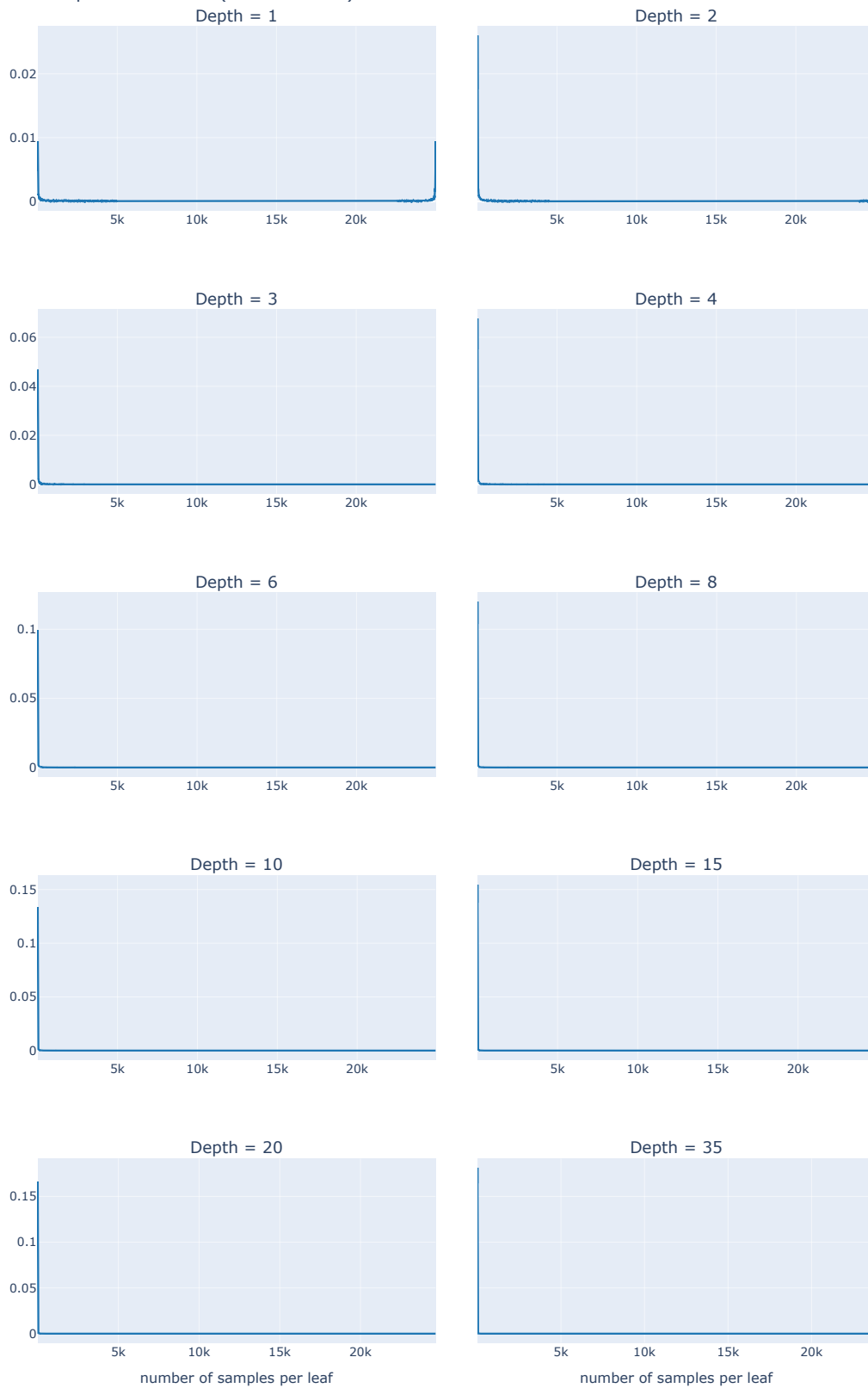


Figure 35: Relative frequency distribution of leaf means for adaptive estimation ($n=25000$).

E.2 Honest estimation with NaN coverages included

Relative frequency distribution of the number of samples per leaf for $n = 1000$ and honest estimation with nan coverages included (bin width=0.5)

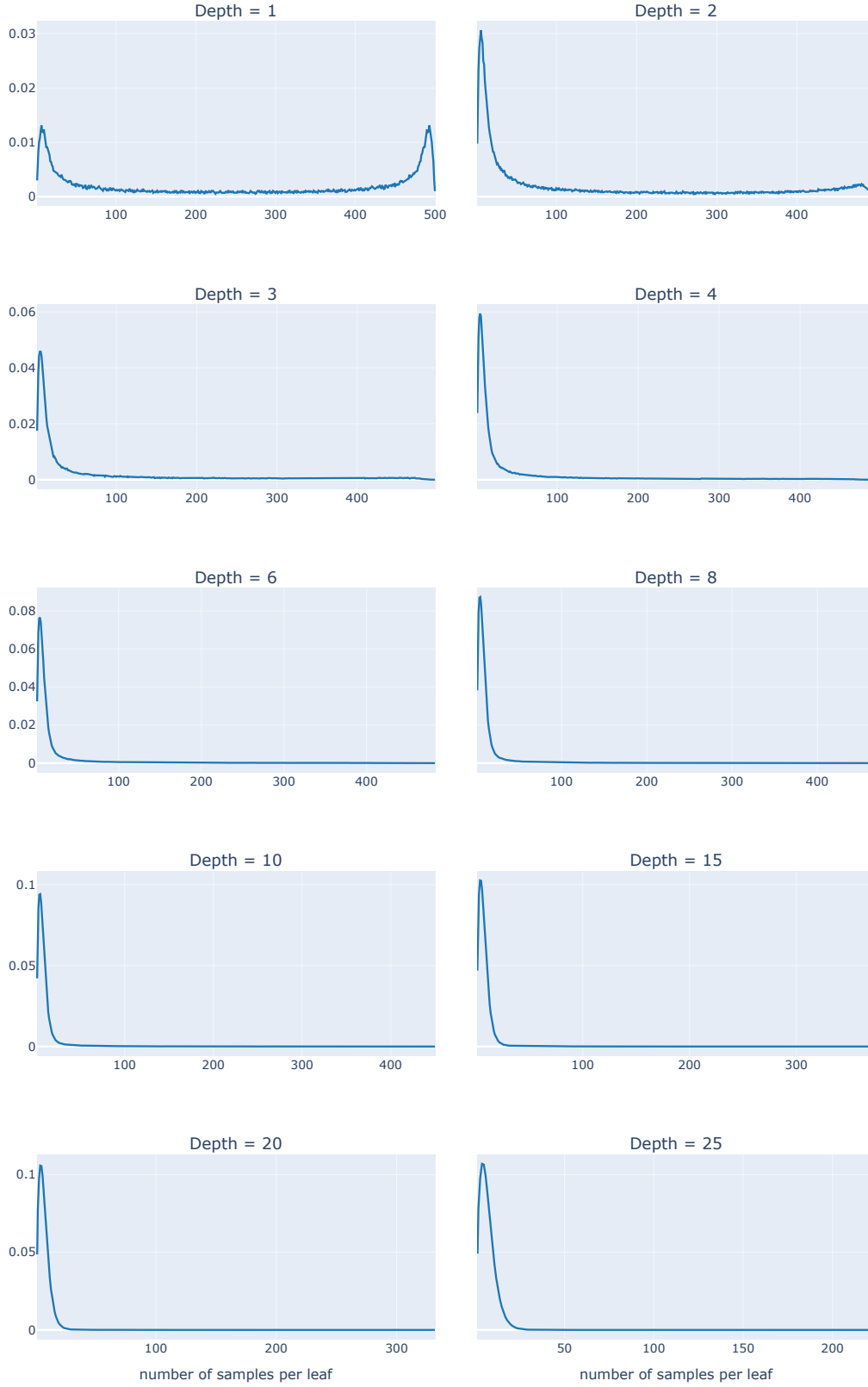


Figure 36: Relative frequency distribution of leaf means for honest estimation with leaf means that have coverage as nan ($n=1000$).

Relative frequency distribution of the number of samples per leaf for $n = 5000$ and honest estimation with nan coverages included (bin width=0.5)

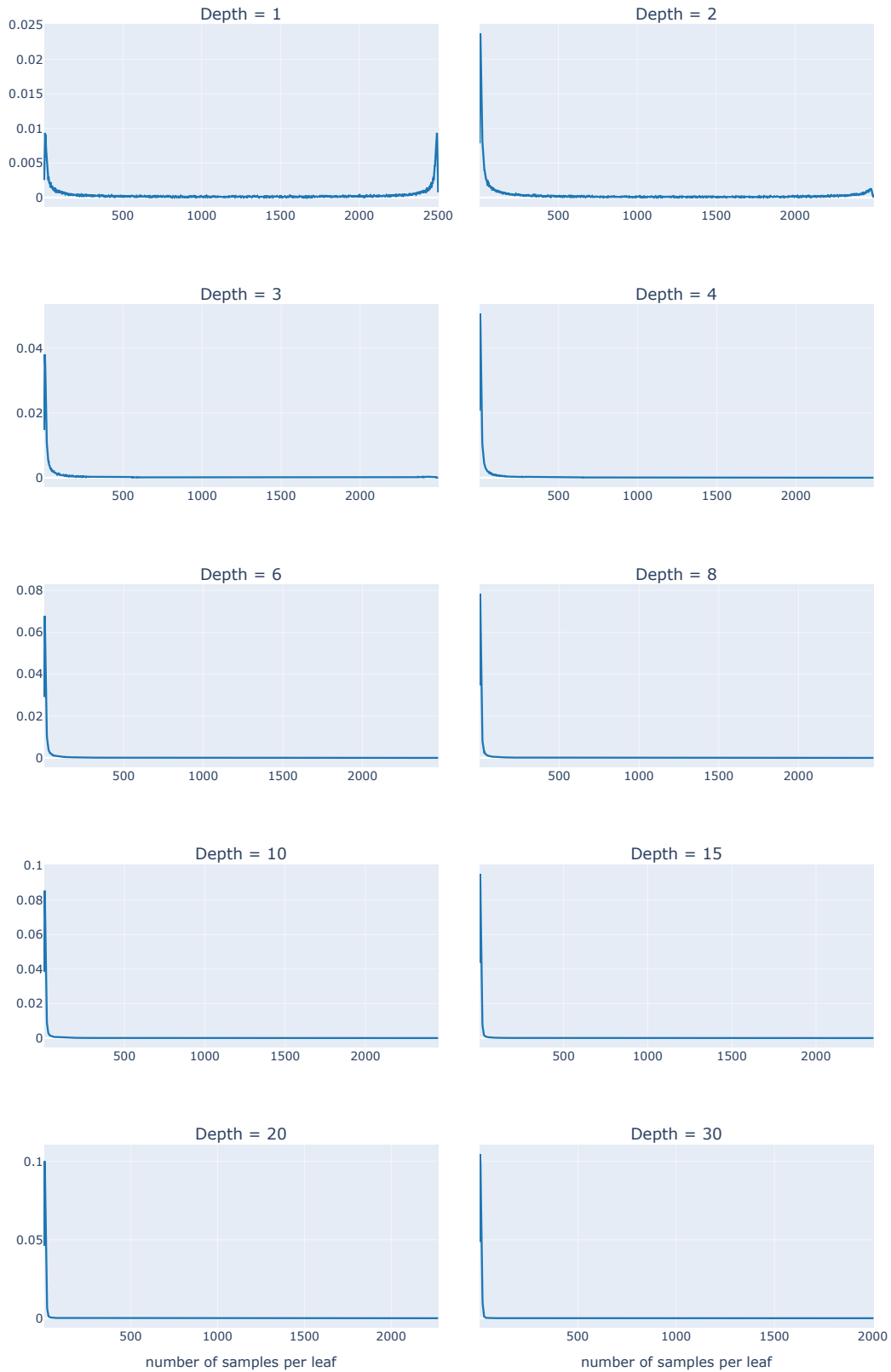


Figure 37: Relative frequency distribution of leaf means for honest estimation with leaf means that have coverage as nan ($n=5000$).

Relative frequency distribution of the number of samples per leaf for $n = 25000$ and honest estimation with nan coverages included (bin width=0.5)

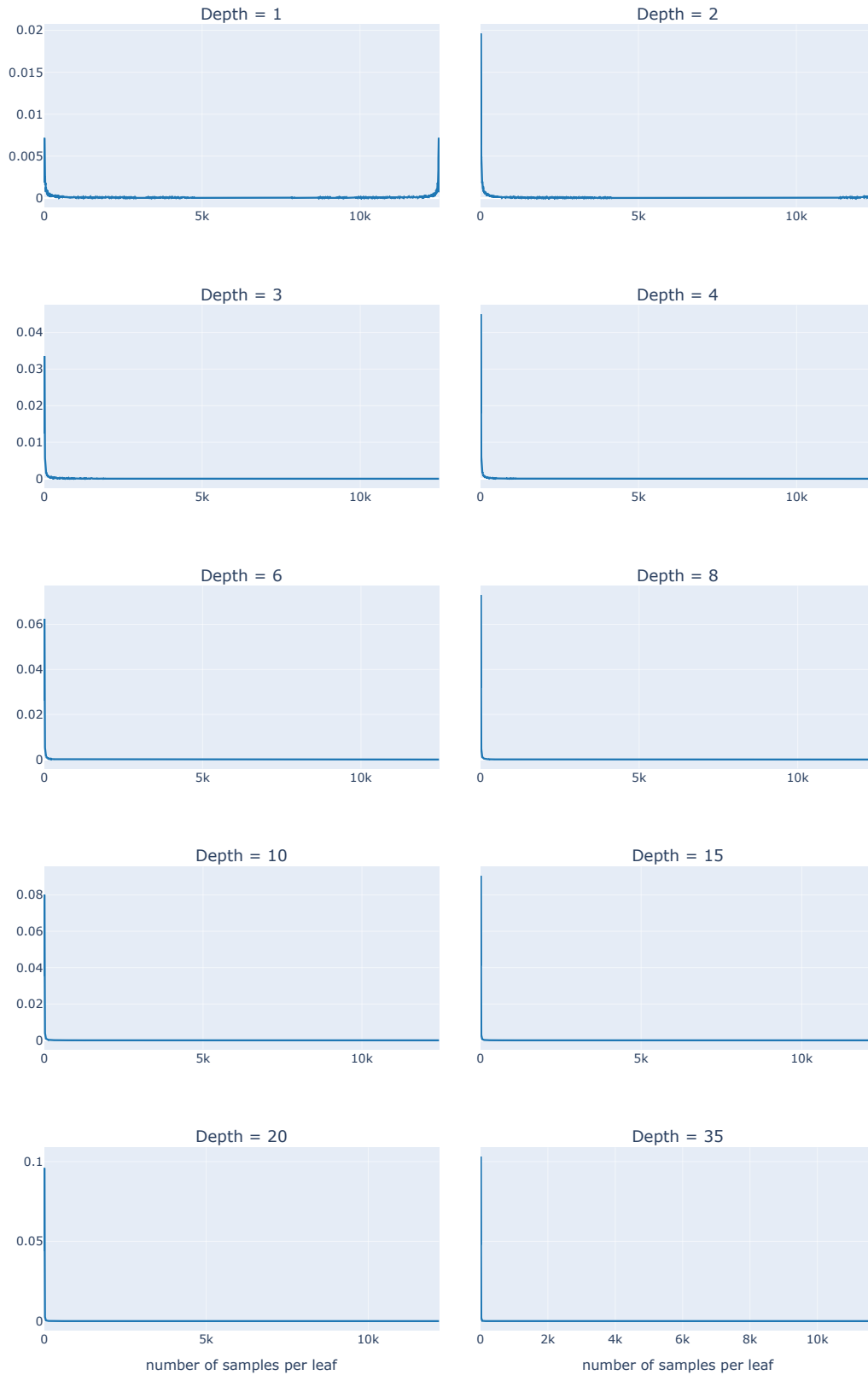


Figure 38: Relative frequency distribution of leaf means for honest estimation with leaf means that have coverage as nan ($n=25000$).

F Mean Leaf Sizes

Table 4: Mean leaf sizes overall by splitting type ($n = 1000$)

Depth	Coverage		Adaptive	Honest
1	0		202.33	218.29
	1		686.32	261.53
	NaN		–	3.76
2	0		97.87	109.92
	1		407.91	148.79
	NaN		–	3.62
3	0		56.89	60.97
	1		233.49	93.05
	NaN		–	3.54
4	0		37.60	39.19
	1		141.38	62.84
	NaN		–	3.49
6	0		19.91	21.67
	1		63.99	34.47
	NaN		–	3.43
8	0		13.21	14.80
	1		36.15	22.59
	NaN		–	3.41
10	0		10.12	11.79
	1		23.77	16.76
	NaN		–	3.39
15	0		7.60	9.06
	1		12.35	11.16
	NaN		–	3.37
20	0		7.01	8.34
	1		8.90	9.64
	NaN		–	3.36
25	0		6.84	8.16
	1		7.60	9.24
	NaN		–	3.36

Table 5: Mean leaf sizes for extreme absolute deviations of leaf means from population mean
(Part 1: 47.0 to 48.0, $n = 1000$)

Abs. Deviation Interval	Depth	Coverage	Adaptive	Honest
(46.999, 47.5]	1	0	54.24	27.12
		1	54.86	27.70
		NaN	–	3.69
	2	0	24.79	16.46
		1	44.89	20.28
		NaN	–	3.69
	3	0	16.31	13.72
		1	29.79	16.35
		NaN	–	3.56
	4	0	12.22	11.95
		1	21.51	13.94
		NaN	–	3.47
	6	0	8.71	9.53
		1	13.71	11.60
		NaN	–	3.38
	8	0	7.52	8.81
		1	10.32	10.44
		NaN	–	3.36
	10	0	7.12	8.52
		1	8.86	9.82
		NaN	–	3.37
(47.5, 48.0]	1	0	43.56	20.57
		1	43.73	22.14
		NaN	–	6.00
	2	0	20.53	15.15
		1	36.82	17.38
		NaN	–	3.60
	3	0	14.76	12.13
		1	24.89	14.97
		NaN	–	3.51
	4	0	10.88	10.81
		1	18.70	13.02
		NaN	–	3.51
	6	0	8.29	9.21
		1	12.26	10.96
		NaN	–	3.42
	8	0	7.41	8.63
		1	9.67	10.19
		NaN	–	3.39
	10	0	7.06	8.30
		1	8.34	9.81
		NaN	–	3.37

Table 6: Mean leaf sizes for extreme absolute deviations of leaf means from population mean
(Part 2: 48.0 to 49.0, $n = 1000$)

Abs. Deviation Interval	Depth	Coverage	Adaptive	Honest
(48.0, 48.5]	1	0	33.22	16.95
		1	34.08	17.33
		NaN	–	7.20
	2	0	16.61	13.64
		1	28.03	15.07
		NaN	–	3.69
	3	0	12.45	12.29
		1	19.20	13.65
		NaN	–	3.44
	4	0	9.65	10.73
		1	14.97	12.51
		NaN	–	3.42
	6	0	7.78	9.61
		1	10.67	11.12
		NaN	–	3.36
	8	0	7.24	9.12
		1	8.83	10.50
		NaN	–	3.34
	10	0	7.00	8.86
		1	8.11	10.20
		NaN	–	3.33
(48.5, 49.0]	1	0	23.27	12.10
		1	23.11	12.79
		NaN	–	5.32
	2	0	12.94	11.24
		1	19.96	12.17
		NaN	–	4.05
	3	0	10.35	10.76
		1	14.99	11.69
		NaN	–	3.72
	4	0	8.56	10.31
		1	12.37	11.32
		NaN	–	3.61
	6	0	7.41	9.97
		1	9.48	10.95
		NaN	–	3.45
	8	0	6.99	9.73
		1	8.27	10.67
		NaN	–	3.40
	10	0	6.87	9.54
		1	7.67	10.54
		NaN	–	3.37

Table 7: Mean leaf sizes for extreme absolute deviations of leaf means from population mean
(Part 3: 49.0 to 50.0, $n = 1000$)

Abs. Deviation Interval	Depth	Coverage	Adaptive	Honest
(49.0, 49.5]	1	0	13.21	8.38
		1	13.78	8.95
		NaN	–	4.23
	2	0	9.98	8.11
		1	12.27	8.74
		NaN	–	4.06
	3	0	8.97	7.95
		1	10.62	8.68
		NaN	–	3.98
	4	0	8.01	7.96
		1	9.65	8.61
		NaN	–	3.95
	6	0	7.37	7.90
		1	8.48	8.57
		NaN	–	3.91
	8	0	7.14	7.88
		1	7.95	8.54
		NaN	–	3.88
	10	0	7.04	7.94
		1	7.65	8.53
		NaN	–	3.86
(49.5, 50.0]	1	0	7.42	6.33
		1	7.35	6.41
		NaN	–	3.03
	2	0	6.74	6.01
		1	6.91	6.38
		NaN	–	3.01
	3	0	6.62	5.97
		1	6.67	6.33
		NaN	–	3.00
	4	0	6.47	5.94
		1	6.53	6.31
		NaN	–	3.01
	6	0	6.35	5.95
		1	6.40	6.31
		NaN	–	3.01
	8	0	6.30	5.93
		1	6.35	6.32
		NaN	–	3.00
	10	0	6.27	5.93
		1	6.33	6.31
		NaN	–	3.00

G Share of Leaves with Coverage for Honest Estimation

Table 8: Percentage of Leaves with Coverage for $n = 1000$

Depth	Proportion of Populated Leaves
1	0.9660
2	0.9075
3	0.8486
4	0.8015
6	0.7376
8	0.6985
10	0.6726
15	0.6411
20	0.6299
25	0.6262

Table 9: Percentage of Leaves with Coverage for $n = 5000$

Depth	Proportion of Populated Leaves
1	0.9755
2	0.9279
3	0.8747
4	0.8310
6	0.7691
8	0.7298
10	0.7036
15	0.6667
20	0.6482
30	0.6321

Table 10: Percentage of Leaves with Coverage for $n = 25000$

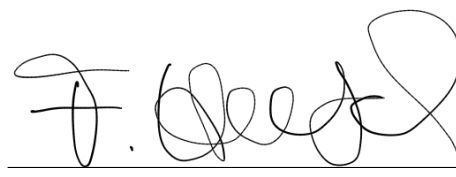
Depth	Proportion of Populated Leaves
1	0.9802
2	0.9421
3	0.8932
4	0.8513
6	0.7886
8	0.7490
10	0.7228
15	0.6843
20	0.6637
35	0.6377

Declaration of Authorship

I hereby confirm that the work presented has been performed and interpreted solely by myself except for where I explicitly identified the contrary. I assure that this work has not been presented in any other form for the fulfillment of any other degree or qualification. Ideas taken from other works in letter and in spirit are identified in every single case.

Bonn, May 26, 2026

Place, Date

A handwritten signature in black ink, appearing to read 'F. Weid', written over a horizontal line.

Signature